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MASTER OF SCIENCE IN CIVIL ENGINEERING

PROJECT II: DESIGN OF 4 STOREYS DUAL SYSTEM R.C BUILDING
ACCORDING TO EUROCODE 8 AND NTC-18

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Table of Contents

LIST OF FIGURES	4
LIST OF TABLES:.....	4
CHAPTER 1	8
INTRODUCTION:	8
1.1 SITE DESCRIPTION:	8
1.2 GEOMETRY:	9
1.3 MATERIAL:	10
CHAPTER 2	16
2.1 PRELIMINARY DESIGN.....	16
2.2 LOAD ANALYSIS.....	18
2.2.1 : Structural Permanent Load	18
CHAPTER 3	40
3.1 MODELLING	40
3.1.1 Beams And Columns:	42
3.1.2 Shear Walls	42
3.1.3 Staircase:	43
3.2 REGULARITY CHECKS:	44
3.2.1. Regularity in plan:.....	44
3.2.2. Regularity in elevation:	46
4. Model Analysis:	47
CHAPTER 5	51
DESIGN REINFORCEMENT:.....	51
5.1 BEAM DESIGN:	51
5.1.1. Local Ductility Check:	54
5.2. Shear Design and Check of Transversal Reinforcement:.....	56
5.3. Shear Reinforcement	58
5.3.1. Critical Area.....	58
5.3.2. Out of Critical Area.....	60
5.3.4. Reinforcement Detailing in Beams:.....	61
5.2 COLUMN DESIGN.....	62
5.2.1. Compression check:	62
5.2.2. Biaxial Bending with axial Compression:.....	63
4.2.1.1. MMN Domain of Columns.....	64
5.2.3. Capacity Check:	64

5.2.4. Shear Check:	65
5.2.5. Local Ductility Check:	67
5.2.6. Stability Check:.....	68
5.2.7. 2 nd Order Effect Check:.....	69
5.2.8. Column Reinforcement Detailing:	69
5.3 SHEAR WALLS.....	70
5.3.1. Geometry Checks:.....	72
5.3.2. Maximum Actions on Shear Walls:	73
5.3.3. Vertical Reinforcement Checks:	74
5.3.4. Horizontal Reinforcement Calculations and Checks:	75
5.3.5. Moment Capacity Check:.....	78
5.3.6. Shear Check:	79
5.3.7. Shear Wall Reinforcement Details:.....	82
CHAPTER 6.....	83
WEIGHT OF REINFORCEMENT:	83
6.1 BEAMS.....	83
6.2 COLUMNS	83
6.3 SHEAR WALLS.....	84
6.4 SAFETY FACOTR:.....	86
CONCLUSION:	86
REFERENCES	87

LIST OF FIGURES:

Figure 1- Arial view of Udine City	8
Figure 2- Structure Layout.....	9
Figure 3- Vertical Cross Section of Structure	10
Figure 4- Stress-Strain relationship.....	11
Figure 5- Stress Strain relationship of Steel	15
Figure 6- Span to depth ratio	17
Figure 7- Typical Floor system.....	17
Figure 8- Floor Slab	19
Figure 9- External Wall.....	20
Figure 10- Stairs.....	20
Figure 11- Inter Story Landing	21
Figure 12-Udine region Snow load characteristics	22
Figure 13- Exposure Coefficient from NTC-2018.....	23
Figure 14- Wind Zoning of Udine Region.....	24
Figure 15- Exposure Coefficient vs Height	26
Figure 16- P+ and P- VS height.....	27
Figure 17-PGA range from INGV	29
Figure 18- PGA range from global with average returning period of 30 years	29
Figure 19- PGA range from Dlubal website with returning period of 475 years	30
Figure 20- Elastic Response Spectrum	31
Figure 21- Identifying the site hazard Phase 1.....	34
Figure 22- Structural working life NTC-8.....	34
Figure 23- phase II Choosing design life and Strategy	35
Figure 24- Determination of Project Action Elastic and Design Response Spectra	36
Figure 25- Spectrum Variable with Spetttri and Simqke,GR software.....	36
Figure 26- Design Response Spectra	37
Figure 27- STRUCTURAL MODEL.....	41
Figure 28-Floor Plan	41
Figure 29- Beams and Columns.....	42
Figure 30- Shear Walls	43
Figure 31- Shear Wall Layout.....	43
Figure 32- Staircase.....	44
Figure 33: Mode 1 (X-direction).....	48
Figure 34: Mode 2 (Y-direction).....	49
Figure 35: Mode 3 (Torsional in Z-axis)	49
Figure 36- Beams, Columns that designed	51
Figure 37: Moment Diagram X-direction Continuous 5 bay beam	52
Figure 38: Moment Diagram y-direction continuous 3 Bay beam	52
Figure 39: Beam in Y-direction Longitudinal Reinf. Check	53
Figure 40: Beam in X-direction Longitudinal Reinf. Check	53
Figure 41: Local Ductility Check - Beam in Y-direction	54
Figure 42: Local Ductility Check - Beam in X-direction	55
Figure 43: Capacity design values of shear forces on beams	57
Figure 44: Shear- Y-Direction	59
Figure 45: Shear- X-Direction	60

Figure 46: MMN Domain of Column C1, C2, C9 Respectively	64
Figure 47- Bending Moment Translation.....	70
Figure 48: Design Envelopes for Shear walls.....	72
Figure 49: Confined boundary element of free-edge wall end	77
Figure 50: MN Interaction Diagrams for shear walls	78
Figure 51: Shear Amplification coefficient.....	80
Figure 52: Shear Envelope for SW-1.....	80

LIST OF TABLES:

Table 1- Concrete Properties.....	10
Table 2- Material Properties.....	11
Table 3: Strength and Deformation Characteristics of Concrete (Ref Eurocode 2)	12
Table 4- Steel Properties	14
Table 5- Permanent Load on beams.....	18
Table 6- Slab Load.....	19
Table 7- External Wall Loads	20
Table 8- Stair Loads	20
Table 9- Inter Story Landing Load	21
Table 10-Italian Characteristics Snow Load Zones[3].....	22
Table 11- Snow Load Shape Coefficient	23
Table 12- Basic Wind Velocities with Italian regions[3].....	25
Table 13- K_r , Z_o , and Z_{min} Values and categories	26
Table 14- Wind Load	27
Table 15- Value of C_u for building Class	35
Table 16-SLS Static and Dynamic Load Combination.....	38
Table 17- SLE-Rare Static Load Combinations.....	38
Table 18- SLE - Frequent static load Combinations.....	39
Table 19- SLE – Quasi- static load Combinations.....	39
Table 20- Weight Center and Shear Center	46
Table 21: Modal Analysis.....	47
Table 22: Modal Participation Masses.....	48
Table 23:Inclined Bar check - Y direction Beam.....	57
Table 24:Inclined Bar check - X direction Beam.....	58
Table 25:Shear Reinforcement design in critical region (Y-Direction).....	59
Table 26: Transversal Reinf outside critical region- Y-direction Beam	60
Table 27:Transversal Reinf outside critical region- Y-direction Beam	61
Table 28: Reinforcement Detailing in Beams.....	61
Table 29: Comparison of Max excitation in Columns.....	62
Table 30: Column Compression check	63
Table 31: Column Check for Biaxial Bending.....	63
Table 32: Column Geometrical Ratio	64
Table 33: Capacity Check in Columns	65
Table 34: Column stirrup design.....	66
Table 35: Shear Check in Column	67
Table 36: Local Ductility of Column.....	68
Table 37: 2nd order effect check.....	69
Table 38: Column Reinforcement Detailing.....	69
Table 39: Critical height h_{cr} determination	72
Table 40: Critical Lengths of Shear walls.....	73
Table 41: Max Excitation in Shear Walls.....	73
Table 42: Compression Check	74
Table 43: Reinforcement in Critical Regions (shear wall)	74
Table 44:Reinforcement outside Critical Regions (shear wall)	75
Table 45:Local Ductility Check.....	77

Table 46: Shear tension and shear compression failure	81
Table 47: Reinforcement Details of Shear Walls	82
Table 44- Beam Longitudinal Reinforcement Weight.....	83
Table 45- Beam Shear Reinforcement Weight	83
Table 46- Column Longitudinal Reinforcement Weight	83
Table 47- Column Shear Reinforcement Weight.....	84
Table 48- Shear Wall Longitudinal Reinforcement Weight	84
Table 49- Shear Wall Shear Reinforcement Weight.....	86
Table 50- Safety Factor Evaluation.....	86

CHAPTER 1

INTRODUCTION:

1.1 SITE DESCRIPTION:

Design the building of a 4-story reinforced concrete structure according to Eurocode 8. Some information is carried out with the Italian national annex NTC-2008. This building is in the city of Udine. Udine is a city and commune in northeastern Italy, in the middle of the Friuli-Venezia Giulia region, between the Adriatic Sea and the Alps. The construction site is 113m above sea level. Given soil type C. Soil C type is constituted by deposits of coarse-grained soils, medium-dense, or fined-grained soils. According to website data, all the towns of the city fall under the category of seismic zone II [1]. Environmental conditions have been considered “aggressive”, so it has been used in the exposure class XC1. Our site is 113m above sea level.



Figure 1- Arial view of Udine City

1.2 GEOMETRY:

The geometry of the building has been assigned. The project deals with 4 floors building, whose inter-floor height is 3.3 m. The total height of the building is 13.2m. The building is not symmetrical concerning both longitudinal and transversal axis, the building dimension is 19.4m x 15.4m. The multi-story building is intended for residential use and is not a cellar. There is a staircase which is located at the center of the building, having internal beam support and a landing. The staircase waist slab is one side supported by a floor beam and the other end supported by an intermediate landing. There are four earthquake-resistant cut walls, two for each main direction of the building; the geometry of the structure remains the same for all floors. The structure is designed, constructed, and tested according to high ductility class (DCH), verified according to Eurocode 8

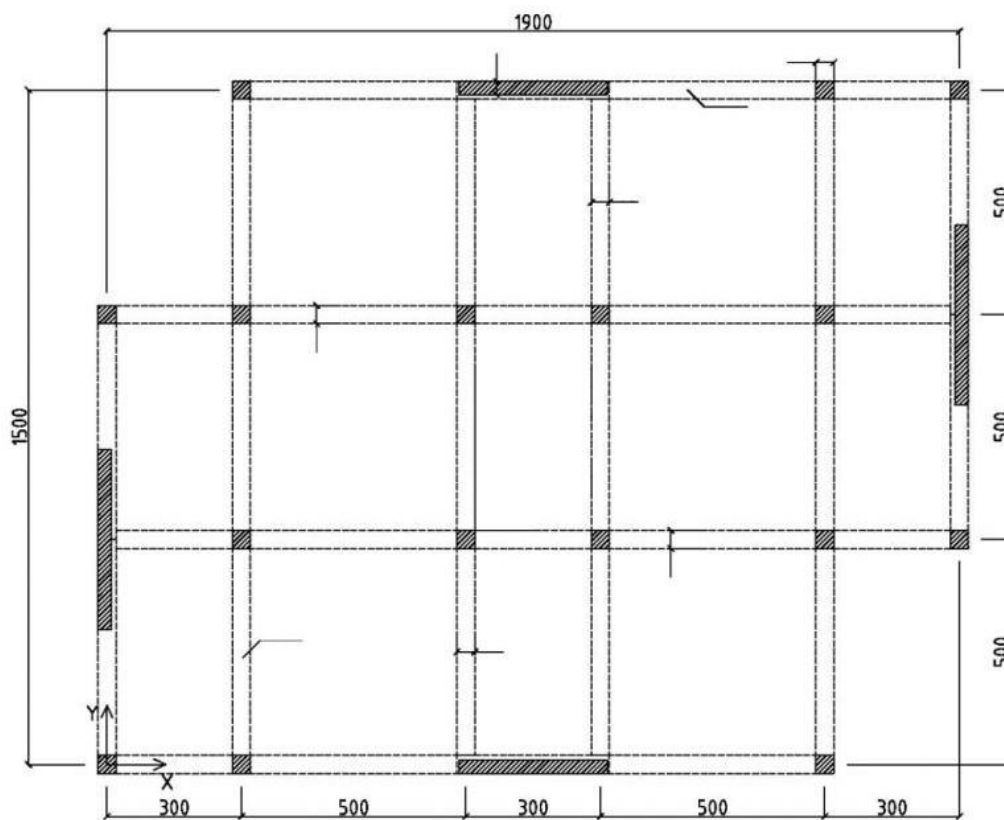


Figure 2- Structure Layout

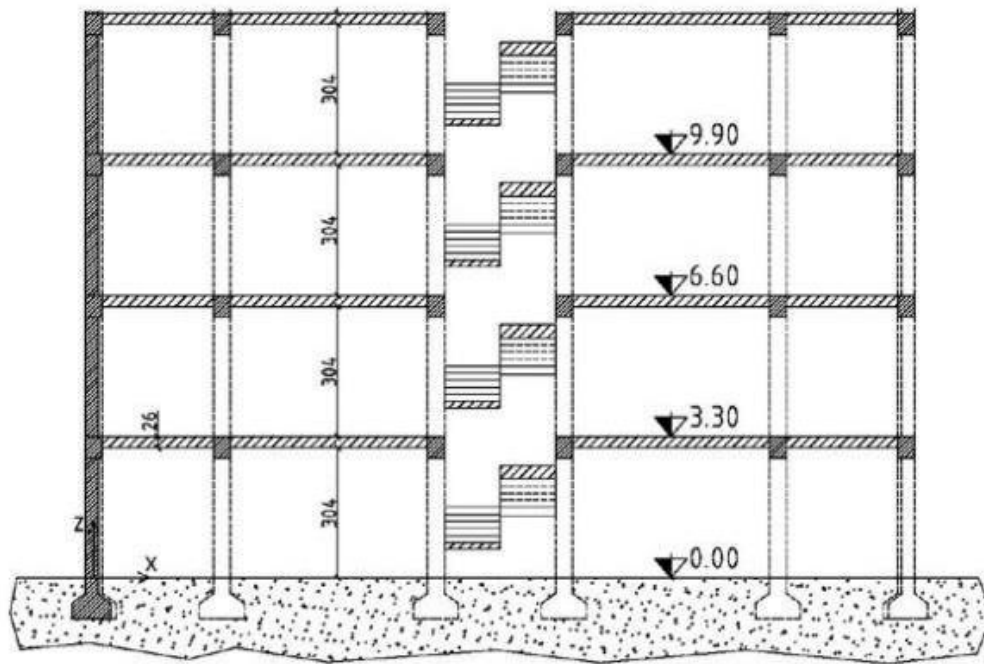


Figure 3- Vertical Cross Section of Structure

The size of beams, columns, shear walls, and slabs are preliminarily designed according to load conditions and will be presented in the next pages.

1.3 MATERIAL:

1.3.1 CONCRETE:

C35/40 has been used as a concrete class, whose main characteristics are

DESCRIPTION	Symbol(Unit)	Values
Characteristic cylinder compressive strength	f_{ck} (MPa)	35
Characteristic cylinder compressive strength	$f_{ck,cube}$ (MPa)	45
Mean cylinder compressive strength	f_{cm} (MPa)	43
Mean tensile strength	f_{ctm} (MPa)	3.21
Elastic modulus	E_{cm} (MPa)	34077
Design compressive strength (for $\alpha_{cc}=1.00$)	f_{cd} (MPa) (for $\alpha_{cc}=1.00$)	23.33
Design compressive strength (for $\alpha_{cc}=0.85$)	f_{cd} (MPa) (for $\alpha_{cc}=0.85$)	19.83
Design tensile strength (for $\alpha_{ct}=1.00$)	f_{ctd} (MPa) (for $\alpha_{ct}=1.00$)	1.5
Minimum longitudinal tension reinforcement ratio	ρ_{min} (%)	0.185
Minimum shear reinforcement ratio	$\rho_{w, min}$ (%)	0.105

Table 1- Concrete Properties

The associated coefficients are:

- $\gamma_c = 1.5$
- $\alpha_{cc} = 0.85$ (For gravity design)
- $\alpha_{cc} = 1$ (For Seismic Design)

The behavior of concrete at ULS has been studied with the stress-block constitutive law. Under the act, the concrete section is expected to be in tension (no reactions in concrete) until the plastic deformation, we use $\varepsilon_{c1} = 0.07\%$ for our concrete class. After the point $\varepsilon_{c1} = 0.07\%$ section has constant tension until fracture, the Corresponding deformation value is equal to $\varepsilon_{cu} = 0.35\%$.

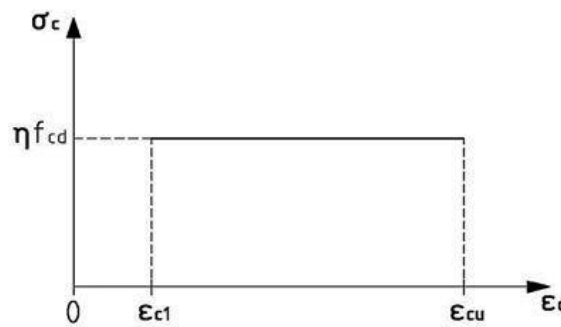


Figure 4- Stress-Strain relationship

The coefficient η , which defines the effective resistance, is a unit value because the characteristic cylindrical compressive strength of the concrete used is less than 50 MPa. According to the stress block model, concrete does not react when it is subjected to traction.

General material properties for reinforced concrete according to Eurocodes.

Material Property	Value
Density ρ	$\approx 2500 \text{ kg/m}^3$
Unit weight γ	$\approx 25.0 \text{ kN/m}^3$
Modulus of elasticity E_{cm}	see table above
Shear modulus G	$G = E / [2 \cdot (1 + \nu)]$
Poisson's ratio ν (uncracked concrete)	0.2
Poisson's ratio ν (cracked concrete)	0.0
Coefficient of linear thermal expansion α	$10 \times 10^{-6} \text{ }^\circ\text{K}^{-1}$

Table 2- Material Properties

Strength classes for concrete															Analytical relation / Explanation	
f_{ck} (MPa)	12	16	20	25	30	35	40	45	50	55	60	70	80	90		
$f_{ck,cube}$ (MPa)	15	20	25	30	37	45	50	55	60	67	75	85	95	105	2.8	
f_{cm} (MPa)	20	24	28	33	38	43	48	53	58	63	68	78	88	98	$f_{cm} = f_{ck} + 8$ (MPa)	
f_{ctm} (MPa)	1,6	1,9	2,2	2,6	2,9	3,2	3,5	3,8	4,1	4,2	4,4	4,6	4,8	5,0	$f_{ctm} = 0,30 \cdot f_{ck}^{(2/3)} \leq C50/60$ $f_{ctm} = 2,12 \cdot \ln(1 + (f_{cm}/10))$ $> C50/60$	
$f_{tk,0.05}$ (MPa)	1,1	1,3	1,5	1,8	2,0	2,2	2,5	2,7	2,9	3,0	3,1	3,2	3,4	3,5	$f_{tk,0.05} = 0,7 \cdot f_{ctm}$ 5% fractile	
$f_{tk,0.95}$ (MPa)	2,0	2,5	2,9	3,3	3,8	4,2	4,6	4,9	5,3	5,5	5,7	6,0	6,3	6,6	$f_{tk,0.95} = 1,3 \cdot f_{ctm}$ 95% fractile	
E_{cm} (GPa)	27	29	30	31	33	34	35	36	37	38	39	41	42	44	$E_{cm} = 22(f_{cm}/10)^{0.3}$ (f_{cm} in MPa)	
ϵ_{t1} (‰)	1,8	1,9	2,0	2,1	2,2	2,25	2,3	2,4	2,45	2,5	2,6	2,7	2,8	2,8	see Figure 3.2 $\epsilon_{t1}(\text{‰}) = 0,7 \cdot f_{cm}^{0.25} \leq 2,8$	
ϵ_{cu1} (‰)	3,5					3,2					3,0	2,8	2,8	2,8	see Figure 3.2 for $f_{ck} \geq 50$ Mpa $\epsilon_{cu1}(\text{‰}) = 2,8 + 27(98 - f_{ck})/100^4$	
ϵ_{t2} (‰)	2,0					2,2					2,3	2,4	2,5	2,6	see Figure 3.3 for $f_{ck} \geq 50$ Mpa $\epsilon_{t2}(\text{‰}) = 2,0 + 0,085(f_{ck} - 50)^{0.53}$	
ϵ_{cu2} (‰)	3,5					3,1					2,9	2,7	2,6	2,6	see Figure 3.3 for $f_{ck} \geq 50$ Mpa $\epsilon_{cu2}(\text{‰}) = 2,6 + 35(90 - f_{ck})/100^4$	
n	2,0					1,75					1,75	1,6	1,45	1,4	1,4	for $f_{ck} \geq 50$ Mpa $n = 1,4 + 23,4(90 - f_{ck})/100^4$
ϵ_{t3} (‰)	1,75					1,8					1,8	1,9	2,0	2,2	2,3	see Figure 3.4 for $f_{ck} \geq 50$ Mpa $\epsilon_{t3}(\text{‰}) = 1,75 + 0,55(f_{ck} - 50)/40$
ϵ_{cu3} (‰)	3,5					3,1					2,9	2,7	2,6	2,6	2,6	see Figure 3.4 for $f_{ck} \geq 50$ Mpa $\epsilon_{cu3}(\text{‰}) = 2,6 + 35(90 - f_{ck})/100^4$

Table 3.1 Strength and deformation characteristics for concrete

BS EN 1992-1-1:2004
EN 1992-1-1:2004 (E)

Table 3: Strength and Deformation Characteristics of Concrete (Ref Eurocode 2)

Design tensile strength f_{ctd} :

The design tensile strength f_{td} is determined according to EN1992-1-1 §3.1.6(2)P:

$$f_{ctd} = \alpha_{ct} \cdot f_{ctk,0.05} / \gamma_c$$

where γ_c is the partial safety factor for concrete for the examined design state, as specified in EN1992-1-1 §2.4.2.4 and the National Annex.

The coefficient α_{ct} considers long-term effects on the tensile strength and unfavourable effects, resulting from the way the load is applied. It is specified in EN1992-1-1 §3.1.6(2)P and the National Annex (for bridges see also EN1992-2 §3.1.6(102)P and the National Annex).

Modulus of elasticity E_{cm} :

The elastic deformation properties of reinforced concrete depend on its composition, especially on the aggregates. Approximate values for the modulus of elasticity E_{cm} (secant value between $\sigma_c = 0$ and $0.4f_{cm}$) for concretes with quartzite aggregates are given in EN1992-1-1 Table

3.1 According to the following formula:

$$E_{cm} [\text{MPa}] = 22000 \cdot (f_{cm}/10\text{MPa})^{0.3}$$

According to EN1992-1-1 §, 3.1.3(2) for limestone and sandstone aggregates the value of E_{cm} should be reduced by 10% and 30% respectively. For basalt aggregates, the value of E_{cm} should be increased by 20%. The values of E_{cm} given in EN1992-1-1 should be regarded as indicative for general applications, and they should be specifically assessed if the structure is likely to be sensitive to deviations from these general values.

The variation of the modulus of elasticity $E_{cm}(t)$ with time t is specified in *EN1992-1-1* §3.1.3(3).

Poisson ratio ν :

According to *EN1992-1-1* §3.1.3(4), the value of Poisson's ratio ν may be taken equal to $\nu = 0.2$ for uncracked concrete and $\nu = 0$ for cracked concrete.

Coefficient of thermal expansion α :

According to *EN1992-1-1* §3.1.3(5), the value of the linear coefficient of thermal expansion α may be taken equal to $\alpha = 10 \cdot 10^{-6} \text{ } ^\circ\text{K}^{-1}$, unless more accurate information is available.

Minimum longitudinal reinforcement ρ_{min} for beams and slabs:

The minimum longitudinal tension reinforcement for beams and the main direction of slabs is specified in *EN1992-1-1* §9.2.1.1(1).

$$A_{s,min} = 0.26 \cdot (f_{ctm} / f_{yk}) \cdot b_t \cdot d$$

Where b_t is the mean width of the tension zone and d is the effective depth of the cross-section, f_{ctm} is the mean tensile strength of concrete, and f_{yk} is the characteristic yield strength of steel.

The minimum reinforcement is required to avoid brittle failure. Typically, a larger quantity of minimum longitudinal reinforcement for crack control is required by *EN1992-1-1* §7.3.2. Sections containing less reinforcement should be considered unreinforced.

According to *EN1992-1-1* §9.2.1.1(1) Note 2 for the case of beams where a risk of brittle failure can be accepted, $A_{s,min}$ may be taken as 1.2 times the area required in ULS verification.

Minimum shear reinforcement $\rho_{w,min}$ for beams and slabs:

The minimum shear reinforcement for beams and slabs is specified in *EN1992-1-1* §9.2.2(5).

$$\rho_{w,min} = 0.08 \cdot (f_{ck}^{0.5}) / f_{yk}$$

Where f_{ck} is the characteristic compressive strength of concrete and f_{yk} is the characteristic yield strength of steel.

The shear reinforcement ratio is defined in *EN1992-1-1* §3.1.3(5) as:

$$\rho_w = A_{sw} / [s \cdot b_w \cdot \sin(\alpha)]$$

where b_w is the width of the web and s is the spacing of the shear reinforcement along the length of the member. The angle α corresponds to the angle between shear reinforcement and the longitudinal axis. For typical shear reinforcement with perpendicular legs $\alpha = 90^\circ$ and $\sin(\alpha) = 1$.

1.3.2 STEEL:

The building is designed considering High Ductility Class, so we need to use the Steel class B or C, section 5.5.1.1 of Eurocode 8, considering Class C of design. The properties are calculated below:

Material Design Properties for Structural Steel S450 according to *EN1993-1-1* §3.2.6

Material Property	Value
Density ρ	$\approx 7850 \text{ kg/m}^3$
Unit weight γ	$\approx 78.5 \text{ kN/m}^3$
Modulus of elasticity E (Young's modulus)	210000 MPa
Shear modulus G	$G = E / [2 \cdot (1 + \nu)] \approx 81000 \text{ MPa}$
Yield strength f_{yd}	391.63 Mpa

Ultimate strength f_u	563 Mpa
Poisson's ratio in elastic range ν	0.30
Coefficient of linear thermal expansion α	$12 \times 10^{-6} \text{ } ^\circ\text{K}^{-1}$

Table 4- Steel Properties

Nominal values of structural steel yield strength and ultimate strength:

For structural design according to Eurocode 3 (EN1993-1-1), the nominal values of the yield strength f_{yd} and the ultimate strength f_u for structural steel are obtained as a simplification from EN1993-1-1 Table 3.1, which is reproduced above in tabular format.

The provided values for f_{yd} and f_u are nominal. To structural design appropriate material safety factors are applicable, such as the factors γ_{M0} , γ_{M1} , γ_{M2} , etc. following EN1993-1-1 §6.1.

The steel classes are defined in the corresponding standards:

EN 10025-2 Hot rolled products of structural steel. Technical delivery conditions for non-alloy structural steels

EN 10025-3 Hot rolled products of structural steel. Technical delivery conditions for normalized/normalized rolled weldable fine-grain structural steels

EN 10025-4 Hot rolled products of structural steel. Technical delivery conditions for thermo mechanical rolled weldable fine grain structural steels

EN 10025-5 Hot rolled products of structural steel. Technical delivery conditions for structural steels with improved atmospheric corrosion resistance

EN 10025-6 Hot rolled products of structural steel. Technical delivery conditions for flat products of high-yield strength structural steels in the quenched and tempered condition

EN 10210-1 Hot finished structural hollow sections of non-alloy and fine grain steels. Technical delivery requirements

EN 10219-1 Cold-formed welded structural hollow sections of non-alloy and fine-grain steels. Technical delivery requirements

Unit weight and density of structural steel

The unit weight of structural steel is specified in the design standard EN 1991-1-1 Table A.4 between 77.0 kN/m^3 and 78.5 kN/m^3 . For structural design, it is standard practice to consider the unit weight of structural steel equal to $\gamma = 78.5 \text{ kN/m}^3$ and the density of structural steel approximately $\rho = 7850 \text{ kg/m}^3$.

Modulus of elasticity of structural steel

The modulus of structural steel's elasticity (Young's modulus) is specified in the design standard EN 1993-1-1 Section 3.2.6. For the structural design, the modulus of elasticity of structural steel is considered as $E = 210000 \text{ MPa}$.

Design values of additional material mechanical properties for structural steel

According to EN1993-1-1 §3.2.6, the design values of the material properties and coefficients for structural steel are:

Modulus of elasticity (Young's modulus):

$$E = 210000 \text{ MPa}$$

Shear modulus:

$$G = E / [2 \cdot (1 + \nu)] = 80769 \text{ MPa} \approx 81000 \text{ MPa}$$

Poisson's ratio in the elastic range:

$$v = 0.30$$

Coefficient of linear thermal expansion:

$$\alpha = 12 \times 10^{-6} \text{ } ^\circ K^{-1}$$

According to EN1991-1-3 §3.2.6, for calculating the structural effects of unequal temperatures in composite concrete-steel structures to EN 1994 the coefficient of linear thermal expansion may be taken as $\alpha = 10 \times 10^{-6} \text{ }^{\circ}\text{K}^{-1}$, i.e., the same as the coefficient of thermal expansion of concrete.

For normal design, either of the following assumptions may be made as shown below in figure 5. an inclined top branch with a strain limit of ϵ_{ud} and maximum stress ($k f_{yk} / \gamma_s$) at, $\epsilon u k$, where $k = (f_t / f_y)_k$. A horizontal top branch without the need to check the strain limit.

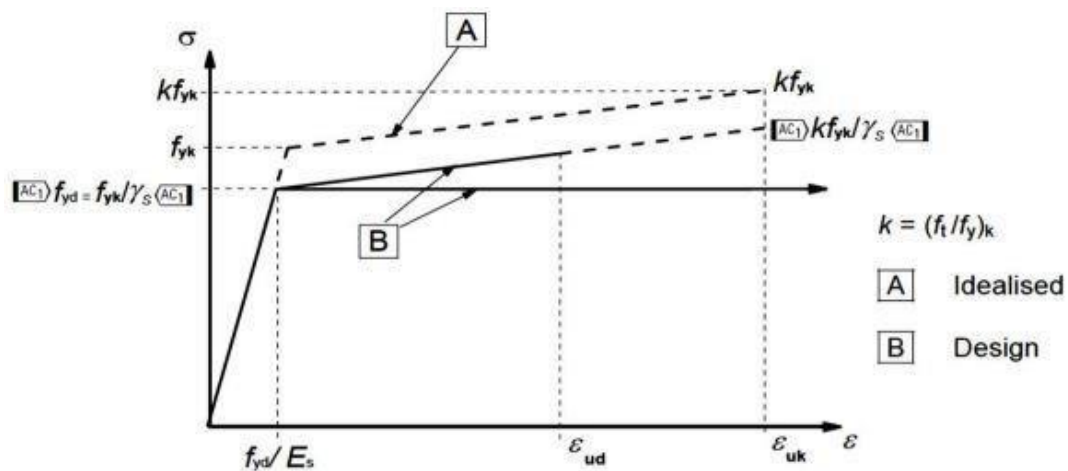


Figure 5- Stress Strain relationship of Steel

CHAPTER 2

2.1 PRELIMINARY DESIGN

In this section, we will preliminarily design our column, beam, slab, and shear wall for the calculation of load analysis. This is a kind of draft design of sizes of different structural elements.

2.1.1 COLUMN:

We will preliminarily design the sizes of columns we will take a square column we will find one side of the column with loads and a tributary area of the column.

Structural loads for residential building = 10 kN/m^2

Tributary area of column = $(250+150) \times (500) = 200000 \text{ cm}^2 = 20 \text{ m}^2$

Total number of floors = 4

Total axial load = loads * No of Floors * T.A + 10% for total Axial load (for column)

$$= 10 \times 20 \times 4 + 10\%(10 \times 20 \times 4) = 880 \text{ KN}$$

800 KN will be axial load on the column then with the formula we will find one side of the column.

As will be taken as 1-2%. We took 1.5%.

$$b = \sqrt{\frac{N_{ed}}{0.8f_{cd} + A_s f_{yd}}}$$

$$b = \sqrt{\frac{880 \times 10^3}{0.8 \times 19.83 + \frac{1.5}{100} \times 391.3}}$$

$$b = 201 \text{ mm}$$

As per Eurocode requirement in seismic zones, the minimum column size is 25×25 so we are designing against earthquake load with high seismicity, so we took a $50 \times 50 \text{ cm}$ column. As our columns are secondary lateral load-resisting elements they should not provide more than 15% of the total stiffness of the system against the lateral load-resisting system.

2.1.2 BEAMS:

With the span-to-depth ratio, we can find the depth of the beam. So, we took beam to depth ratio as 15, and from the chart “Adapted from Goodchild, C. H., Webster, RM, and Elliott, Kim S.2009, “Economic Concrete Frame Elements.” British Cement Association” it came like 35mm so for safety reason we took 50mm depth with 40mm width.

$$\text{Span/depth} = 15$$

$$500/15 = \text{depth}$$

$$\text{Depth} = 33.33 \text{ cm}$$

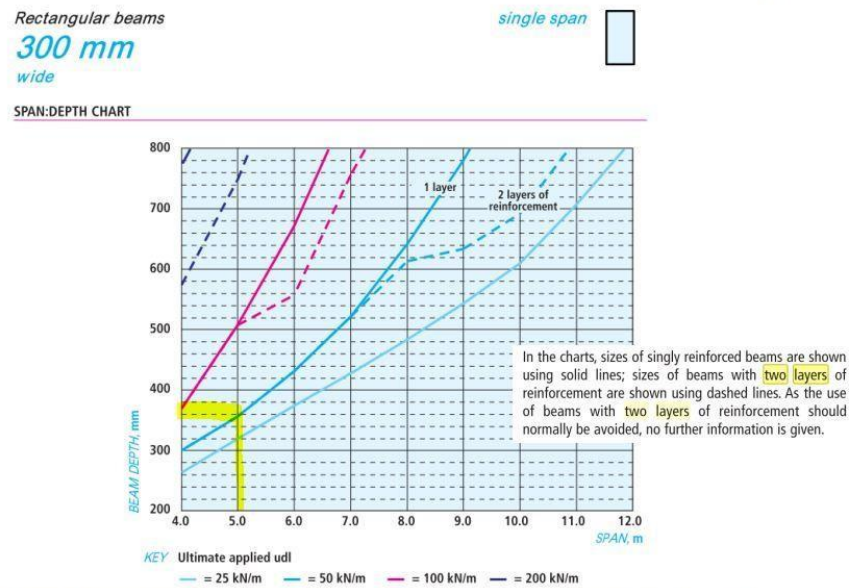


Figure 6- Span to depth ratio

So, we adopted beam size as 40cm*50cm.

2.1.3 SHEAR WALLS:

In the structure, there is only one type of wall, a thickness of 30 cm. Given the geometry, which is characterized by the main development in length and height, it is considered a distributed load per unit area.

2.1.4 SLABS:

The projected floors are of the traditional type, which is typically being used in Italy with bricks of a thickness of 22 cm and slab finishing works thickness of 4 cm. The 10 cm wide bricks are arranged with 50 cm spacing. The total thickness of the deck is equal to 26 cm. Items brought from the lower deck are plaster, a lightweight subfloor, and a tiled floor, as illustrated.

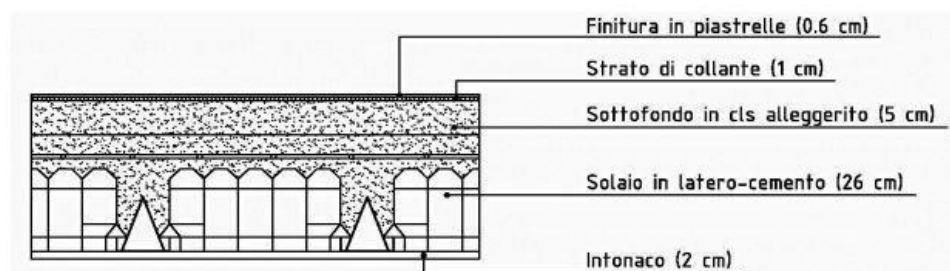


Figure 7- Typical Floor system

2.2 LOAD ANALYSIS:

In this part, the loads acting on the structure are defined; they are divided, in general, into permanent loads - the weight function of the materials used - and variable loads - imposed by the legislation for the different types of environments.

2.2.1 : STRUCTURAL PERMANENT LOAD:

The test of the building structure is made entirely of reinforced concrete, therefore, according to the standard, it has a specific weight equal to 25.00 kN/m^3 . Conventionally they are followed by the symbol G_1 and, therefore, the consequent loads for unit length or surface area are defined by the symbol g_1 .

BEAMS:

In the structure, there is a single type of beam, of section $40 \times 50 \text{ cm}$; the linear load calculation consists in multiplying the specific weight of the concrete for the area of the section:

$$\text{Beam: } g_1 = 25.00 \times 0.40 \times 0.50 = 5.00 \text{ kN/m}$$

COLUMNS:

In the structure, there is only one type of column of section $40 \times 40 \text{ cm}$; the linear load calculation consists of multiplying the specific weight of the concrete for the area of the section:

$$\text{Column: } g_1 = 25.00 \times 0.50 \times 0.50 = 6.25 \text{ kN/m}$$

SHEAR WALLS:

The calculation consists in multiplying the specific weight of the concrete by the wall thickness:

$$\text{Wall: } g_1 = 25.00 \times 0.30 = 7.50 \text{ kN/m}^2$$

2.2.2 : NON-STRUCTURAL PERMANENT LOAD:

According to rules, the non-structural dead loads are those not removable during the normal operation of the building. Conventionally they are followed by the G_2 symbol and, therefore, the consequent overall loads per unit length or surface area are defined by the g_2 symbol. Considering the loads on the beams, the weight of the floors, the external walls, stairs, and landings between floors.

LOAD ON BEAMS:

For loads on the beams, it means all the finishing packages on the beam excluding the slab; permanent loads are the plaster soffit, the lighter the background, a tiled floor, and the incidence of barriers. As they are defined, they only affect the internal beams of the intermediate floors and all those of the plan; the total load is shown in Table 4.

Material	Load	Units
Plastic	0.4	KN/m ²
Lightweight Subfloor	0.65	KN/m ²
Ceramic Floor	0.4	KN/m ²
Partitions	1.0	KN/m ²
Total- G_{beam}	2.45	KN/m²

Table 5- Permanent Load on beams

$$q_{b,int} = 2.0 \text{ KN/m}^2 \text{ (Category A)}$$

$$q_{b,roof} = 0.5 \text{ KN/m}^2 \text{ (Category H1)}$$

With this, the values are also 2.5 as we took 2.45 KN/m^2

FLOOR SLAB:

Floor Slabs The projected floors are of the traditional type, with bricks of a thickness of 22 cm and slab finishing works thickness of 4 cm. The 10 cm wide bricks are arranged with 50 cm spacing. The total thickness of the deck is equal to 26 cm. Items brought from the lower deck are plaster, a lightweight subfloor, and a tiled floor, as illustrated.

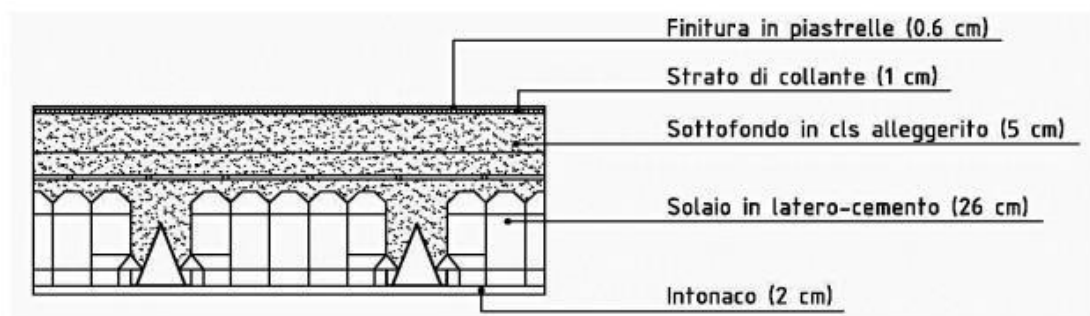


Figure 8- Floor Slab

It is assumed that the floors are all equal, including that of coverage, whereas the equivalent weight incidence rate consists of the partitions and the layer of more screed waterproofing. In fact, in addition, concerning figure 3 also considers the presence of the partition's interior of the premises. Since these may change over time, it is possible to estimate their incidence as a distributed load. Table 4 shows the permanent loads of the slabs.

Material	Value	Units
Slab	3.6	KN/m ²
Plaster	0.4	KN/m ²
Lightweight Subfloor	0.65	KN/m ²
Ceramic Floor	0.4	KN/m ²
Partitions	1.0	KN/m ²
Total G-slab	6.05	KN/m²

Table 6- Slab Load

EXTERNAL WALL LOAD:

The external walls have a total thickness of about 40 cm and are made up of external plaster, brick blocks, plaster, insulating, hollow brick, and interior plaster, as shown in figure 7, where the loads are shown in Table 6.



Figure 9- External Wall

Material	Value	Units
Exterior Plaster	0.6	KN/m ²
Brick block	2.6	KN/m ²
Rough Plaster	0.4	KN/m ²
Insulation	0.01	KN/m ²
Hollow Blocks	0.64	KN/m ²
Interior plaster	0.4	KN/m ²
Total-External	4.65	KN/m²

Table 7- External Wall Loads

STAIRS:

The rampant staircase to the slab is made of a slab 15 cm thick reinforced concrete soffit is plastered and extrados weigh the stairs and the marble covering, as shown in figure 8. Table 6 shows the values of permanent loads about the stairs.

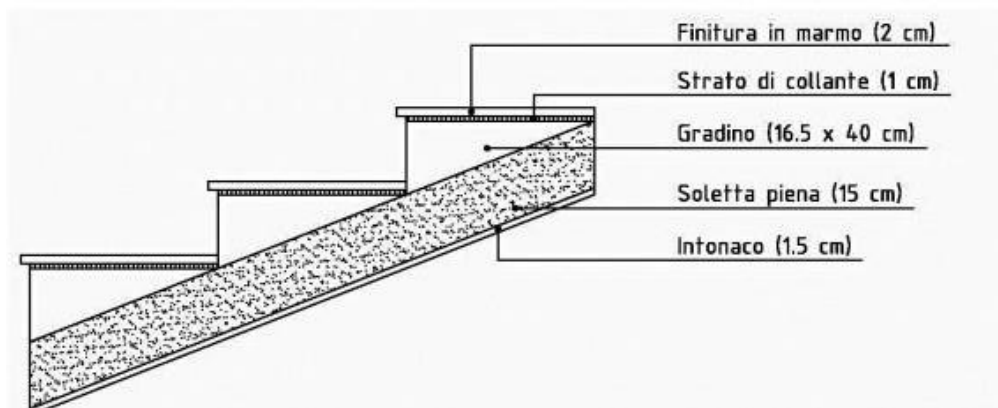


Figure 10- Stairs

Material	Value	Units
Slab	3.75	KN/m ²
Plaster	0.3	KN/m ²
Step	2.06	KN/m ²
Marble	0.8	KN/m ²
Total- G_{stairs}	6.91	KN/m²

Table 8- Stair Loads

LANDING INTER-STORY:

They are made of reinforced concrete and have a thickness of 20 cm. The soffit is plastered while extrados rests the marble covering, as shown in figure 11.



Figure 11- Inter Story Landing

Material	Value	Units
Slab	5.0	KN/m ²
Plaster	0.3	KN/m ²
Marble Floor	0.8	KN/m ²
Total- $G_{interstory\ landing}$	6.10	KN/m²

Table 9- Inter Story Landing Load

2.2.3 VARIABLE LOAD:

The variable loads include loads of the work related to the intended use. Conventionally they are marked with Q_v and, therefore, the consequent overall loads per unit length or surface are defined by the symbol q_v .

BEAMS:

For beams of intermediate floors lie into category A, while the beams of cover belong to environment category H1, so the variable loads evenly distributed apply:

- Beam intermediate level: $Q_{inter_beam} = 2.00 \text{ kN/m}^2$
- roof beam: $Q_{roof_beam} = 0.50 \text{ kN/m}^2$

FLOOR SLABS:

For beams of intermediate floors lies in the category A, while the beams of Cover belong to environment category H1, so the variable loads evenly distributed apply:

- Slabs intermediate level: $Q_{inner_slab} = 2.00 \text{ kN/m}^2$
- roof slab: $Q_{roof_slab} = 0.50 \text{ kN/m}^2$

STAIRS AND LANDING:

Both the stairs for the landing inter-story one fall into the category C2, so the variable loads evenly apply:

- stairway: $Q_{stair} = 4.00 \text{ kN/m}^2$
- Landing: $Q_{landing} = 4.00 \text{ kN/m}^2$

2.2.4 SNOW LOADS:

According to “Eurocode 1 - Actions on structures - Part 1-3: General actions - Snow loads”, the total snow load acting on the roof is evaluated with the following expression [2]

$$S = S_k \cdot \mu_i \cdot C_e \cdot C_t$$

Where,

- S is the snow load on the roof
- μ_i is the snow load shape coefficient.
- S_k is the characteristic value of the snow load on the ground reference for a period of return of 50 years
- C_e is the exposure coefficient
- C_t is the thermal coefficient.

d, it has been considered that our area lies in zone 1-A of the Alpine region at 113 m above the sea level, lower than (200 m) as shown in the figure below.

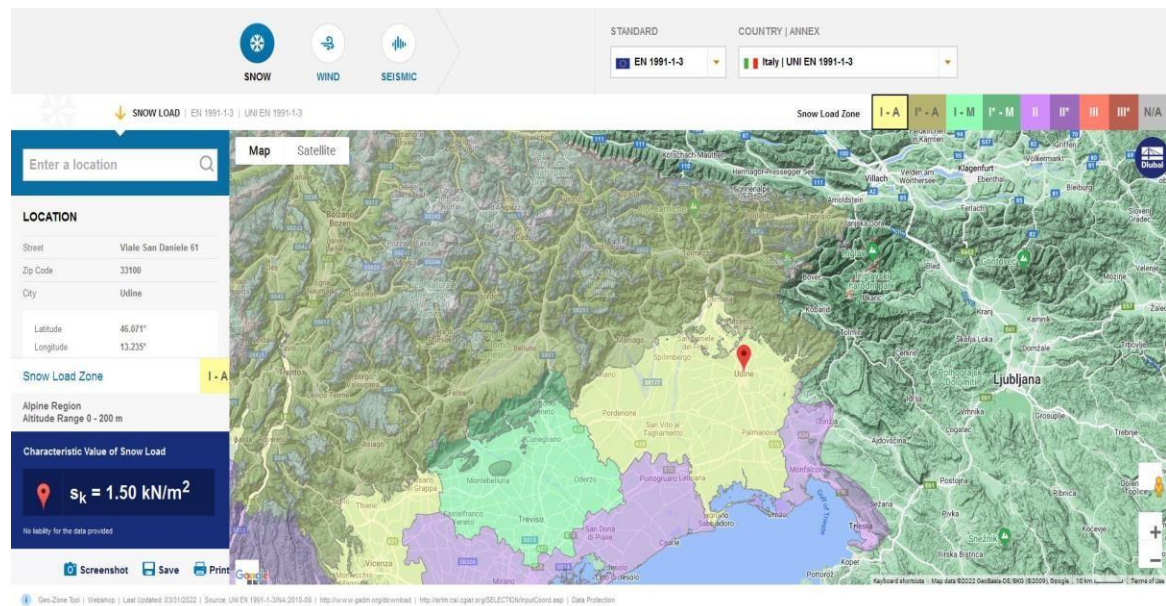


Figure 12-Udine region Snow load characteristics

Conventionally, the effects due to the action of the snow are followed by the S symbol and then, the resulting load per unit area is defined by the symbol S_k .

CHARACTERISTICS VALUE OF SNOW LOAD:

The Italian snow map is divided into 4 different homogeneous climatic zone; in each climatic region we have a given load-altitude correlation formula, and this is given in the following table.

Zone	Values	ASL
Zone I Alpine	$S_k = 1.5 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/728)^2] \text{ kN/m}^2$	$a_s \leq 200\text{m}$ $a_s > 200\text{m}$
Zone II Mediterranean	$S_k = 1.5 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/602)^2] \text{ kN/m}^2$	$a_s \leq 200\text{m}$ $a_s > 200\text{m}$
Zone III	$S_k = 1.0 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/481)^2] \text{ kN/m}^2$	$a_s \leq 200\text{m}$ $a_s > 200\text{m}$
Zone IV	$S_k = 0.6 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/481)^2] \text{ kN/m}^2$	$a_s \leq 200\text{m}$ $a_s > 200\text{m}$

Table 10-Italian Characteristics Snow Load Zones[3]

As our building area is in Zone 1 and it is 113m above sea level, our characteristics snow load coefficient is S_k is 1.5 kN/m^2 . As we can see from the above picture taken from a website.

EXPOSURE COEFFICIENT:

The exposure coefficient C_e considers the place in which the structure is located: dealing with a “normal” zone, the code Table 5.1 suggests C_e equals to 1. The exposure coefficient C_e should be used for determining the snow load on the roof. The choice for C_e should consider the future development around the site and it should be taken as 1.0 unless otherwise specified for different topographies.

Topography	C_e
Windswept ^a	0,8
Normal ^b	1,0
Sheltered ^c	1,2

^a *Windswept topography*: flat unobstructed areas exposed on all sides without, or little shelter afforded by terrain, higher construction works or trees.

^b *Normal topography*: areas where there is no significant removal of snow by wind on construction work, because of terrain, other construction works or trees.

^c *Sheltered topography*: areas in which the construction work being considered is considerably lower than the surrounding terrain or surrounded by high trees and/or surrounded by higher construction works.

Figure 13- Exposure Coefficient from NTC-2018

SNOW LOAD SHAPE COEFFICIENT:

Since there are no specific indications regarding the roof's geometry, it has been used the worst-case scenario (Table 5.2) of roof pitch inclination on the horizontal plane ($0^\circ \leq \alpha \leq 30^\circ$), which sets μ_i equal to 0.8.

Angle of pitch of roof α	$0^\circ \leq \alpha \leq 30^\circ$	$30^\circ < \alpha < 60^\circ$	$\alpha \geq 60^\circ$
μ_1	0,8	$0,8(60 - \alpha)/30$	0,0
μ_2	$0,8 + 0,8 \alpha/30$	1,6	—

Table 11- Snow Load Shape Coefficient

THERMAL COEFFICIENT:

The thermal coefficient C_t should be used to account for the reduction of snow loads on roofs with high thermal transmittance ($> 1 \text{ W/m}^2\text{K}$), but in our case, we can assume a value of $C_t = 1$ as temperature in the case of snow is usually low so we are assuming 1.

So, our total snow load comes out to be,

$$S = S_k \cdot \mu_i \cdot C_e \cdot C_t$$

$$S = 1.5 \times 0.8 \times 1 \times 1$$

$$S = 1.2 \text{ kN/m}^2$$

2.2.5 WIND LOAD:

Wind direction is considered to be predominantly horizontal. The load is computed according to Eurocode -1(2005) in both x and y directions, Wind load will be calculated with this formula.

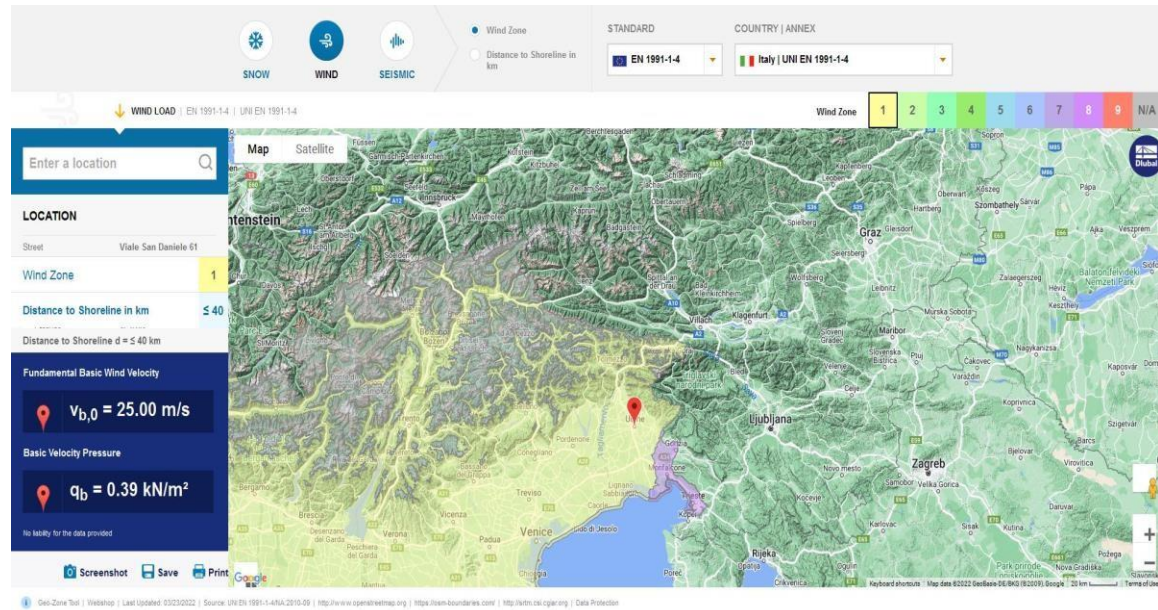


Figure 14- Wind Zoning of Udine Region

Udine is in wind zone 1 and the distance to the shoreline is less than 40km.

$$p = q_b \cdot c_e \cdot c_p \cdot c_d$$

Where,

- q_b is the peak velocity pressure
- c_e is the exposure coefficient
- c_p is the aerodynamic coefficient depending on the geometry and orientation of the building. This value is obtained by wind tunnel test
- c_d is the dynamic coefficient which takes account of the reductive effects associated with the non-contemporaneity of maximum local pressures and amplification effects due to structural vibration

BASIC WIND VELOCITY:

The Basic wind velocity, V_b , is the characteristic value of the wind speed at 10 m above the ground on a plot of category II exposure, averaged over ten minutes and referred to a period of return of 50 years.

According to NTC-18, we can use this table to verify the basic wind velocity given by the website as attached in the screenshot above and by this formula we can find V_b .

$$V_b = V_{b,0} \quad a_s \leq a_0$$

$$V_b = V_{b,0} + k_a(a_s - a_0) \quad a_0 < a_s \leq 1500\text{m}$$

Zona	Descrizione	$v_{b,0}$ [m/s]	a_0 [m]	k_s [1/s]
1	Valle d'Aosta, Piemonte, Lombardia, Trentino Alto Adige, Veneto, Friuli Venezia Giulia (con l'eccezione della provincia di Trieste)	25	1000	0,010
2	Emilia Romagna	25	750	0,015
3	Toscana, Marche, Umbria, Lazio, Abruzzo, Molise, Puglia, Campania, Basilicata, Calabria (esclusa la provincia di Reggio Calabria)	27	500	0,020
4	Sicilia e provincia di Reggio Calabria	28	500	0,020
5	Sardegna (zona a oriente della retta congiungente Capo Teulada con l'Isola di Maddalena)	28	750	0,015
6	Sardegna (zona a occidente della retta congiungente Capo Teulada con l'Isola di Maddalena)	28	500	0,020
7	Liguria	28	1000	0,015
8	Provincia di Trieste	30	1500	0,010
9	Isole (con l'eccezione di Sicilia e Sardegna) e mare aperto	31	500	0,020

Table 12- Basic Wind Velocities with Italian regions[3]

As we are in Udine city, so our region is Friuli Venezia Giulia and our site is located 113m above sea level which is lower than 1000m so we can take 25m/s $V_{b,0}$.

$$V_b = V_{b,0} \quad a_s \leq a_0$$

$$V_b = 25 \text{ m/s}$$

As data collected from the website refer to the same value of wind velocity.

PEAK VELOCITY PRESSURE:

Peak Velocity pressure can be found with this formula.

$$q_b = 1/2 \cdot \rho \cdot V_b^2$$

Where ρ is the density of air which is 1.25kg/m³ and V_b is the basic wind velocity.

$$q_b = 1/2 \cdot \rho \cdot V_b^2$$

$$q_b = 1/2 \cdot 1.25 \cdot 25^2$$

$$q_b = 390.625 \text{ N/m}^2$$

As the website refers the same amount of velocity pressure 0.39 KN/m²

EXPOSURE COEFFICIENT:

The coefficient c_e depends on the height z exposure on the soil considered, from the topography of the land, and the exposure category of the construction site. In the absence of specific analysis, we consider the wind direction, the actual roughness and topography of the land surrounding the building, the height of the ground not greater than $z = 200$ m, it is given by:

$$c_e(z) = k_r^2 c_t \ln(z/z_0) [7 + c_t \ln(z/z_0)] \quad Z \geq Z_{min}$$

$$c_e(z) = c_e[Z_{min}] \quad Z < Z_{min}$$

Where: k_r, z_0, c_t function of the exposure category of the site is located in the construction; They are marked in Table 11

Categoria di esposizione del sito	k_r	z_0 [m]	z_{min} [m]
I	0,17	0,01	2
II	0,19	0,05	4
III	0,20	0,10	5
IV	0,22	0,30	8
V	0,23	0,70	12

Table 13- K_r , Z_0 , and Z_{min} Values and categories

Assuming that the construction of the exposure category in Study IV is, the above-listed parameters take the values: c_t is the topography coefficient, generally which is set equal to 1. The variation of (z) concerning height Z, is shown in the below graph

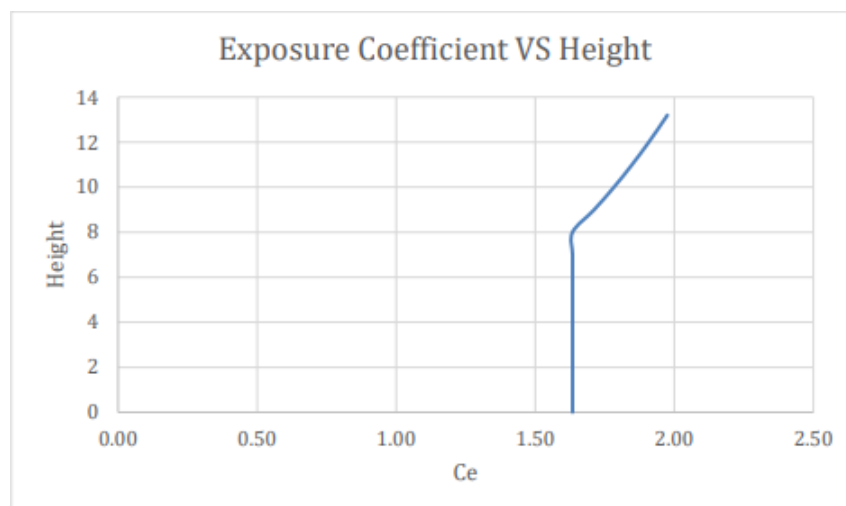
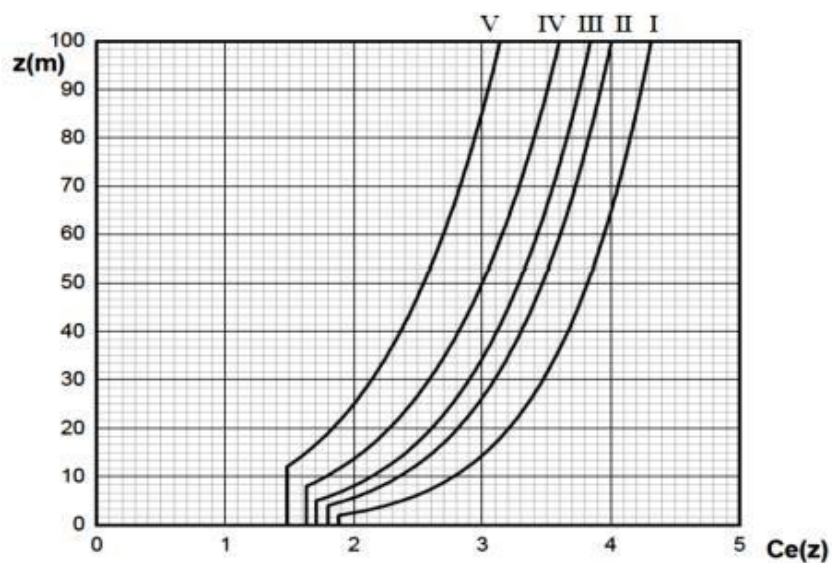


Figure 15- Exposure Coefficient vs Height

As our building's total height is around 13.2m our exposure coefficient at that point is around 1.9.

COEFFICIENT OF AERODYNAMICS:

For the evaluation of the coefficient of form or aerodynamics, it considers only the inclination of the facades of 90 °, because they are the only parts of the building exposed to the wind. For the surface, that is directly acted by the wind, it is considered a value of shape coefficient of $c_p = 0.8$; for the leeward surfaces, that are not directly invested by the wind, it is considered a value of the form factor of $c_p = 0.4$.

COEFFICIENT OF DYNAMICS:

The dynamic coefficient c_d takes account of the reducing effects associated with contemporaneity the maximum local pressures and amplification effects due to the dynamic response of the structure. It is assumed to be equal to 1.

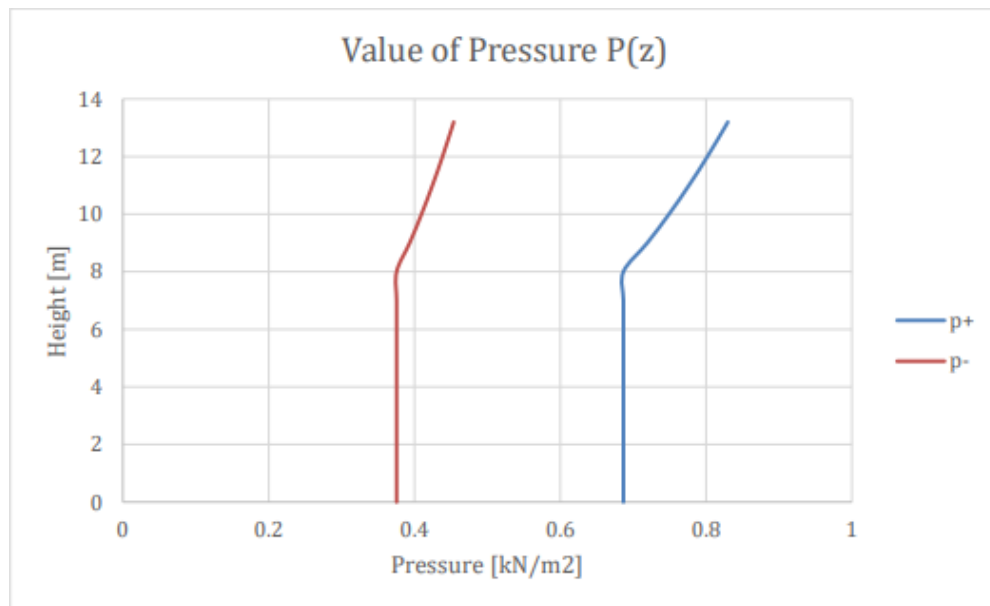


Figure 16- P+ and P- VS height

WIND LOAD				
a_s [m]	113			
a_0 [m]	1000	→	v_b [m/s]	25
$V_{b,0}$ [m/s]	25			
k_a	0			
ρ_{aria} [N/m³]	12.26	→	q_b [kN/m²]	0.39
k_r	0.22			
c_t	1			
z_0 [m]	0.3			
z [m]	13.2	→	p [kN/m²]	0.53
c_p	0.8			
c_d	1			
c_e	1.70			
α	1	→	q_w [kN/m²]	0.53

Table 14- Wind Load

2.2.5 SEISMIC LOAD:

Seismic action on the buildings is induced by the non-uniform motion of the ground due to seismic wave propagation. Usually indicated with the letter E, seismic action is considered through the Response Spectrum method. Firstly, it has been defined as the Elastic Response Spectrum; secondly, after having scaled it with the q factor (dealing with post-elastic behavior), it has been possible to obtain the Design Response Spectrum.

IDENTIFICATION OF SEISMIC ZONE:

The seismic classification in Italy (Italian Classificazione sismica dell'Italia) is the subdivision of the territory of Italy into specific areas, characterized by a common seismic risk. Currently, the seismic classification of the Italian territory into zones has remained exclusively for statistical and administrative aspects. With the legislation that came into force in 2009 (NTC08), after the earthquake that affected the city of L'Aquila, a new calculation methodology based on a point-like statistical approach is used for anti-seismic design. Each point of the Italian territory is characterized by a precise ground acceleration value PGA as a function of a return time (a probabilistic value).

- **Zone 1** : high seismicity (PGA over 0.25 g), includes 708 municipalities.
- **Zone 2** : medium-high seismicity (PGA between 0.15 and 0.25 g), includes 2,345 municipalities (in Tuscany and Piedmont some municipalities are classified in Zone 3S, medium seismicity, which requires the calculation of the identical seismic action to Zone 2).
- **Zone 3** : medium-low seismicity (PGA between 0.05 and 0.15 g), includes 1,560 municipalities.
- **Zone 4** : low seismicity (PGA less than 0.05 g), includes 3,488 municipalities.

As data taken from different websites like INGV - National Institute of Geophysics and Volcanology Italia it shows our PGA ranging from 0.2-0.225g in our site location and this has been verified from global as well it also shows the same value here attaching the screenshots of PGA range for our site location. So, we are in seismic zone II.

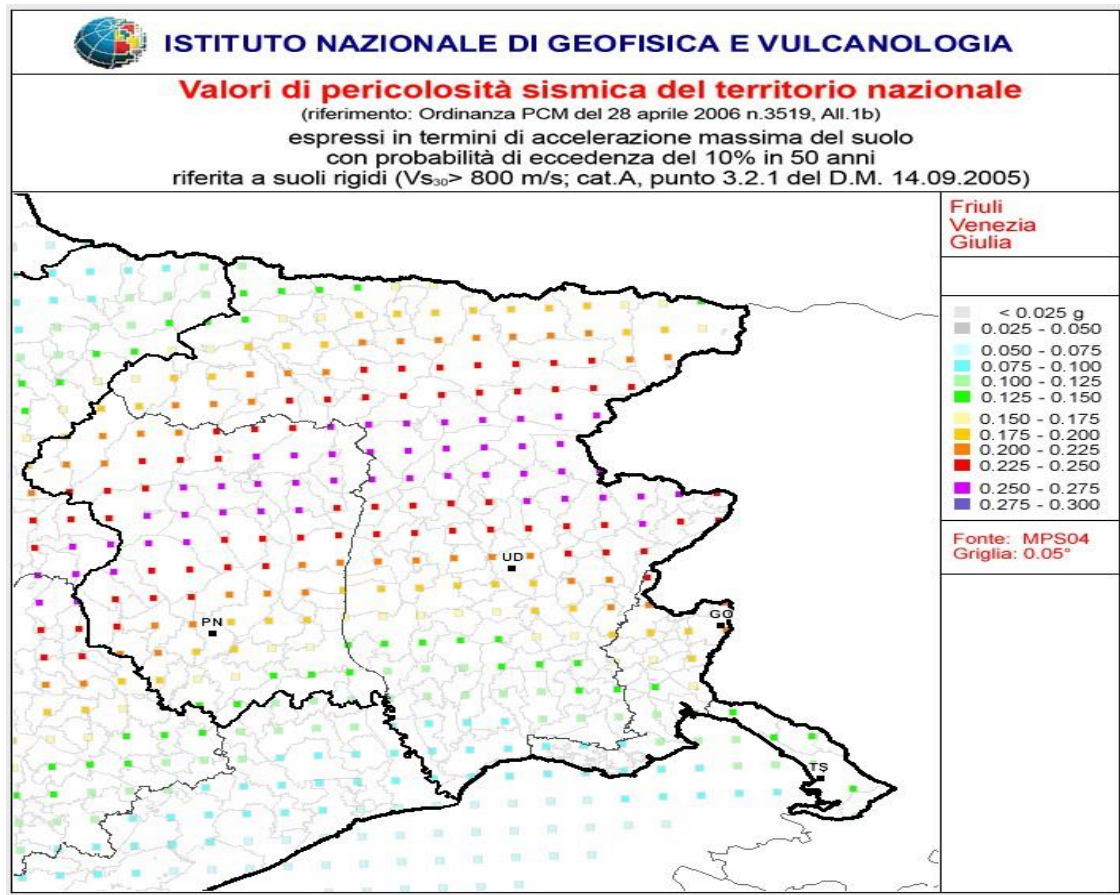


Figure 17-PGA range from INGV

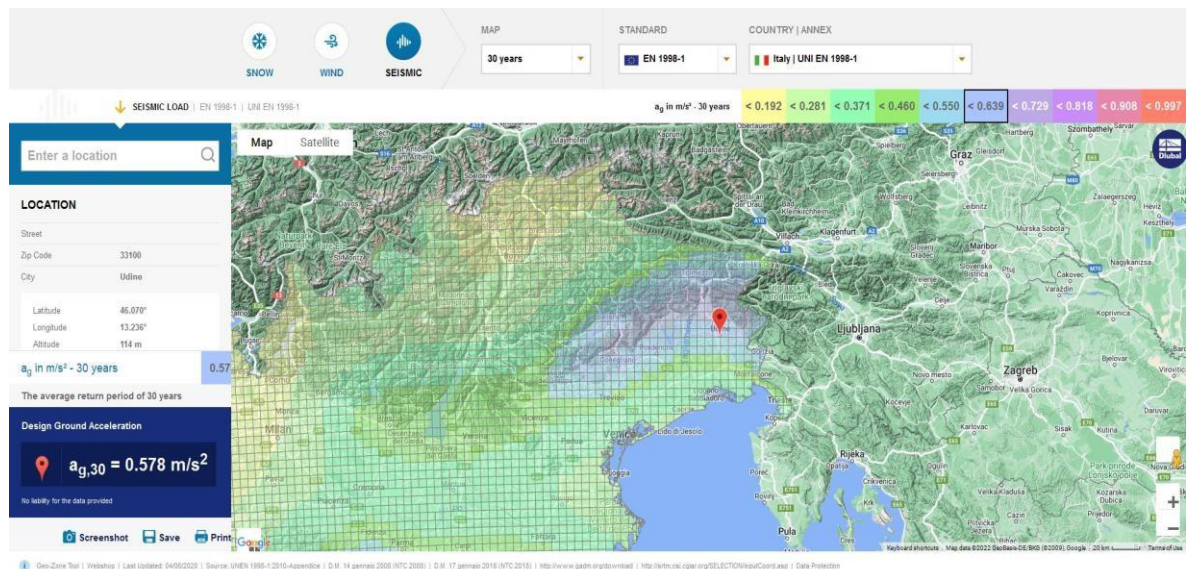


Figure 18- PGA range from glubal with average returning period of 30 years

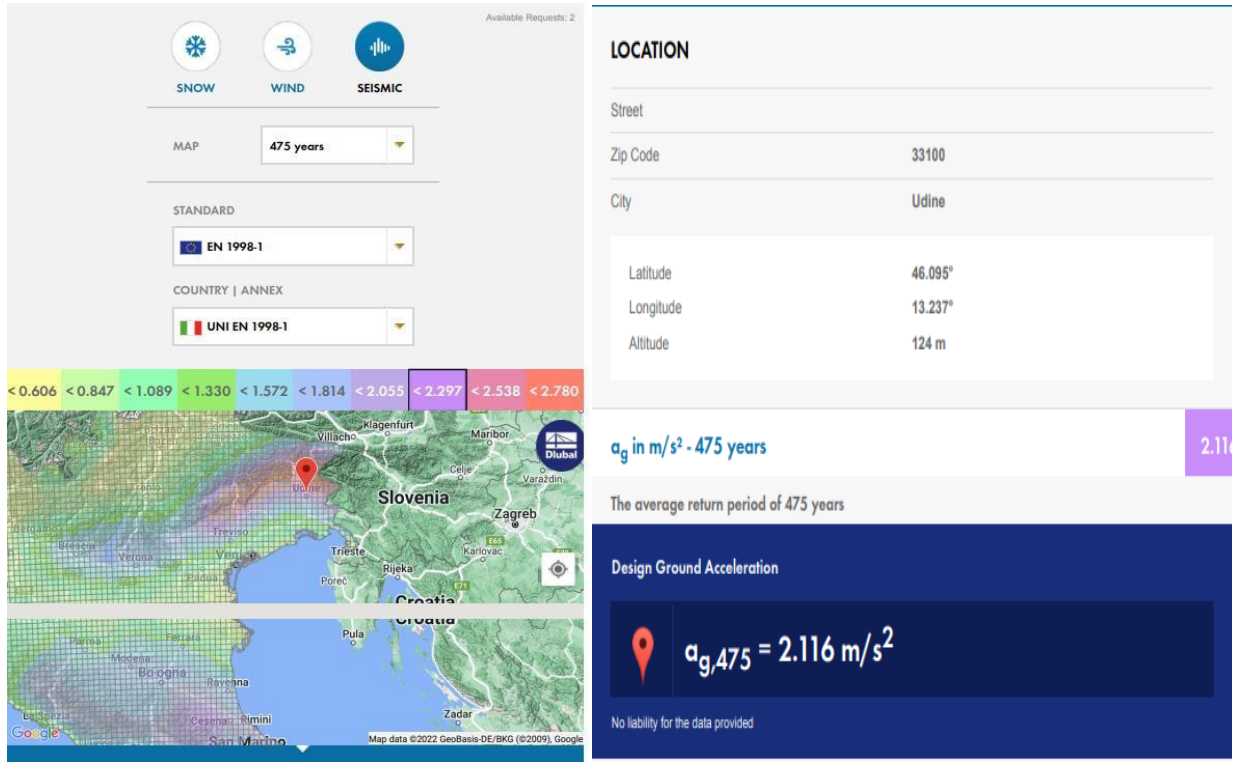


Figure 19- PGA range from Dlubal website with returning period of 475 years

CLASSIFICATION OF TYPE OF SOIL:

It has already been suggested in Chapter 1 that the building arises on a type C: Deep deposits of dense or medium dense sand, gravel, or clay consistent with thickness varying from tens of meters to several hundred meters. From this assumption, it follows that the coefficient of the ground is equal $t_0 = 1.15$

ELASTIC RESPONSE SPECTRUM:

Appropriate investigations were carried out to identify the ground conditions, associated with the types given. It is worth noting that the ground classification could include values of the parameters S , T_B , T_C and T_D which define the horizontal and vertical elastic response spectra. Dealing with the seismic action (3.2.1), the hazard is described by a single parameter: the reference peak ground acceleration, PGA which in our case is 0.206g with soil type C. The earthquake motion at a given point on the surface is represented by an elastic ground acceleration response spectrum, called elastic response spectrum (3.2.2.1). The elastic response spectrum $S_e(T)$ is defined according to the following expressions (31-34); its construction is divided in different subintervals, differentiated in terms of T_B , T_C , T_D .

$$0 \leq T \leq T_B : S_e(T) = a_g \cdot S \cdot [1 + (T / T_B) \cdot (\eta \cdot 2.5 - 1)]$$

$$T_B \leq T \leq T_C : S_e(T) = a_g \cdot S \cdot \eta \cdot 2.5$$

$$T_C \leq T \leq T_D : S_e(T) = a_g \cdot S \cdot \eta \cdot 2.5 \cdot [T_C / T]$$

$$T_D \leq T \leq 4s : S_e(T) = a_g \cdot S \cdot \eta \cdot 2.5 \cdot [T_C \cdot T_D / T^2]$$

Where,

- $S_e(T)$ = elastic response spectrum.
- T = period of vibration of a linear single-degree-of-freedom system,
- a_g = design ground acceleration on type C ground ($a_g = \gamma \cdot a_{gR}$),
- T_B = lower limit of the period of constant spectral acceleration branch
- T_C = upper limit of constant spectral acceleration branch
- T_D = value which defines the beginning of the constant displacement response range of the spectrum,
- S = soil factor,
- η = damping correction factor.

The values of the periods characterizing the shape of the response spectrum depend on the type of soil, which in this case is type C: From NTC calculated at Midas,

$$T_B = 0.17s; T_C = 0.5s; T_D = 2.42s$$

In conclusion, the diagram of the elastic response spectrum for a site with the characteristics listed so far is shown in the Figure below.

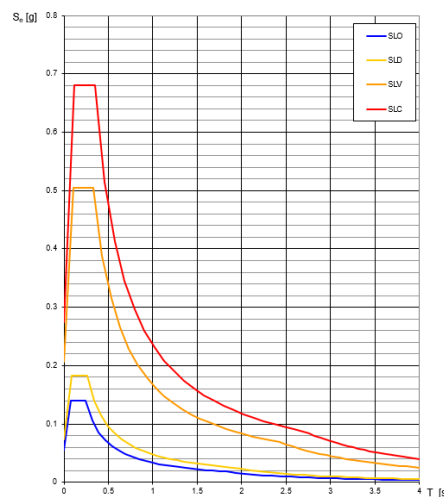


Figure 20- Elastic Response Spectrum

BEHAVIOUR FACTOR:

The behavior factor q accounts for the capacity of the structure to dissipate energy, through mainly ductile behavior of its elements and/or other mechanisms. To avoid explicit inelastic structural analysis in design, an elastic analysis is performed based on a response spectrum reduced by dividing the spectral acceleration with the behavior factor q , as compared to the elastic spectrum. The value of the behavior factor q approximates the ratio of the seismic forces corresponding to a 5% damped elastic response spectrum to the actual seismic forces that the structure would experience considering its actual inelastic response.

The building is designed according to the requirements for the High ductility class (DCH). The upper limit value of the behavior factor q to account for energy dissipation capacity shall be derived for each design direction as follows:

$$q = q_0 * k_w$$

Where,

- q_0 is the basic value of the behavior factor, dependent on the type of the structural system and on its regularity in elevation.
- k_w is the factor reflecting the prevailing failure mode in structural systems with wall.

The building belongs to a High Ductility Class (CD “A”), considering the structural type “frame system, dual system, coupled wall system” (most of the shear actions is taken by the walls), q_0 is evaluated through the following expression.

$$q_0 = 4.5 * (\alpha_u/\alpha_1)$$

Where,

- α_1 is the value for which is multiplied by the seismic action of the horizontal project to reach the first flexural strength in a steel member in any structure, while all other design actions remain constant.
- α_u is the value which is multiplied by the seismic action of a horizontal project to form a plastic hinge in several sections sufficient for the development of overall structural instability, while all other design actions remain constant.

Since there are not specific calculations to compute the two multipliers (α_1 , α_u), considering our structural systems, the ratio α_u/α_1 is equal to 1.3.

$$q_0 = 4.5 * (\alpha_u/\alpha_1)$$

$$q_0 = 4.5 * (1.2)$$

$$q_0 = 5.4$$

Dealing with k_w , the Code (5.2.2.2) provides the expression.

$$k_w = 0.5 \leq (1 + \alpha_0)/3 \leq 1$$

here α_0 = ratio between the height of the walls and their length, computed through the following expression (9):

$$\alpha_0 = \sum h_w / \sum L_w$$

Where:

- h_w is the height of the wall.
- l_w is the length of the section of the wall.

Considering, therefore, that all the walls are 13.2m high and two walls of 3m and two walls of 4m in length, the coefficient α_0 applies

$$\alpha_0 = 13.2 * 8 / 22.8$$

$$\alpha_0 = 4.63$$

So,

$$k_w = (1 + \alpha_0)/3$$

$$k_w = (1 + 4.63)/3$$

$$k_w = 1.87$$

As k_w cannot be greater than 1 so we used $k_w = 1$.

In conclusion, the behavior factor is:

$$q = q_0 * k_w$$

$$q = 5.4 * 1$$

$$q = 5.4$$

To define the response spectra of the building used the program SPETTRI-NTC which provides the representative response spectra of the horizontal and vertical components of the seismic design for the generic site of the Italian national territory.

The definition of spectra is divided into three phases:

- Identifying of site hazards.
- Choosing of design strategy.
- Determination of the project action.

The following picture shows the screen for selecting the site concerned.

FASE 1. INDIVIDUAZIONE DELLA PERICOLOSITÀ DEL SITO

☐ Ricerca per coordinate

LONGITUDINE
13.2378

LATITUDINE
46.0619

☒ Ricerca per comune

REGIONE
Friuli-Venezia Giulia

PROVINCIA
Udine

COMUNE
Udine

Elaborazioni grafiche

Grafici spettri di risposta

Variabilità dei parametri

Elaborazioni numeriche

Tabella parametri

Nodi del reticolo intorno al sito

Reticolo di riferimento

Controllo sul reticolo

● Sito esterno al reticolo

● Interpolazione su 3 nodi

● Interpolazione corretta

Interpolazione

superficie rigata

La "Ricerca per comune" utilizza le coordinate ISTAT del comune per identificare il sito. Si sottolinea che all'interno del territorio comunale le azioni sismiche possono essere significativamente diverse da quelle così individuate e si consiglia, quindi, la "Ricerca per coordinate".

INTRO

FASE 1

FASE 2

FASE 3

Figure 21- Identifying the site hazard Phase 1

In the 2° phase, we insert the value of the working life V_N which is the period for which a structure is to be used for its intended purpose with anticipated maintenance but without major repair being necessary. According to the table below (NTC 2008) we use the value of V_N for common structures as importance factor 1:

TIPI DI COSTRUZIONE		Vita Nominale V_N (in anni)
1	Opere provvisorie – Opere provvisionali - Strutture in fase costruttiva ¹	≤ 10
2	Opere ordinarie, ponti, opere infrastrutturali e dighe di dimensioni contenute o di importanza normale	≥ 50
3	Grandi opere, ponti, opere infrastrutturali e dighe di grandi dimensioni o di importanza strategica	≥ 100

Figure 22- Structural working life NTC-8

$$V_N \geq 50 \text{ years}$$

Buildings are also classified in 4 importance classes, depending on the consequences of collapse for human life, on their importance for public safety and civil protection in the immediate post-earthquake period, and on the social and economic consequences of collapse:

- Importance class I: Buildings of minor importance for public safety, e.g., agricultural buildings, etc.
- Importance class II: Ordinary buildings, not belonging in the other categories.
- Importance class III: Buildings whose seismic resistance is of importance in view

of the consequences associated with a collapse, e.g., schools, assembly halls, cultural institutions etc.

- Importance class IV: Buildings whose integrity during earthquakes is of vital importance for civil protection, e.g., hospitals, fire stations, power plants, etc.

According to the project the second class should be selected.

CLASSE D'USO	I	II	III	IV
COEFFICIENTE C_U	0,7	1,0	1,5	2,0

Table 15- Value of C_U for building Class

So, the reference return period of the reference seismic action for the no-collapse requirement is given by:

Class of use is II.

$$V_R = V_N \cdot C_U = 50 \text{ years}$$

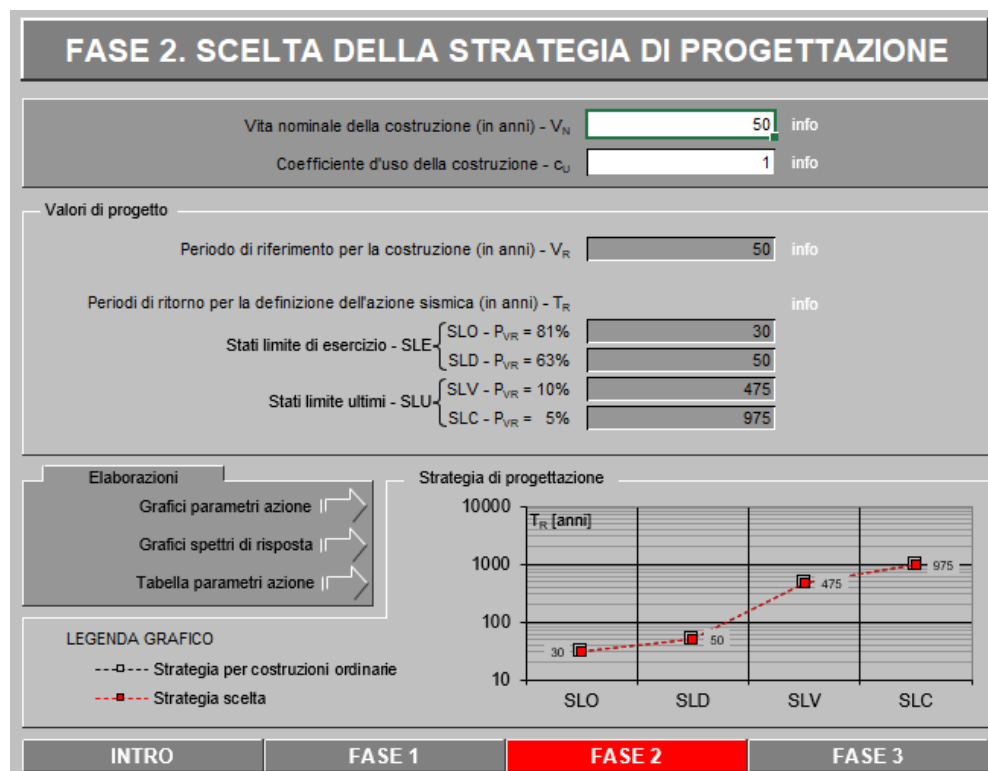


Figure 23- phase II Choosing design life and Strategy

In the last phase can be inserted the various limit states, soil and topographical class, service limit states or ultimate limit states as shown in the following figure. As mentioned, before it was used a soil class C and a topographical class T1.

FASE 3. DETERMINAZIONE DELL'AZIONE DI PROGETTO

Stato Limite
 Stato Limite considerato **SLV** info

Risposta sismica locale
 Categoria di sottosuolo **C** info
 Categoria topografica **T1** info
 $S_S = 1.397$ $C_C = 1.511$ info
 $h/H = 0.000$ $S_T = 1.000$ info
(h=quota sito, H=altezza rilievo topografico)

Compon. orizzontale
☐ Spettro di progetto elastico (SLE)
☒ Spettro di progetto inelastico (SLU)
 Smorzamento ξ (%) **5** $\eta = 1.000$ info
 Fattore q_0 **5.4** Regol. in altezza **no** info

Compon. verticale
 Spettro di progetto
 Fattore q **1.5** $\eta = 0.667$ info

Elaborazioni
 Grafici spettri di risposta |>
 Parametri e punti spettri di risposta |>

Spettri di risposta

— Spettro di progetto - componente orizzontale
 — Spettro di progetto - componente verticale
 — Spettro elastico di riferimento (Cat. A-T1, $\xi = 5\%$)

INTRO **FASE 1** **FASE 2** **FASE 3**

Figure 24- Determination of Project Action Elastic and Design Response Spectra

The values of the spectrum parameters T_R , a_g , F_0, T_c^* , for the seismic analysis limit states are reported in the following table:

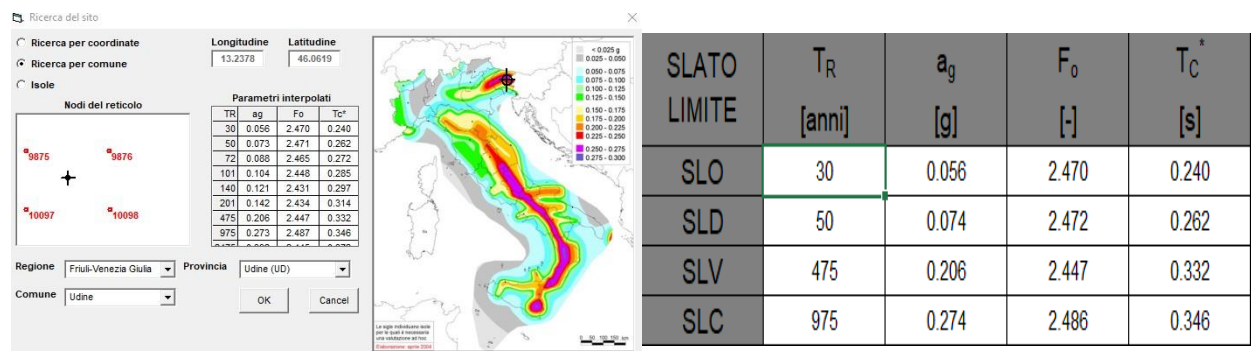


Figure 25- Spectrum Variable with Spettrti and Simqke,GR software

Where:

- T_R is the reference return period of the reference seismic action for the no-collapse requirement.
- a_g is the design ground acceleration on type C ground, expressed in relation with the acceleration of gravity.
- F_0 is the maximum value of the amplification factor of the spectrum in horizontal acceleration.
- T_c^* is the corner period at the upper limit of the constant acceleration region of the elastic spectrum.

DESIGN RESPONSE SPECTRA:

The structure's ability to dissipate energy (which depends on the ductile behavior of its members), is considered with an elastic analysis based on a reduced Response Spectrum with respect to the elastic one: it is called Design Response Spectrum. This reduction is considered with the behavior factor, applying the same characterization in subintervals followed for the Elastic Response Spectrum: $0 \leq T < T_B$, $T_B \leq T < T_C$, $T_C \leq T < T_D$ and $T_D \leq T$, where T is the period of vibration.

The horizontal components of the seismic action of the Design Spectrum $S_d(T)$ are defined in the following way, considering the same division in subintervals of the Elastic Spectrum

$$0 \leq T \leq T_B : S_d(T) = a_g \cdot S \cdot [2/3 + (T/T_B) \cdot (2.5/q - 2/3)]$$

$$T_B \leq T \leq T_C : S_d(T) = a_g \cdot S \cdot 2.5/q$$

$$T_C \leq T \leq T_D : S_e(T) = a_g \cdot S \cdot (2.5/q) \cdot (T_C/T) \geq \beta \cdot a_g$$

$$T_D \leq T \leq 4s : S_e(T) = a_g \cdot S \cdot (2.5/q) \cdot (T_C \cdot T_D / T^2) \geq \beta \cdot a_g$$

Where q = behavior factor (computed as value equals 5.4) β = lower bound factor for the horizontal design spectrum (its values can be found in the National Annex of the considered country, but the recommended value is 0.2). With all the values previously computed, it is possible to build the Design Response Spectrum, which is presented in the figure below.

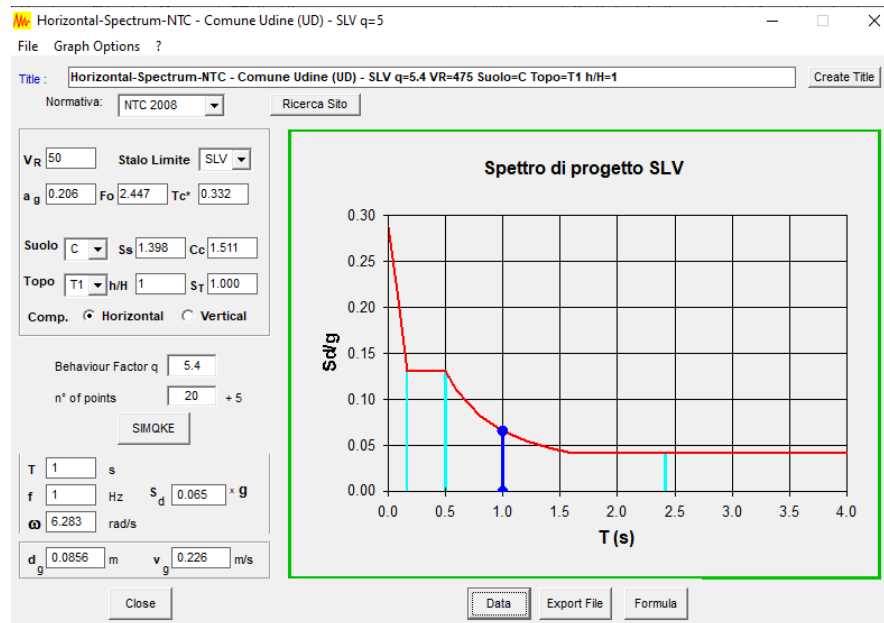


Figure 26- Design Response Spectra

2.2.6 LOAD COMBINATION:

There are two different load combinations which must be implemented in the model: SLU and SLV. For analyzing the structures according to the codes, it is necessary to consider different load combinations. Due to the Eurocode and national annex, the following tables show all combinations that should be considered for designing the structure.

SLU STATIC AND DYNAMIC COMBINATION:

Static and dynamic SLU load combinations are calculated after using the below formula. The values are shown in the table below.

$$F_d = \gamma_{G1}G_1 + \gamma_{G2}G_2 + \gamma_{Q1}Q_{k1} + \sum_{i=2}^n \gamma_{Qi}\psi_{0i}Q_{ki}$$

Combination	Dead Load		Variable Load	Snow	Wind Load		Seismic Load	
	Structural	Non-Structural						
	G₁	G₂	Q_v	Q_s	Q_{wx}	Q_{wy}	S_x	S_y
SLU 1	1.3	1.3	1.5	0.75	±0.9	0	0	0
SLU 2	1.3	1.3	1.05	1.05	±0.9	0	0	0
SLU 3	1.3	1.3	1.05	0.75	±1.5	0	0	0
SLU 4	1.3	1.3	1.5	0.75	0	±0.9	0	0
SLU 5	1.3	1.3	1.05	1.5	0	±0.9	0	0
SLU 6	1.3	1.3	1.05	0.75	0	±1.5	0	0
SLV 1	1	1	0.6	0	0	0	1	0.3
SLV 2	1	1	0.6	0	0	0	0.3	1

Table 16-SLS Static and Dynamic Load Combination

2.2.6.1 SLE RARE STATIC LOAD COMBINATION:

SLE-Rare static load combinations are calculated after using below formula. The values are shown in table below.

$$F_d = G_1 + G_2 + Q_{k1} + \sum_{i=2}^n \psi_{0i}Q_{ki}$$

Combination	Dead Load		Variable Load	Snow	Wind Load	
	Structural	Non-Structural				
	G₁	G₂	Q_v	Q_s	Q_{wx}	Q_{wy}
SLE - Rare 1	1	1	1	0.5	±0.6	0
SLE - Rare 2	1	1	0.7	1	±0.6	0
SLE - Rare 3	1	1	0.7	0.5	±1	0
SLE - Rare 4	1	1	1	0.5	0	±0.6
SLE - Rare 5	1	1	0.7	1	0	±0.6
SLE - Rare 6	1	1	0.7	0.55	0	±1

Table 17- SLE-Rare Static Load Combinations

SLE FREQUENT STATIC LOAD COMBINATION:

SLE-Rare static load combinations are calculated after using the below formula. The values are shown in table below.

$$F_d = G_1 + G_2 + \psi_{11}Q_{k1} + \sum_{i=2}^n \psi_{2i}Q_{ki}$$

Combination	Dead Load		Variable Load	Snow Load	Wind Load	
	Structural	Non-Structural				
	G_1	G_2			Q_{w_x}	Q_{w_y}
SLE - Frequent 1	1	1	0.5	0	0	0
SLE - Frequent 2	1	1	0.3	0.2	0	0
SLE - Frequent 3	1	1	0.3	0	± 0.2	0
SLE - Frequent 4	1	1	0.3	0	0	± 0.2

Table 18- SLE - Frequent static load Combinations

2.2.6.2 SLE QUASI STATIC LOAD COMBINATION:

SLE - Quasi permanent static load Combinations are calculated after using below formula. The values are shown in table below.

$$F_d = G_1 + G_2 + \psi_{21}Q_{k1} + \sum_{i=2}^n \psi_{2i}Q_{ki}$$

Combination	Dead Load		Variable Load	Snow	Wind Load	
	Structural	Non-Structural				
	G_1	G_2			Q_{w_x}	Q_{w_y}
SLE - Quasi Permanent	1	1	0.3	0	0	0

Table 19- SLE – Quasi- static load Combinations

CHAPTER 3

3.1 MODELLING:

The structure has been modelled through the commercial software MIDAS Gen. Firstly, it has been set a grid of nodes for the two main directions X and Y (X-Grid Lines and Y-Grid Lines). Secondly, structural members (beams and columns) were created linking the nodes. Shear walls in both directions have been modelled as Wall elements, sub-type “membrane”, with no out-of plane stiffness.

The material used to define such sections is concrete C35/45, according to the given data. In addition, three basic hypotheses have been considered:

- No soil-structure interaction
- Each floor is modelled as infinitely rigid in its own plane
- Each floor mass is concentrated entirely at the floor centroid.

Because the foundations are supposed to be endlessly rigid, the beam ends at ground level are treated as clamps in the first hypothesis. The one that defines the structure's vertical parts and allows for their simplification. Each floor will behave as a hard body, according to the model. The result is that the All points belonging to a certain floor will have their displacements defined explicitly depending on its three degrees of freedom: floor centroid displacement components in the horizontal directions (x and y axes), floor rotation in vertical direction (z axis).

The third hypothesis is that it is possible to concentrate the entire mass of each floor in specific nodes (master nodes) located at the floor centroid (which coincides with the mass center of the slab) to reduce computational effort and have more direct interaction with the masses involved in the various analyses (modal analysis and structural response to great excitations). This will obviously result in a loss of accuracy, but it may be worth it. Each master node is constrained in terms of vertical translation and the sections of the "DIAPHRAGM" type constraint (previously described) are applied to each floor's nodes.

Loads have been computed as previously described, following the subdivision in permanent structural loads, permanent non-structural loads, and variable loads. Permanent structural loads are computed automatically by the program (through the command Self Weight, in the Load – Structure Loads / Masses menu). Dealing with the other two categories of loads, each beam has been loaded through the command Beam Load, subtype Element. As regards the loads acting on ramps and landings of the staircases, it has been used the command Pressure Loads in the Pressure. The description of the modelling of the various structural members is presented in the following paragraphs.

The following figure shows the model created by MIDAS.

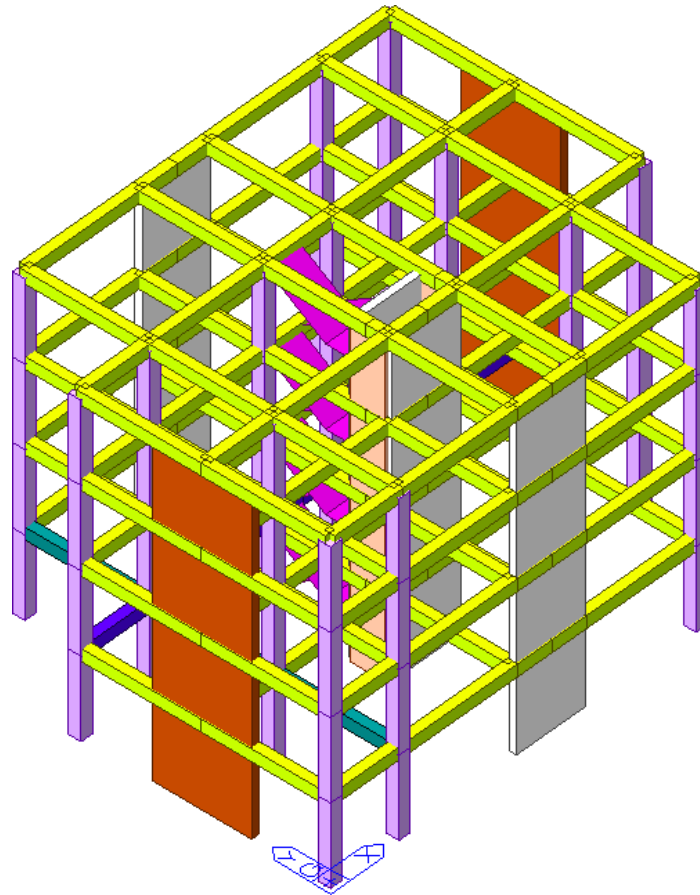


Figure 27- STRUCTURAL MODEL

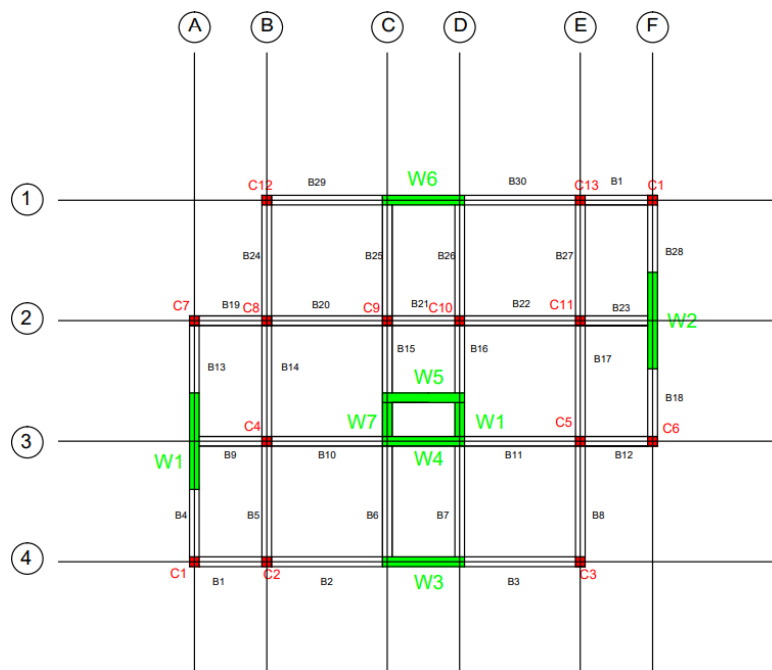


Figure 28-Floor Plan

3.1.1 BEAMS AND COLUMNS:

Beams and columns were represented using beam elements, with each node having six degrees of freedom. As previously indicated, the software applied permanent structural loads to both automatically with the command Self Weight. Rectangular sections are seen in beams and columns. Permanent non-structural loads have been applied to beams using the command Beam Load, subtype Element, which creates a uniform distributed load throughout the specified length of the beam. Because the half width of the columns has not been included, the uniform distributed load does not span the complete length of the beam when taking into consideration the overlap between the several structural parts that make up the node. As a result, internal subdivision allows for the evaluation of the loads that operate on the system.

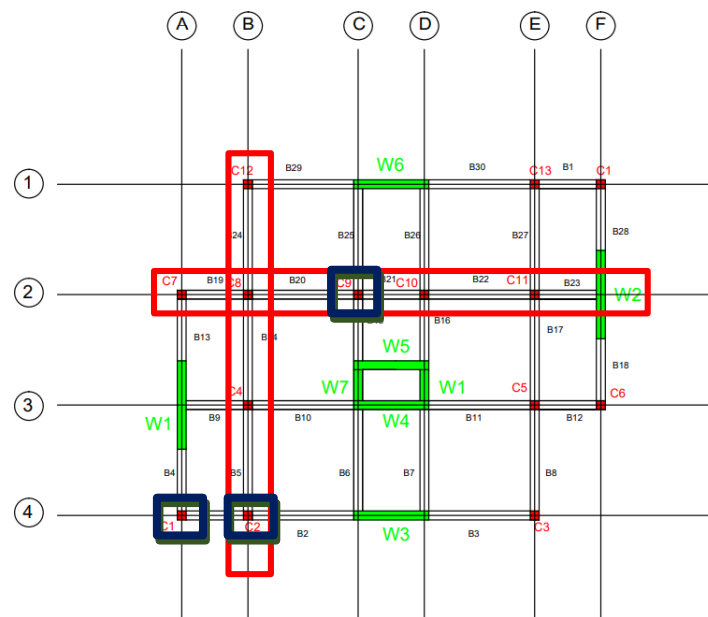


Figure 29- Beams and Columns

3.1.2 SHEAR WALLS:

As previously stated, walls have been modelled as Wall elements, sub-type “membrane”, with no out-of-plane stiffness. Walls thickness is 0.3m and their length is 3m, 4m and 1.4m. Horizontal walls are 3m long, while vertical walls are 4m long and for elevator shaft vertical walls are 1.4m long. (For both walls along the longitudinal direction X transversal direction Y). Through the command Wall element, the software generates different “walls” between the various floors of the building (inter-floor height of 3.3 m). This means that, it creates four different walls: two walls belong to the longitudinal direction, the others to the transversal one. The following figure shows the Wall elements.

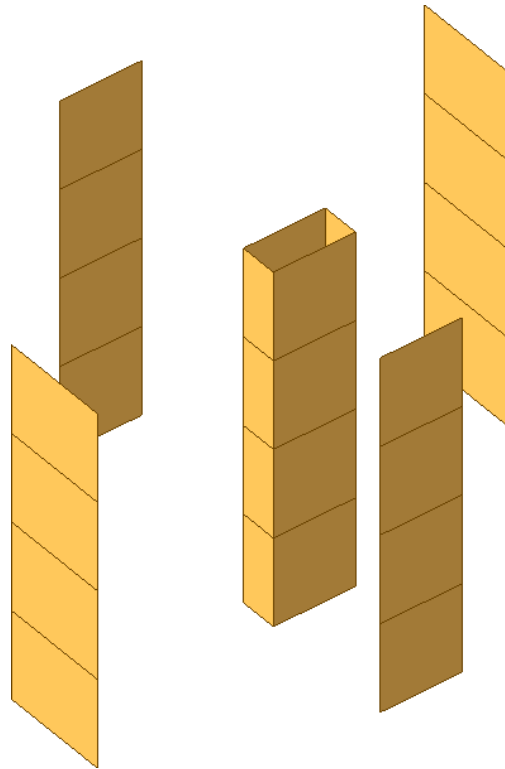


Figure 30- Shear Walls

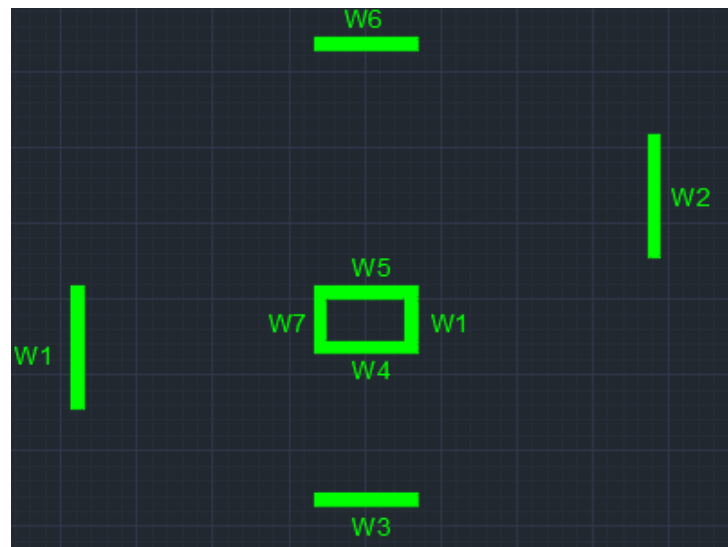


Figure 31- Shear Wall Layout

3.1.3 STAIRCASE:

Staircase ramps and mid-floor slabs have been modelled through plate elements, subtype “thick”. Ramps and mid-floor slabs are loaded with pressure plane loads, command Pressure Loads / Assign Pressure Loads, under the pressure Load menu. Uniform loads act along the direction of local z to consider the real projection of the load acting on both elements. The thickness of the floor is assumed to be 180mm thick. It is considered a bit higher than the assumed thickness of other slabs which have a thickness of 150mm.

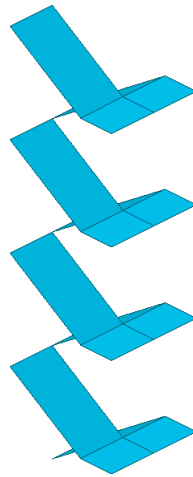


Figure 32- Staircase

3.2 REGULARITY CHECKS:

According to EC8 section 4.2.3.1 the method of analysis, which can either be a simplified response spectrum analysis (lateral force method) or a response spectrum analysis, depends on whether the building is regular in plan or not. The equivalent lateral force analysis and the response spectrum analysis are two types of linear elastic analysis that are commonly used to analyze buildings, the difference between them is that the first one is a static analysis while the second is a dynamic analysis.

The equivalent lateral force analysis is used when the structure is symmetric (with minimum vertical or horizontal irregularities) and has no discontinuities in the system. This means the torsional effects are minimal. This kind of analysis leads to accurate results for short and regular buildings.

On the other hand, the response spectrum analysis is a dynamic analysis by considering the mode shapes and modal mass participation of the structure. This kind of analysis is based on the natural periods of the building so the response will be more accurate and realistic. It should be applied for buildings with reentrant corners or irregularities in plan and elevation, and where torsional effects are not minimum.

3.2.1. Regularity in plan:

For a plan to be considered regular all the conditions listed in 4.2.3.2 of EC8 should be satisfied.

The first condition requires that concerning the lateral stiffness and mass distribution, the building structure shall be approximately symmetrical in plan concerning two orthogonal axes. It is satisfied.

Even if this condition is not fully satisfied, it was decided to consider the building as regular in plan because all the remaining conditions for the regularity in plan were accomplished. In the following all these checks are described.

The slenderness $\lambda = L_{max}/L_{min}$ of the building in the plan shall be not higher than 4, where L_{max} and L_{min} are respectively the larger and smaller in plan dimension of the building, measured in orthogonal directions.

$$\lambda = 19/15 = 1.267 < 4$$

At each level and for each direction of analysis X and Y, the structural eccentricity e_0 and the torsional radius r_i shall follow the two conditions below, which are expressed for the direction of analysis Y

$$e_{0x} \leq 0.30 * r_x \quad r_x \geq l_s$$

Where:

e_{0x} – The distance between the centre of stiffness and the centre of mass, measured along the x direction, which is normal to the direction of analysis considered

r_x – Square root of the ratio of the torsional stiffness to the lateral stiffness in the y direction (“torsional radius”)

$$r_x = \sqrt{\frac{\sum x^2 * I_y + y^2 * I_x}{\sum l_x}}$$

l_x – radius of gyration of the floor mass in plan (square root of the ratio of the polar moment of inertia of the floor mass in plan concerning the center of mass of the floor to the floor mass).

$$l_x = l_y = \sqrt{\frac{l_x^2 + l_y^2}{12}} = 6.988 \text{ m}$$

The reference system for the calculation of the centre of gravity is placed on the axis of column "1", located at the bottom left of the building plan. The formulas used for the calculation of the coordinates of the centre of gravity and the rotational inertias are the following.

$$x_g = \frac{\sum P_i \cdot x_i}{\sum P_i}$$

$$y_g = \frac{\sum P_i \cdot y_i}{\sum P_i}$$

$$I_{0i} = \frac{M_i}{12} \cdot (L_{xi}^2 + L_{yi}^2)$$

$$I_{zi} = I_{0i} + M_i \cdot d_i^2$$

Where:

- P_i is the element weight
- M_i is the mass of the element

- I_{0i} is the moment of inertia with respect to the centre of gravity of the single element
- d_i is the distance of the centre of gravity of the single element from the floor's centre of gravity
- I_{zi} is the moment of inertia with respect to the centre of gravity of floor.

After modelling all the values like weight center, stiffness center, eccentricities, torsional stiffness, El radius were found from MIDAS and presented below in table.

Story	Level (m)	Weight Center		Stiffness Center		Ecc. Dist.		El. Radius		Torsional Radius		Check
		X (m)	Y (m)	X (m)	Y (m)	X (m)	Y (m)	X (m)	Y (m)	Rx (m)	Ry (m)	
Roof	13.2	9.5	7.48	9.63	7.04	0.14	0.44	9.86	9.37	2.958	2.811	OK
4F	9.9	9.5	7.43	9.54	6.77	0.04	0.65	9.06	9.34	2.718	2.802	OK
3F	6.6	9.5	7.43	9.53	6.56	0.03	0.86	9.46	9.28	2.838	2.784	OK
2F	3.3	9.5	7.43	9.5	6.56	0	0.87	9.59	9.54	2.877	2.862	OK

Table 20- Weight Center and Shear Center

3.2.2. Regularity in elevation:

The structure in the analysis is considered regular in elevation since there are no present any changes for adjacent storeys in elevation. The same is valid for the lateral stiffness and the mass of the individual storeys, which remain constant from the base to the top of the building.

4. Model Analysis:

For the evaluation of the seismic response of the structure the modal response spectrum analysis has been used. It is a type of analysis where the dynamic behaviour of a structure is separated in distinctive modes and each of them contributes to the global behaviour with a certain level of contribution. The response of all the modes of vibration contributing significantly to the global response shall be taken into account. To use this analysis the following rules must be followed (EC8 Sec.4.3.3.3):

- The sum of the effective modal masses for the modes taken into account amounts to at least 90% of the total mass of the structure
- All modes with effective modal masses greater to 5% of the total mass are taken into account

When a spatial model is adopted, the above conditions should be verified for each relevant direction. The modal analysis of the model previously described was carried out in the calculation program and the natural periods of the structure were obtained. It is worth noting that the layout of the structure was modified to avoid a torsional mode of the structure in the first and second modes. This was done to be able to compare the results obtained by the software with the ones obtained by simplified hand calculation model. To verify the accuracy of the model, the period of the first mode of vibration was compared with the empirical formula, provided by EC8 (Par. 4.3.3.2.2), which estimates the first vibration period of the structure:

$$T_1 = C * H^{3/4}$$

Where,

$$C = \frac{0.075}{\sqrt{A_c}} = \frac{0.075}{\sqrt{0.3 \times 22.8}} = 0.0287$$

A_c = Area of shear walls at ground floor = Length of wall * thickness of wall

H = total height of building =13.2 m

$$T_1 = 0.0287 * 13.2^{0.75} = 0.199 \text{ s}$$

EIGEN VALUE ANALYSIS			
Mode No	Direction	Frequency (cycle/sec)	Period (sec)
1	TX	4.2955	0.2328
2	TY	4.4558	0.2244
3	R	5.9634	0.1677
4	TX	11.9746	0.0835
5	TY	15.4766	0.0646
6	TY	15.912	0.0628
7	TY	15.9957	0.0625
8	TX	19.4374	0.0514
9	TY	19.5546	0.0511
10	TY	21.0037	0.0476
11	R	25.2288	0.0396

Table 21: Modal Analysis

According to the table 21 it is sufficient to consider 11 modes in order to have as sum at least 90% of the participation masses of the structure.

MODAL PARTICIPATION MASSES						
Mode No	TRAN-X		TRAN-Y		ROTN-Z	
	MASS (%)	SUM (%)	MASS (%)	SUM (%)	MASS (%)	SUM (%)
1	70.4891	70.4891	0.0625	0.0625	0.9661	0.9661
2	0.0624	70.5515	72.4069	72.4694	0.0012	0.9673
3	0.8844	71.4359	0	72.4694	71.4554	72.4227
4	0.1795	71.6154	0.0829	72.5523	0.0002	72.4229
5	0.0002	71.6157	0.0092	72.5615	0.0035	72.4264
6	0.019	71.6346	0.0343	72.5959	0.0007	72.4271
7	0.0019	71.6365	0.0063	72.6022	0.0005	72.4276
8	17.4232	89.0597	1.7156	74.3178	0.5076	72.9352
9	1.9033	90.963	17.6593	91.977	0.0545	72.9897
10	0.3033	91.2663	0.3863	92.3633	0.0233	73.0131
11	0.7403	92.0066	0.0001	92.3634	18.6334	91.6464
Total Mass Participation	92.0066		92.3634		91.6464	

Table 22: Modal Participation Masses

The first mode of vibration occurs in X-direction, the second in Y-direction and the third is a torsional one, as shown below.

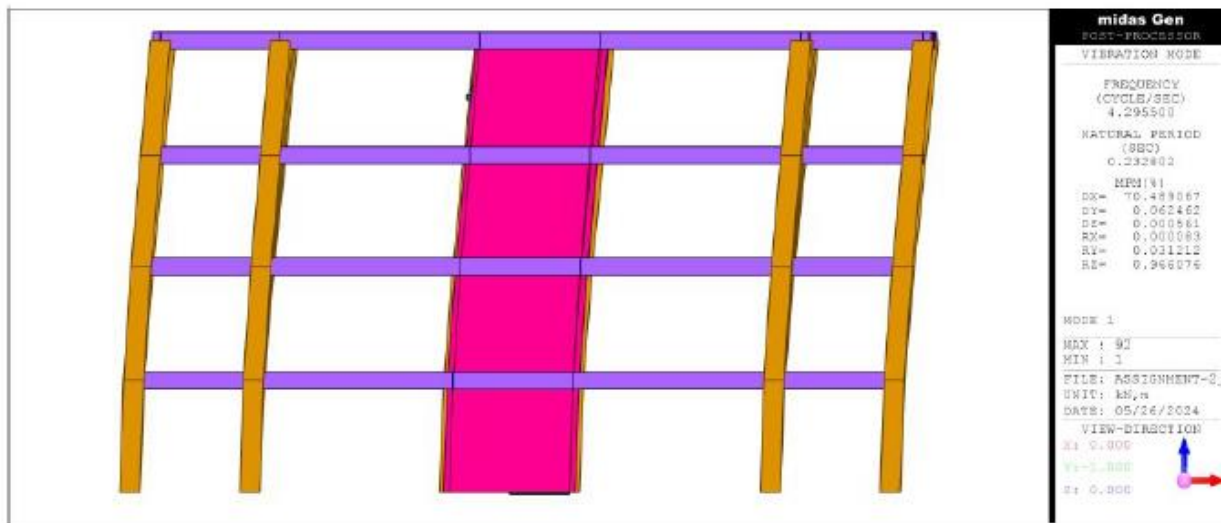


Figure 33: Mode 1 (X-direction)

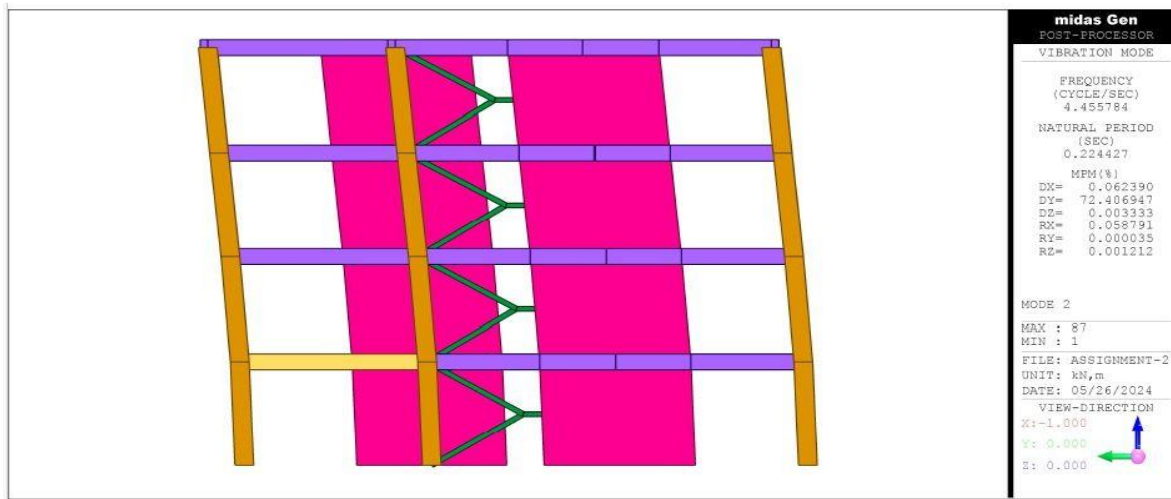


Figure 34: Mode 2 (Y-direction)

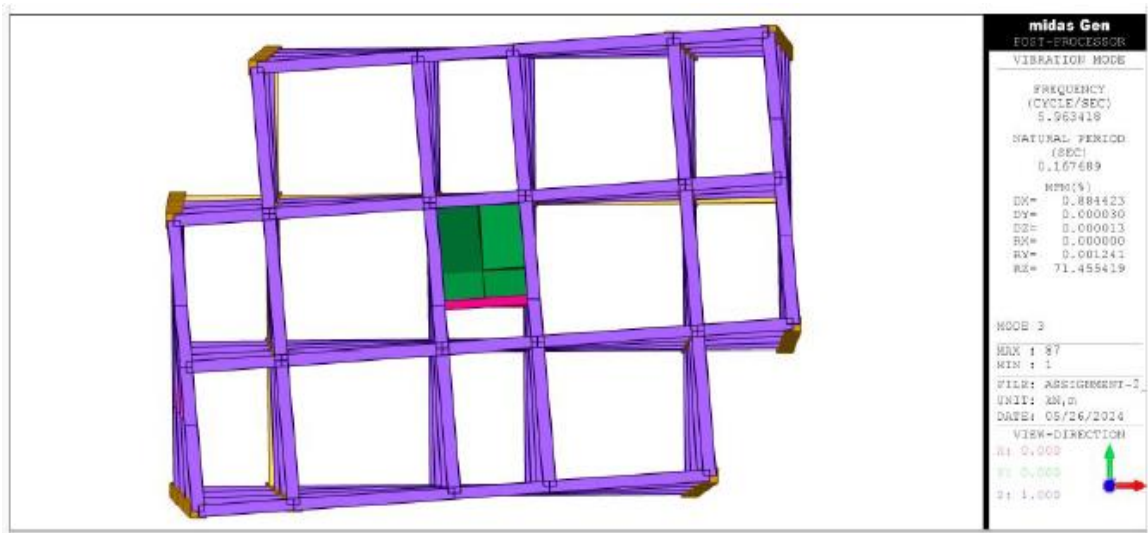


Figure 35: Mode 3 (Torsional in Z-axis)

Now since in our case $T_b < T_1 < T_c$ ($0.17 < 0.199 < 0.5$)

$$S_d(T) = a_g \cdot S \cdot \frac{2.5}{q} = 0.133 \text{ m/s}^2$$

Where, $a_g = 0.206$, $S = 1.397$, $q = 5.4$

From EC8 Equation 4.5,

$$F_b = \lambda m S_d(T)$$

V_b – base shear

$S_d(T_1)$ – ordinate of the design spectrum at period T_1

T_1 - the fundamental period of vibration of the building for lateral motion in the direction considered

m – the total mass of the building, above the foundation or above the top of a rigid basement

λ - the correction factor, the value of which is equal to: 0,85 if $T1 < 2TC$ and the building has more than two storeys, or $\lambda = 1,0$ otherwise.

	X	Y
λ	0.85	0.85
T1 (s)	0.199	0.199
$S_d(T1)$	0.133	0.133
Weight (KN)	10981.30049	10981.3
V_b (KN)	1241.436021	1241.44

$$\text{Ratio of Base shear to total building weight} = \frac{1241.436}{10981.3} \times 100 = 11.3\%$$

CHAPTER 5

DESIGN REINFORCEMENT:

In this chapter, we will discuss about the design of the following elements:

- All walls
- One girder in the X direction
- One girder in the Y direction
- One corner column, one exterior column and one interior column

The results of the analysis for static and seismic loads carried out with MIDAS, have shown that the most stressed beams are present in the floor between the first floor and the second floor. Therefore, it was chosen to design and consequently check at ULS internal girders shown in next plan and column C1,C2 and C9.

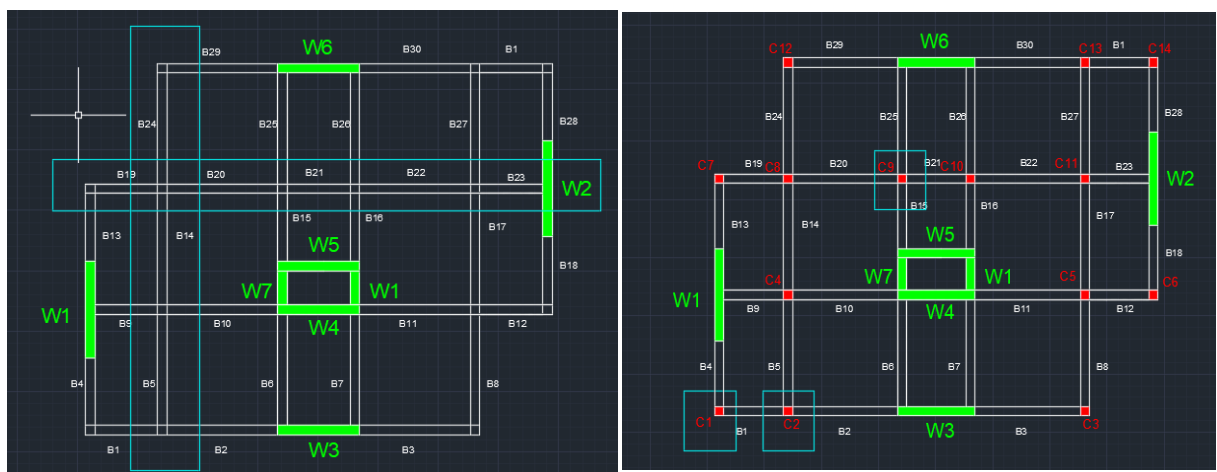


Figure 36- Beams, Columns that designed

5.1 BEAM DESIGN:

Nowadays, the current philosophy behind seismic design is based on the method of capacity design, which is well reflected in the present codes and guidelines. To this extent and to understand the mechanism behind the design process in this assignment a remark about this procedure must be made. Capacity design is based, roughly speaking, on the creation of a hierarchy within the structural elements, so that the design is done in such a way that some elements fail before others (more “important”). In other words, the dissipation needed to sustain earthquake loads is done through the yielding of sections strategically chosen so that does not catastrophically compromise the entire building. This yielding is strictly related to the ductility of the material in the elements in the selected zones and it is idealized by the formation of plastic hinges. Following this reasoning, the concept of the strong column–weak beam gains meaning, in a context where the main objective is to impose a beam-sway mechanism, forcing plastic hinges at the beam ends and out of the columns, avoiding in this way soft story.

We Considered 8 beams named as B5, B14, B24, B19, B20, B21, B22, and B23. For all beams acting bending moment and resisting bending moment were found reinforcement in the longitudinal direction at the top and bottom was provided and all the shear stirrups were provided.



Figure 37: Moment Diagram X-direction Continuous 5 bay beam



Figure 38: Moment Diagram y-direction continuous 3 Bay beam

Initially, the amount of the required reinforcement in each beam was obtained with this formula:

$$A_s = \frac{M_{ed}}{0.9 \cdot f_{yd} \cdot d}$$

The analysis takes both positive and negative bending moments (top and bottom reinforcement). After that the reinforcement was defined and subsequently modified considering all code prescriptions. The resisting bending moment was computed with this formula:

$$M_{rd} = A_s \cdot f_{yd} \cdot z$$

The minimum amount of the reinforcement given from EC2 par 9.2.11 is:

$$A_{s,min} = 0.26 \cdot \frac{f_{ctm}}{f_{yk}} \cdot b_t \cdot \quad \& \quad A_{s,min} \geq 0.0013 \cdot b_t \cdot d$$

	PARAMETER	SPAN 1			SPAN 2			SPAN 3			Units
		M-left	M-Mid	M-Right	M-left	M-Mid	M-Right	M-left	M-Mid	M-Right	
ACTING	Med	-41.30	23.70	-41.30	-68.10	37.10	-68.10	-67.40	37.80	-67.40	KNm
	As Top	275.94	63.34	275.94	454.99	99.15	454.99	450.32	101.02	450.32	mm ²
	As Bot	110.37	158.35	110.37	182.00	247.87	182.00	180.13	252.55	180.13	mm ²
	As,min	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	mm ²
DESIGN	Mrd	-70.77	70.77	-70.77	-106.16	70.77	-106.16	-106.16	70.77	-106.16	KNm
	As'	401.92	401.92	401.92	602.88	401.92	602.88	602.88	401.92	602.88	mm ²
	As	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	mm ²
	Total Bars N	4 Ø 16	4 Ø 16	4 Ø 16	5 Ø 16	4 Ø 16	5 Ø 16	5 Ø 16	4 Ø 16	5 Ø 16	
	Check	OK	OK	OK	OK	OK	OK	OK	OK	OK	

Figure 39: Beam in Y-direction Longitudinal Reinf. Check

	PARAM.	SPAN 1			SPAN 2			SPAN 3			SPAN 4			SPAN 5			Units
		M-left	M-Mid	M-Right	M-left	M-Mid	M-Right	M-left	M-Mid	M-Right	M-left	M-Mid	M-Right	M-left	M-Mid	M-Right	
ACTING	Med	-29.50	9.10	-29.50	-69.90	43.40	-69.90	-16.40	1.90	-16.40	-72.50	43.20	-72.50	-8.40	9.50	-8.40	KNm
	As Top	197.10	24.32	197.10	467.02	115.99	467.02	109.57	5.08	109.57	484.39	115.45	484.39	56.12	25.39	56.12	mm ²
	As Bot	78.84	60.80	78.84	186.81	289.97	186.81	43.83	12.69	43.83	193.76	288.63	193.76	22.45	63.47	22.45	mm ²
	As,min	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	332.80	mm ²
DESIGN	Mrd	-70.77	70.77	-70.77	-106.16	70.77	-106.16	-70.77	70.77	-70.77	-106.16	70.77	-106.16	-70.77	70.77	-70.77	KNm
	As'	401.92	401.92	401.92	602.88	401.92	602.88	401.92	401.92	401.92	602.88	401.92	602.88	401.92	401.92	401.92	mm ²
	As	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	mm ²
	Total Bars N	4 Ø 16	4 Ø 16	4 Ø 16	5 Ø 16	4 Ø 16	5 Ø 16	4 Ø 16	4 Ø 16	4 Ø 16	5 Ø 16	4 Ø 16	5 Ø 16	4 Ø 16	4 Ø 16	4 Ø 16	
	Check	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	

Figure 40: Beam in X-direction Longitudinal Reinf. Check

5.1.1. Local Ductility Check:

For the high ductility class structure, by par 5.5.3.1.3 of EC8, the critical area is defined as an area of a seismic beam extending up to a distance

$$l_{cr} = 1.5 * h_w = 1.5 * 50 = 75 \text{ cm}$$

Where the beam is connected to a beam-column node. In this critical area plasticization can occur under the seismic action of the project. In accordance with the provisions of EC8 p 5.5.3.1.3, the following checks must be carried out on the geometric ratio of the tensioned reinforcement ρ and on the geometric ratio of the compressed reinforcement

$$\rho = \frac{A_s}{bd} \qquad \rho' = \frac{A'_s}{bd}$$

Those ratios must satisfy following checks:

- $\rho' \geq 1/2 * \rho$
- $\rho \leq \rho_{max} = \rho' + \frac{0.0018}{\mu_\phi * \epsilon_{sy}} * \frac{f_{cd}}{f_{yd}}$
- $\rho \geq \rho_{min} 0.15$; $50 f_{ct}/f_y$

Where: μ_ϕ is the ductility factor of curvature. Since $T1 < \dots$, is computed as

$$\mu_\phi = 1 + 2(q_0 - 1)TC/T = 23.4$$

with $q_0 = 5.4$

Below tables show verifications for the beams,

	SPAN 1			SPAN 2			SPAN 3		
	I	J	K	I	J	K	I	J	K
As'	401.92	401.92	401.92	602.88	401.92	602.88	602.88	401.92	602.88
As	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92
ρ'	0.0020	0.0020	0.0020	0.0030	0.0020	0.0030	0.0030	0.0020	0.0030
ρ	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020
$\rho/2$	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010
$\rho' > \rho/2$	OK	OK	OK	OK	OK	OK	OK	OK	OK
$\rho(\max)$	0.0041	0.0041	0.0041	0.0051	0.0041	0.0051	0.0051	0.0041	0.0051
$\rho \leq \rho(\max)$	OK	OK	OK	OK	OK	OK	OK	OK	OK
$\rho(\min)$	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015
$\rho \geq \rho(\min)$	OK	OK	OK	OK	OK	OK	OK	OK	OK

Figure 41: Local Ductility Check - Beam in Y-direction

	SPAN 1			SPAN 2			SPAN 3			SPAN 4			SPAN 5		
	I	J	K	I	J	K	I	J	K	I	J	K	I	J	K
As'	401.92	401.92	401.92	602.88	401.92	602.88	401.92	401.92	401.92	602.88	401.92	602.88	401.92	401.92	401.92
As	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92	401.92
ρ'	0.0020	0.0020	0.0020	0.0030	0.0020	0.0030	0.0020	0.0020	0.0020	0.0030	0.0020	0.0030	0.0020	0.0020	0.0020
ρ	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020
$\rho/2$	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010
$\rho' > \rho/2$	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
$\rho(\max)$	0.0041	0.0041	0.0041	0.0051	0.0041	0.0051	0.0041	0.0041	0.0041	0.0051	0.0041	0.0051	0.0041	0.0041	0.0041
$\rho \leq \rho(\max)$	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
$\rho(\min)$	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015
$\rho \geq \rho(\min)$	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK

Figure 42: Local Ductility Check - Beam in X-direction

5.2. Shear Design and Check of Transversal Reinforcement:

In order to avoid brittle failure, the shear stresses calculation was made. Shear actions were computed with the equilibrium of beams under the action of gravitational loads in the seismic design condition ($G_k + \psi_{ik} * Q_k$) and with the resisting bending moments of the beams ($M_{d,Rd}$) of the end sections in correspondence with the formation of the plastic hinges. For each direction of seismic actions, two values of shear stress are considered, maximum and minimum, assuming respectively the presence and absence of variable loads and resisting moment's $M_{d,Rd}$ to be assumed on the beam in each case in agreement with the seismic direction. The capacity criterion finds shear values from the following expressions:

$$V_{ed,min,i} = \frac{(G_k + \psi_{ik} * Q_k) * l_{beam}}{2} + \frac{M_{1d} + M_{2d}}{l_{beam}}$$

$$V_{ed,max,i} = \frac{G_k * l_{beam}}{2} + \frac{M_{1d} + M_{2d}}{l_{beam}}$$

Where:

- G_k is the gravitational load;
- Q_k is the live load;
- ψ_{ik} is the combination coefficient (for residential use is 0.3);
- l_{beam} is the length of the beam
- M_{id} are end moments of beams

The end moments can be computed with this formula (EC8 5.4.2.2):

$$M_{id} = \gamma_{rd} * M_{Rd,i} * \min \left(1, \frac{\Sigma M_{rc}}{\Sigma M_{rb}} \right)$$

Where:

γ_{rd} – coefficient takes into account the possible over resistance of steel coming from the hardening (=1.2)

M_{Rb} , – end resisting bending moment of i-th end of beam in the sense of the seismic bending moment under the considered sense of the seismic action.

ΣM_{rc} and ΣM_{rb} – are respectively the sum of design bending moment of columns and beams in the considered node.

The static scheme used in accordance with the EC8 is reported in the following figure:

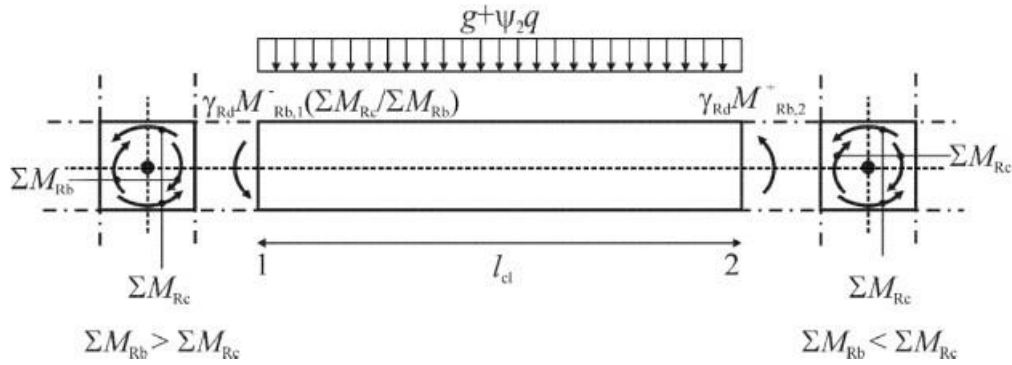


Figure 43: Capacity design values of shear forces on beams

In case if there is the possibility of inversion of the shear at the ends of the beams, it may be necessary arrange diagonal shear reinforcement inclined in two directions. To verify the necessity to design the diagonal reinforcement, the parameter ζ must be calculated with the ratio between the minimum shear and the maximum one.

$$\zeta = \frac{V_{Ed,min}}{V_{Ed,max}}$$

- For $\zeta \geq -0.5$ the inversion of the shear doesn't occur in the critical area;
- For $\zeta \leq -0.5$ the inversion of the shear does occur in the critical area.
- In this case, proceed as follows:
- If $|V_{Ed,max}| \leq (2 + \zeta) * f_{ctd} * b_w * d$, is the same as for $\zeta \geq -0.5$
- If $|V_{Ed,max}| > (2 + \zeta) * f_{ctd} * b_w * d$, it is necessary to have diagonal reinforcements inclined in the two directions at 45° with respect to the beam axis. Furthermore, half of the maximum cutting action must be carried by the stirrups and the remaining part must be carried by the inclined bars.

In the case under consideration, the first condition $\zeta \geq -0.5$ is always satisfied, therefore it is not necessary to insert inclined bars. The results obtained are reported in the following tables:

	SPAN 1		SPAN 2		SPAN 3	
M_{rd}	$M_{rd} (-ve)$	$M_{rd} (+ve)$	$M_{rd} (-ve)$	$M_{rd} (+ve)$	$M_{rd} (-ve)$	$M_{rd} (+ve)$
	-70.77	70.77	-106.16	70.77	-106.16	70.77
g_k	44.91		44.91		44.91	
Ψ_q	4.42		4.42		4.42	
M_{id}	M_{1d}	M_{2d}	M_{1d}	M_{2d}	M_{1d}	M_{2d}
	-84.93	84.93	-127.39	84.93	-127.39	84.93
$V_{ed} (KN)$	$V_{ed} Max$	$V_{ed} Min$	$V_{ed} Max$	$V_{ed} Min$	$V_{ed} Max$	$V_{ed} Min$
	123.32	112.28	114.83	103.78	114.83	103.78
ζ	0.91		0.90		0.90	
Check	OK not req		OK not req		OK not req	

Table 23: Inclined Bar check - Y direction Beam

	SPAN 1		SPAN 2		SPAN 3		SPAN 4		SPAN 5	
M_{rd}	$M_{rd}(-ve)$	$M_{rd}(+ve)$	$M_{rd}(-ve)$	$M_{rd}(+ve)$	$M_{rd}(-ve)$	$M_{rd}(+ve)$	$M_{rd}(-ve)$	$M_{rd}(+ve)$	$M_{rd}(-ve)$	$M_{rd}(+ve)$
	-70.77	70.77	-106.16	70.77	-70.77	70.77	-106.16	70.77	-70.77	70.77
g_k	44.91		44.91		44.91		44.91		44.91	
Ψ_q	4.42		4.42		4.42		4.42		4.42	
M_{id}	M_{1d}	M_{2d}	M_{1d}	M_{2d}	M_{1d}	M_{2d}	M_{1d}	M_{2d}	M_{1d}	M_{2d}
	-84.93	84.93	-127.39	84.93	-84.93	84.93	-127.39	84.93	-84.93	84.93
$V_{ed} (KN)$	$V_{ed} Max$	$V_{ed} Min$	$V_{ed} Max$	$V_{ed} Min$	$V_{ed} Max$	$V_{ed} Min$	$V_{ed} Max$	$V_{ed} Min$	$V_{ed} Max$	$V_{ed} Min$
	123.32	112.28	114.83	103.78	123.32	112.28	114.83	103.78	123.32	112.28
ζ	0.91		0.90		0.91		0.90		0.91	
Check	OK not req		OK not req		OK not req		OK not req		OK not req	

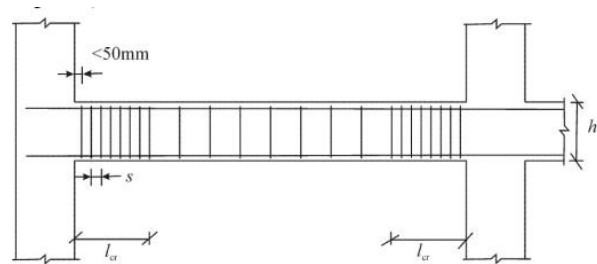
Table 24: Inclined Bar check - X direction Beam

5.3. Shear Reinforcement

5.3.1. Critical Area

In accordance with the requirements of EC8, in critical areas the arrangement of the stirrups must meet the following requirements:

- The first containment stirrup should be no more than 5 cm away from the end section as showed in the following figure:



- Stirrups must be placed with a spacing not greater than the smaller of the following quantities:
 - A quarter of the useful height of the cross section
 - 175 mm for high ductility class
 - Six times the minimum diameter of the longitudinal bars considered for the purposes of the checks for high ductility class
 - 24 times the diameter of the transverse reinforcement

$$s = \min \left(\frac{h_w}{4}; 175 \text{ mm}; 6 * d_{bl}; 24 d_{bw} \right)$$

$$= \min(125 \text{ mm}; 175 \text{ mm}; 96 \text{ mm}; 192 \text{ mm})$$

$$s = 96 \text{ mm}$$

- Minimum diameter of the stirrups is 6mm.

In critical areas of primary beams, the angle of the shear reinforcement with respect to the longitudinal axis is equal to $\theta = 45^\circ$, as required by EC2. Respecting the previous prescription, the resistant shear was then calculated as:

$$V_{rd} = \min (; c)$$

Where,

$$V_{rd,s} = \frac{A_{sw}}{s} * 0.9 * d * f_{ywd} * \cot \theta$$

$$V_{rc,c} = \frac{f_{cd}}{\cot \theta + \tan \theta} * 0.9 * d * b_w * \alpha_{cw} * v_1$$

$$v_1 = v = 0.6 * \left[1 - \frac{f_{ck}}{250} \right] = 0.516$$

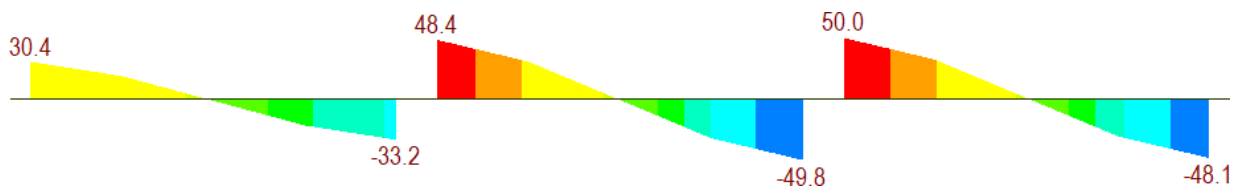


Figure 44: Shear- Y-Direction

	Span 1		Span 2		Span 3	
Location	i	j	i	j	i	j
V_{ed} (KN)	30.4	33.2	48.4	49.8	50	48.1
f_{cd} (KN/m ²)	19.8					
$\emptyset$ (mm)	8mm dia stirrups					
d (mm)	450					
A_{sw} (mm ²)	100.48					
Spacing, s (mm)	90	90	90	90	90	90
V_{rsd} (KN)	111.66	111.66	111.66	111.66	111.66	111.66
V_{rcd} (KN)	832.32	832.32	832.32	832.32	832.32	832.32
V_{rd} (KN)	111.66	111.66	111.66	111.66	111.66	111.66
Safety Index	3.67	3.36	2.31	2.24	2.23	2.32

Table 25: Shear Reinforcement design in critical region (Y-Direction)

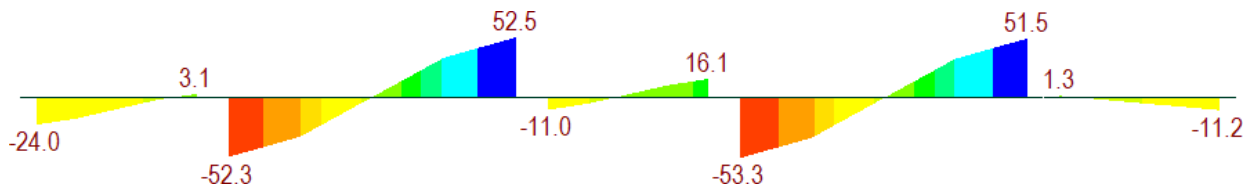


Figure 45; Shear- X-Direction

	Span 1		Span 2		Span 3		Span 4		Span 5	
Location	i	j	i	j	i	j	i	j	i	j
V_{ed} (KN)	24	3.1	52.3	52.5	11	16.1	53.3	51.5	1.3	11.2
f_{cd} (KN/m ²)	19.8									
$\emptyset$ (mm)	8mm dia stirrups									
d (mm)	450									
A_{sw} (mm ²)	100.48									
Spacing, s (mm)	90	90	90	90	90	90	90	90	90	90
V_{rsd} (KN)	111.66	111.66	111.66	111.66	111.66	111.66	111.66	111.66	111.66	111.66
V_{rcd} (KN)	832.32	832.32	832.32	832.32	832.32	832.32	832.32	832.32	832.32	832.32
V_{rd} (KN)	111.66	111.66	111.66	111.66	111.66	111.66	111.66	111.66	111.66	111.66
Safety Index	4.65	36.02	2.13	2.13	10.15	6.94	2.09	2.17	85.89	9.97

5.3.2. Out of Critical Area:

$$l_{cr} = 1.5 * h_w = 1.5 * 50 = 75 \text{ cm}$$

$S_{w,max}$ outside critical regions

$$\min(0.8d, 0.3m) = \min(0.8*450, 300) = 300 \text{ mm}$$

	Span 1		Span 2		Span 3	
Location	1.5	3.5	1.5	3.5	1.5	3.5
V_{ed} (KN)	11.32	14.12	18.94	20.34	20.57	18.67
f_{cd} (KN/m ²)	19.8					
$\emptyset$ (mm)	8mm dia stirrups					
d (mm)	450					
A_{sw} (mm ²)	100.48					
Spacing, s (mm)	200	200	200	200	200	200
V_{rsd} (KN)	50.25	50.25	50.25	50.25	50.25	50.25
V_{rcd} (KN)	832.32	832.32	832.32	832.32	832.32	832.32
V_{rd} (KN)	50.25	50.25	50.25	50.25	50.25	50.25
Safety Index	4.44	3.56	2.65	2.47	2.44	2.69

Table 26: Transversal Reinf outside critical region- Y-direction Beam

	Span 1		Span 2		Span 3		Span 4		Span 5		
Location	Mid span		i	j	Mid span		i	j	Mid span		
V _{ed} (KN)	15.87		20.86	21.06	2.87		21.86	20.06	4.95		
f _{cd} (KN/m ²)	19.8										
Ø (mm)	8mm dia stirrups										
d (mm)	450										
A _{sw} (mm ²)	100.48										
Spacing, s (mm)	200		200	200	200		200	200	200		
V _{rsd} (KN)	50.25		50.25	50.25	50.25		50.25	50.25	50.25		
V _{rcd} (KN)	832.32	832.32	832.32	832.32	832.32	832.32	832.32	832.32	832.32	832.32	
V _{rd} (KN)	50.25		50.25	50.25	50.25		50.25	50.25	50.25		
Safety Index	3.17		2.41	2.39	17.51		2.30	2.50	10.15		

Table 27: Transversal Reinf outside critical region- Y-direction Beam

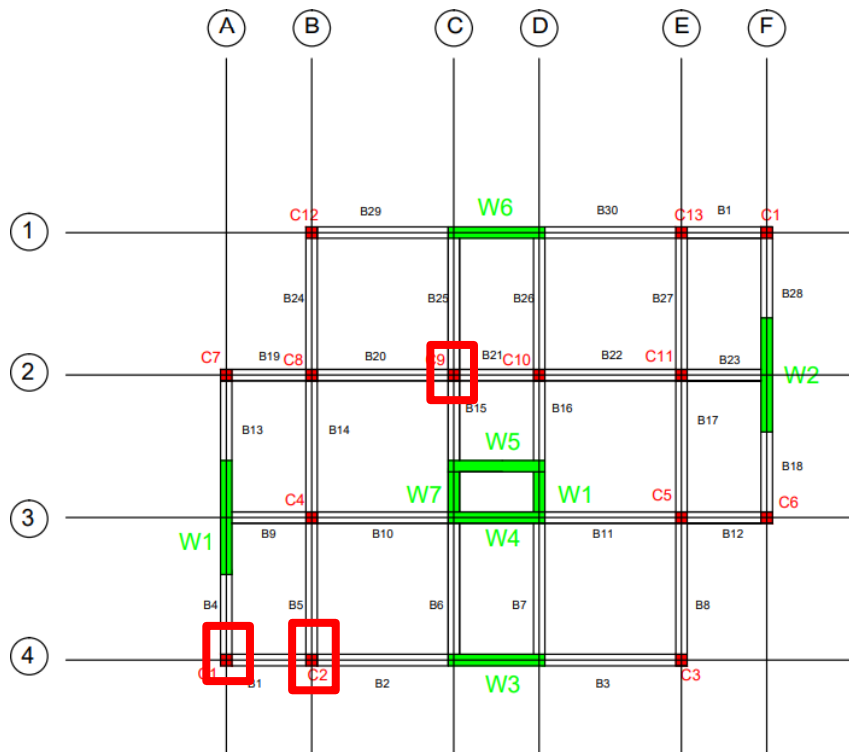
5.3.4. Reinforcement Detailing in Beams:

Beams	Horizontal Reinf.	Stirrups	
		Critical Region ($l_{cr}=0.75m$)	Non Critical Region
1	4/5- $\emptyset 16$	$\emptyset 8@90$	$\emptyset 8@200$
2	4/5- $\emptyset 16$	$\emptyset 8@90$	$\emptyset 8@200$

Table 28: Reinforcement Detailing in Beams

5.2 COLUMN DESIGN:

In this chapter design of internal, external (Corner and non-corner) columns related to frames in X and Y directions was performed in detail. Chosen columns are demonstrated in figure.



First longitudinal reinforcement was designed and check, afterward the shear reinforcement was completed. Maximum actions on columns were taken from Midas Gen, using load combinations prescribed before. There was done a comparison between seismic and gravitational actions, it is summarized for each column below

Column	M _{max} (KNm)			N _{max} (KN)			V _{max} (KN)
	Seismic	ULS (G)	Max	Seismic	ULS (G)	Max	Seismic
C1	33.17	20.05	33.17	131.41	254.72	254.72	20.69
C2	33.8	48.96	48.96	20.24	659.17	659.17	39.54
C3	24.86	69.61	69.61	124.71	1079.33	1079.33	36.98

Table 29: Comparison of Max excitation in Columns

According to EC8 Section 5.5 certain parameters must be met in order to satisfy DCH. EC8 Sec. 5.5.1.2.2 Prescribes geometric constraint that cross section sides must be > 250mm. The dimensions of all the columns is 50 cm*50 cm. N.B.: The same reinforcement layout was assigned to the rest of columns with same sectional properties and symmetric position in structural plan.

5.2.1. Compression check:

According to Eurocode 8, In primary seismic columns the value of the normalized axial force v_d shall not exceed 0,55.

$$v_d = \frac{N_{ed}}{A_c f_{cd}} \leq 0.55$$

Assuming the fundamental combination of the static loads, the values for the axial load for columns C1, C2 and C9 at the second floor and the base floor are given in the table above. The value for normalized axial force is not exceeded, therefore further changes in the design of the building are not required.

Column	h (m)	b(m)	N _{max} (KN)	v _d	Status
C1	0.5	0.5	254.72	0.05	OK
C2	0.5	0.5	659.17	0.13	OK
C9	0.5	0.5	1079.33	0.22	OK

Table 30: Column Compression check

5.2.2. Biaxial Bending with Axial Compression:

To design the longitudinal reinforcement of the columns the biaxial interaction surface shall be exploited, which is based on the sectional equilibrium under axial force and bending moment, and to optimize the design, the interaction plane associated to the resultant action effect of a biaxial flexure shall be considered in determining the longitudinal reinforcement. EC2 Sec. 5.8.9 requires to check the following in the equation:

$$\left(\frac{M_{Edz}}{M_{Rdz}}\right)^a + \left(\frac{M_{Edy}}{M_{Rdy}}\right)^a \leq 1$$

Where a coefficient is determined using

$\frac{N_{Ed}}{N_{Rd}}$	0.1	0.7	1
a	1	1.5	2

$$N_{Rd} = A_c f_{cd} + A_s f_{yd}$$

This check was done using Gelfi VCASLU software and summarized below. Chosen reinforcement bars are decided to put from base to top of the building. Minimum diameter for longitudinal bars for DCH is 8mm.

Columns	N _{max} (KN)	M _{ed,y} (KNm)	M _{ed,z} (KNm)	n	Ø (mm)	M _{rd,y} (KNm)	M _{rd,z} (KNm)	N _{rd} (KN)	$\frac{(M_{ed,y}/M_{rd,y}) + (M_{ed,z}/M_{rd,z})}{1}$	Check
C1	254.7	-20.87	33.17	8	20	-126.30	190.00	5940.45	0.34	OK
C2	659.2	-33.8	-61.75	8	20	-104.70	-196.90	5940.45	0.64	OK
C9	1079	-35	-69.61	8	20	-99.58	-202.30	5940.45	0.70	OK

Table 31: Column Check for Biaxial Bending

MN Domain of Columns:

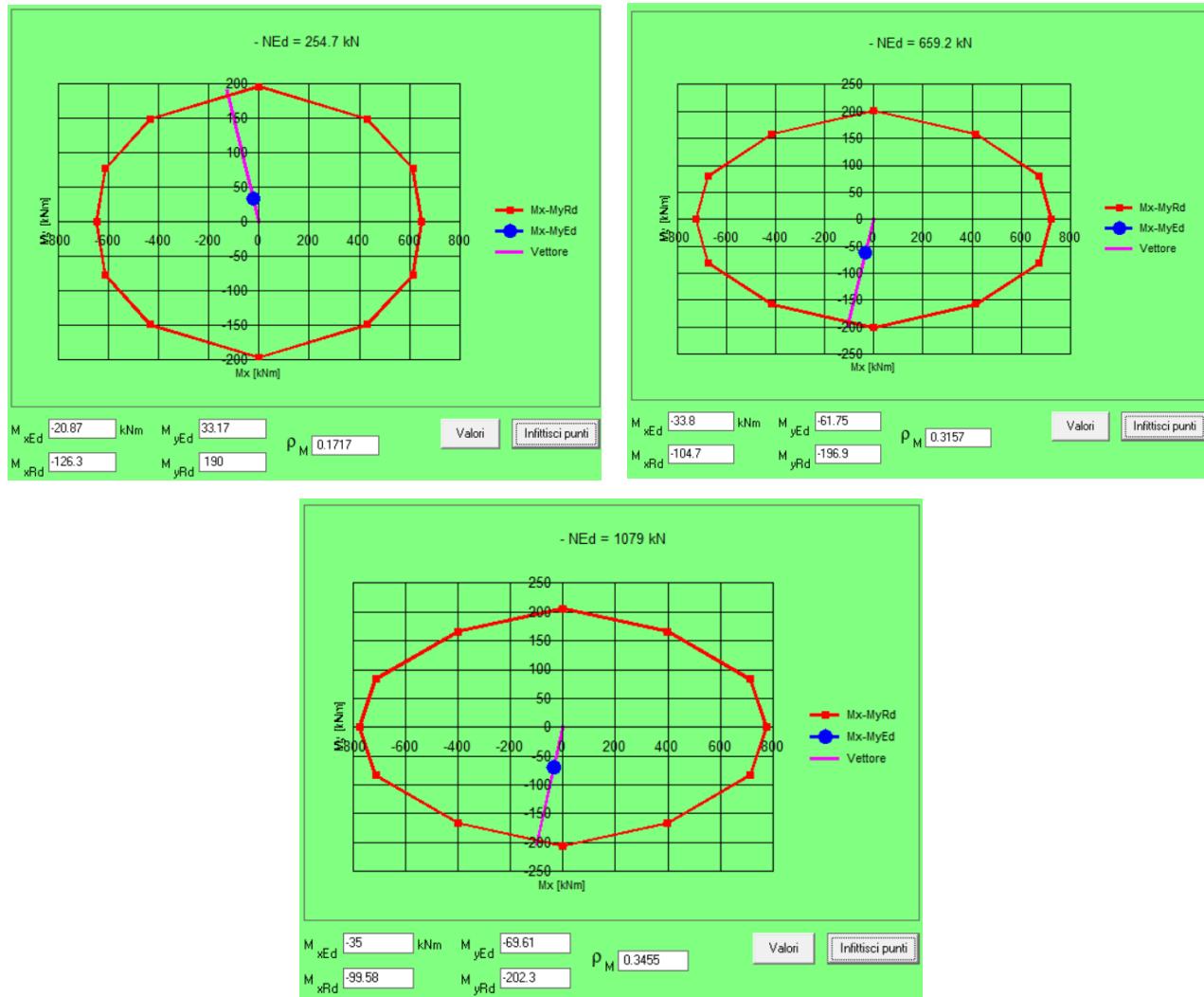


Figure 46: MMN Domain of Column C1, C2, C9 Respectively

After that check on the geometrical steel ratio was performed. EC8 Sec. 5.4.3.2.2 requires:

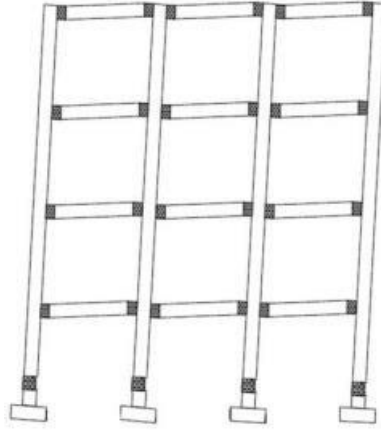
$$1\% \leq \rho = A_s/A_c \leq 4\%$$

Columns	ρ (%)	Check
C1	1.0048	ok
C2	1.0048	ok
C9	1.0048	ok

Table 32: Column Geometrical Ratio

5.2.3. Capacity Check:

Columns shall be designed in such a way to guarantee their complete integrity by avoiding the formation of plastic hinges over their critical region except for their bases, where a plastic hinge may develop to reach a possible global collapse mechanism as shown



With the aim at channeling such plastic mechanism over the entire height of the building the so-mentioned hierarchy of resistance is explicitly considered and enforced by EC8 Sec. 4.4.2.3 by checking in all beam column joints up to the uppermost story a total column capacity 30% greater than crossing-beams resistance, namely:

$$\sum M_{Rc} \geq \gamma_{Rd} \sum M_{Rb}$$

Where:

- $\gamma_{Rd} = 1.3$, is an over strength value for the resistance
- $\sum M_{Rc}$ – sum of the moment resistance of the columns conveying at the node;
- $\sum M_{Rb}$ – sum of moment resistance of the corresponding connecting beams at that node

Columns	Interconnect Beams	$M_{rd}(\text{Col}), x$ (KNm)	$M_{rd}(\text{Col}), y$ (KNm)	$M_{rd}(\text{Beam}), x$ (KNm)	$M_{rd}(\text{Beam}), y$ (KNm)	Check
C1	2	190.00	126.30	70.77	70.77	ok
C2	3	196.90	104.70	35.39	70.77	ok
C9	4	202.30	99.58	35.39	70.77	ok

Table 33: Capacity Check in Columns

5.2.4. Shear Check:

The same capacity-design philosophy used for the shear design of beams shall be exploited for the columns to prevent a brittle shear-based failure which the columns might be prone to if not controlled by flexural capacities. Design of shear should be performed respecting the criteria of resistance hierarchy. EC8 Sec. 5.5.2.2 prescribes rules for avoiding the formation of brittle mechanisms of failure for DCH structures. As developed for beams, an equilibrium analysis based on the incipient possible failure of the column is performed to obtain the design shear demand that ensued from the plastic hinging of the adjacent beams. However, since the hierarchy of strength is ensured, plastic hinges are most likely to develop in beams, hence, the local collapse situation entails a moment distribution at the nodal zone based on the beams' resisting bending moments. So, the shear is constant throughout the column height and is equal to:

$$V_{ed,i} = \frac{M_{1d} + M_{2d}}{H_{cl}}$$

Where: H_{cl} – clear interstorey height

M_{1d} and M_{2d} – resisting bending moments applied on the ends of column

$$M_{id} = \gamma_{Rd} * M_{Rc,i} * \min \left(1; \frac{\sum M_{Rb,i}}{\sum M_{Rc,i}} \right)$$

$\gamma_{Rd} = 1.3$ – factor accounting for the over strength related to the strain hardening of the steel and confinement of the concrete

M_{Rc} , and M_{Rb} , – design bending moments of columns and beams respectively in i -node.

Columns	$M_{rd, x1}$ (KN.m)	$M_{rd, x2}$ (KN.m)	$M_{rd, y1}$ (KN.m)	$M_{rd, y2}$ (KN.m)	$V_{ed,x}$ (KN)	$V_{ed,y}$ (KN)
C1	92.00	92.00	92.00	92.00	55.75982	55.75982
C2	46.01	46.01	46.01	92.00	27.88303	41.82143
C9	46.01	46.01	46.01	92.00	27.88303	41.82143

According to EC8 Sec. 5.5.3.2.2 (4)P, critical zone length is determined as:

$$l_{cr} = \max \left\{ 1.5 * h; \frac{1}{6} * H_{cl}; 0.6m \right\}$$

Additional limits to satisfy as required in EC8 Sec. 5.5.3.2:

$$s_{max} = \min \left\{ \frac{1}{3} * b; 12.5cm; 6 * d_{bL} \right\} \quad \text{in critical zones}$$

$$s_{max} = \min \{ 20 * d_{bL}; h_c; b_c; 40 \text{ cm} \} \quad \text{outside critical zones}$$

$$s_{max} = \min \{ 12 * d_{bL}; 0.6 * h_c; 0.6 * b_c; 24 \text{ cm} \} \quad \text{at lap splices, if } d_{bL} > 14mm$$

$$d_{bw} \geq \min \left\{ 6mm; \frac{d_{bL}}{4} \right\}$$

Columns	l_{cr} (m)	Transverse Bars Non Critical Zone				Transverse Bars Critical Zone		
		ϕ_s (mm)	s_{max} (mm)	s_{max} (splice)	S	ϕ_s (mm)	s_{max} (mm)	S
C1	0.75	8	400	240	200	8	120	100
C2	0.75	8	400	240	200	8	120	100
C9	0.75	8	400	240	200	8	120	100

Table 34: Column stirrup design

For Column Resistant shear,

$$V_{Rd} = \min \{ V_{Rsd}; V_{Rcd} \}$$

$$V_{Rsd} = \frac{A_{sw}}{s} * 0.9 * d * f_{ydw} * (ctg \alpha + ctg \theta) * \sin \alpha$$

$$V_{Rcd} = \frac{f_{cd}}{ctg \theta + tg \theta} * 0.9 * d * b_w * \alpha_{cw} * v_1$$

Where:

A_{sw} – area of hoops

α – inclination angle of steel bars, 90°

θ – inclination angle of compressed struts, 45°

α_{cw} – coefficient that considers the interaction between the stress of the compressed strut and the total axial compressive stress applied.

For non-prestressed structures it is taken equal to 1.

v_1 – resistance reduction factor for concrete cracked due to shear, it is taken equal to 0.516.

Once V_{Rd} computed, this check should be satisfied:

$$V_{Rd} \geq V_{Ed}$$

Computations were performed at the base region, where acting shear has the maximum value. Results are reported.

Columns	$V_{Ed, x}$ (KN)	$V_{Ed, y}$ (KN)	$V_{Rd, s}$ (KN)	$V_{Rd, c}$ (KN)	V_{Rd} (KN)	Check
C1	4.7	6.2	397.52	926.20	397.52	ok
C2	6.4	3.6	397.52	926.20	397.52	ok
C9	5.8	5.2	397.52	926.20	397.52	ok

Table 35: Shear Check in Column

5.2.5. Local Ductility Check:

In order to provide the columns with an overall ductile behavior, EC8 (compliant with EC2) gives detailing rules that serve this purpose to avoid, to a satisfactory extent, buckling of longitudinal rebars and guarantee a confined behavior of the member under large compression actions. This may be accomplished by a proper distribution of longitudinal rebars over the entire cross-section and transverse reinforcement layout of cross ties and/or closed rectangular-shaped hoops with an adequate spacing over the critical and outside the critical regions. These provisions allow the core enclosed by transverse reinforcement to attain its ultimate strain, attributed to the sectional curvature, under compression without spalling of concrete within the confined concrete and consequently permitting a controlled spread of plasticity – if eventually develops – over the potential critical regions of the members

EC8 Sec. 5.4.3.2.2 requires:

$$\alpha * \omega_{wd} \geq 30 * \mu_{\phi} * v_d * \varepsilon_{sy,d} * \frac{b_c}{b_o} - 0.035$$

Where,

- $v_1 = v = 0.6 * [1 - \frac{f_{ck}}{250}] = 0.516$
- $\mu_{\phi} = 1 + 2(q_0 - 1)TC/T = 23.4$
- ω_{wd} – mechanical ratio of hoops
- $\varepsilon_{sy,d} = 0.001957$ – steel deformation at yielding
- b_o – dimension of concrete core in column
- $\alpha = \alpha_s * \alpha_n$ – confinement coefficient:

$$\alpha_n = 1 - \sum_n \frac{b_i^2}{6 * b_o * h_o}$$

$$\alpha_s = \left(1 - \frac{s}{2 * b_o}\right) * \left(1 - \frac{s}{2 * h_o}\right)$$

n – number of longitudinal bars laterally contained by the hoops

b_i -distance between two consecutive bars

Geometrical parameters are represented in Figure below:

Column	$\varnothing_s$ (mm)	s(mm)	ω_{wd}	check
C1	8	100	0.1238	ok
C2	8	100	0.1238	ok
C9	8	100	0.1238	ok

Table 36: Local Ductility of Column

5.2.6. Stability Check:

The slender elements subject to compression must be suitably checked for instability because the effects of the second order could be significant. EC2 (Par. 5.8.3.1) allows you to ignore the effects of the second order if the slenderness of the element is less than a certain limit value, or when:

$$\lambda_{cl} \leq \lambda_{lim}$$

$$\lambda_{lim} = \frac{20 * A * B * C}{\sqrt{v_{d,max}}}$$

$$\lambda_{cl} = \frac{l_o}{r_0} = \frac{\frac{L}{2}}{\sqrt{\frac{I_c}{A_c}}}$$

5.2.7. 2nd Order Effect Check:

The criterion for taking into account the second order effect is based on the interstorey drift sensitivity coefficient θ , which is defined with equation (EN 1998-1/4.4.2.2(2))

$$\theta = \frac{P_{tot} * d_r}{V_{tot} * h} \leq 0.1$$

where

- d_r is the interstorey drift
- h is the storey height
- V_{tot} is the total seismic storey shear obtained by modal response spectrum analysis
- P_{tot} is the total gravity load at and above the storey considered in the seismic design situation ($G + 0.3Q$)

The sensitivity coefficients along the elevation for both directions are determined

Story	Story Height (m)	Maximum Drift of All Vertical Elements	P _{tot} (KN)	V _{tot} (KN)		Θ		Check
				X	Y	X	Y	
4F	3.3	0.0045	10981	1241.436	1241.44	0.012062	0.012062	ok
3F	3.3	0.0044				0.011794	0.011794	ok
2F	3.3	0.0036				0.00965	0.009649	ok
1F	3.3	0.0017				0.004557	0.004557	ok

Table 37: 2nd order effect check

5.2.8. Column Reinforcement Detailing:

Columns	N _{rd} (KN)	M _{rd} (Col), x (KNm)	M _{rd} (Col), y (KNm)	Vertical Reinf.	Stirrups	
					Critical Region (l _{cr} = 0.75m)	Non Critical Region
1	5940.45	190.00	126.30	8Ø20	Ø8@100	Ø8@200
2	5940.45	196.90	104.70	8Ø20	Ø8@100	Ø8@200
3	5940.45	202.30	99.58	8Ø20	Ø8@100	Ø8@200

Table 38: Column Reinforcement Detailing

5.3 SHEAR WALLS:

Walls thickness is 0.3 m and their length is 3 m per floor for both walls along the longitudinal direction X transversal direction Y. Walls' verifications have been generated automatically by the program, The final drawings for the walls are presented in Appendix A. It is worth noting that, having applied the Capacity Design requirements and, after having studied all the possible symmetries in the structure (two types of walls), the construction drawings will be better organized.

Different solutions in terms of diameter ($\Phi 16$) have been adopted for longitudinal and vertical rebar arrangements, considering dissipative and non-dissipative regions. Walls resistance checks (RES) have been performed according to Code prescriptions. One of the main aspects which distinguishes resistance checks of columns (based on the Capacity Design, so the Resistance Hierarchy) from the one of walls is that, owing to the cantilever behaviour of walls, the plasticization will interest only a certain critical length h_{cr} , above the base, which is equal to the average floor height 3.3m. In terms of transversal section, that phenomenon is localized in the two border regions of the element of length l_c . Furthermore, since for both walls the ratio between the total height h_w and length l_w is higher than 2, they can be considered as slender.

For slender walls belonging to CD "A", the bending demand M_{Ed} along the height of the wall (line c of the following graph) is obtained with an upward translation of the envelope of the bending moment diagram (line b), which comes from the bending moments obtained by the analysis (line a); the envelope can be assumed linear if the structure does not present large discontinuities in terms of mass, stiffness and resistance along the height. The translation should be performed following the inclination of the compressed members in the shear resistant mechanism; it can be assumed equal to h_{cr} , which represents the height of the inelastic dissipative region. The graph below (Fig. 26) shows all the aspects previously mentioned:

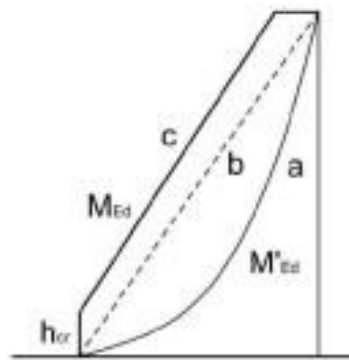
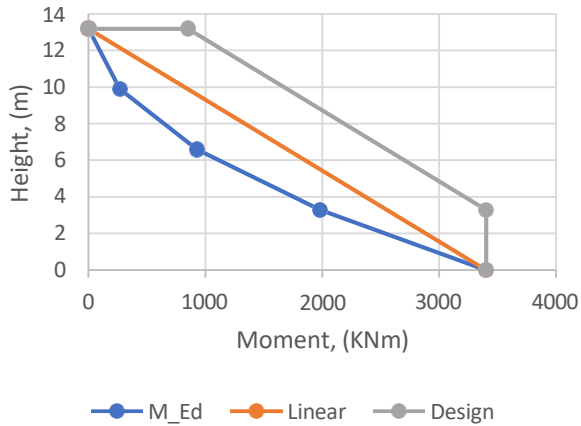
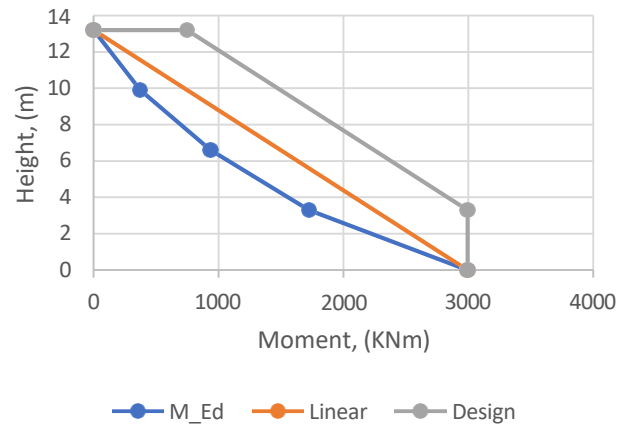


Figure 47- Bending Moment Translation

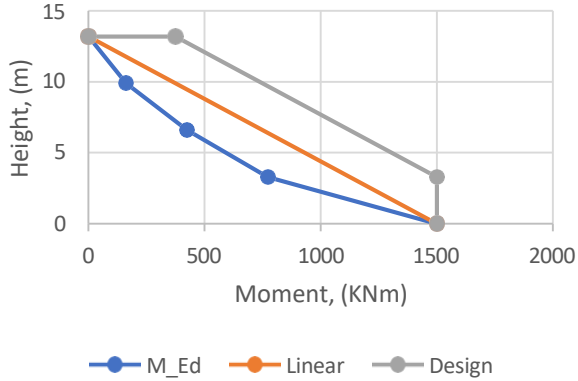
BENDING MOMENT, (SW-1)



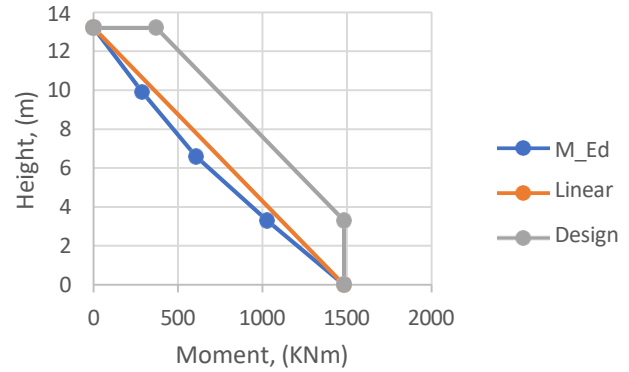
BENDING MOMENT, (SW-2)



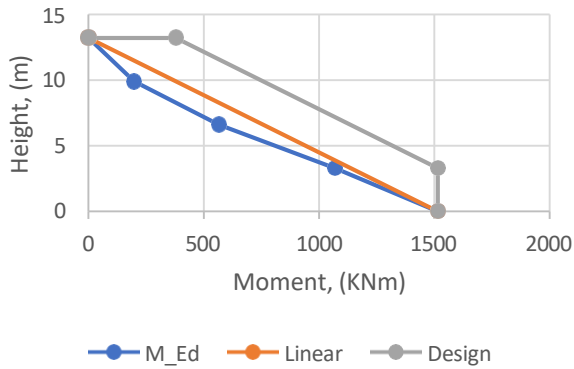
BENDING MOMENT, (SW-3)



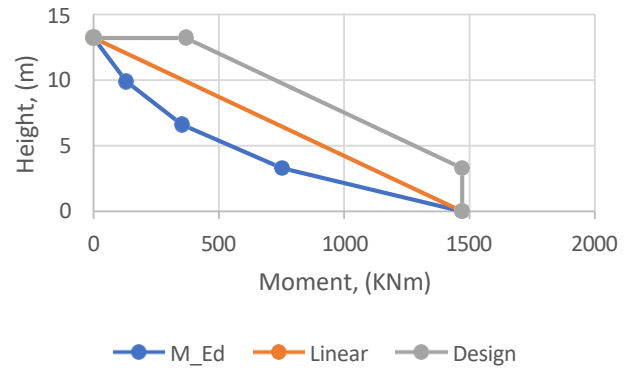
BENDING MOMENT, (SW-4)



BENDING MOMENT, (SW-5)



BENDING MOMENT, (SW-6)



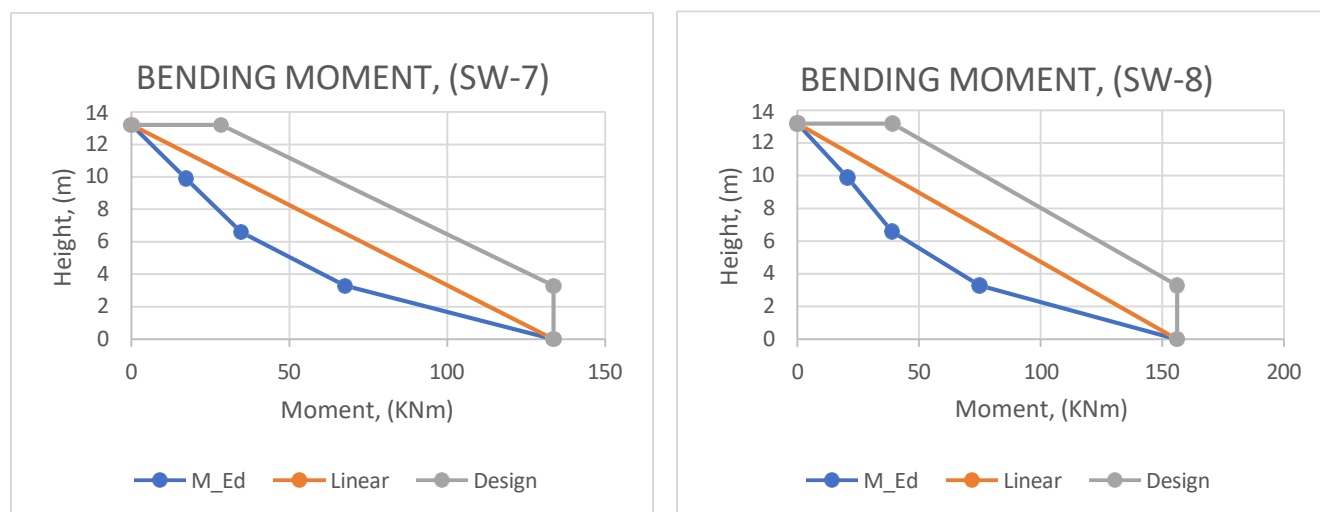


Figure 48: Design Envelopes for Shear walls

5.3.1. Geometry Checks:

Wall Thickness:

$$\text{Web thickness, } b_{w0} \geq \max(150\text{mm}, h_{\text{storey}}/20)$$

$$\text{Web thickness, } b_{w0} \geq \max(150\text{mm}, 3300/20)$$

$$\text{Web thickness, } b_{w0} \geq 165\text{mm}$$

Web thickness considered = 300 mm

For critical region height, h_{cr}

$$\geq \max(l_w, H_w/6)$$

$$\leq \min(2l_w, h_{\text{storey}}) \text{ if wall } \leq 6 \text{ storeys}$$

$$\leq \min(2l_w, 2h_{\text{storey}}) \text{ if wall } > 6 \text{ storeys}$$

Shear Walls	Length (m)	Total Height of wall (m)	Storey Height (m)	h_{cr} (m)		Considered, h_{cr} (m)
				Max	Min	
1	4	13.2	3.3	4	3.3	3.3
2	4	13.2	3.3	4	3.3	3.3
3	3	13.2	3.3	3	3.3	3.3
4	3	13.2	3.3	3	3.3	3.3
5	3	13.2	3.3	3	3.3	3.3
6	3	13.2	3.3	3	3.3	3.3
7	1.4	13.2	3.3	2.2	2.8	2.8
8	1.4	13.2	3.3	2.2	2.8	2.8

Table 39: Critical height h_{cr} determination

For critical region length from edge, l_{cr}

length l_c from edge $\geq 0.15l_w$, $1.5b_w$, length over which $\varepsilon_c > 0.0035$ thickness

b_w over l_c

$$\geq 0.2m;$$

$$h_{st}/15 \text{ if } l_c \leq \max(2b_w, l_w/5),$$

$$h_{st}/10 \text{ if } l_c > \max(2b_w, l_w/5)$$

In our case,

Shear Walls	Length (m)	Length, l_{cr} (m)	Thickness, Req. (m)	Thickness Considered b_w , (m)
1	4	0.6	0.22	0.3
2	4	0.6	0.22	0.3
3	3	0.45	0.22	0.3
4	3	0.45	0.22	0.3
5	3	0.45	0.22	0.3
6	3	0.45	0.22	0.3
7	1.4	0.45	0.22	0.3
8	1.4	0.45	0.22	0.3

Table 40: Critical Lengths of Shear walls

5.3.2. Maximum Actions on Shear Walls:

Shear walls	M_{max} (KN.m)	N_{max} (KN)	V_{max} (KN)
1	3400.64	577.454	395.386
2	2992.77	602.009	397.004
3	1499.67	946.667	254.609
4	1479.91	314.351	336.353
5	1513.75	542.099	335.123
6	1469.62	918.657	234.569
7	86.9079	574.62	133.634
8	120.672	603.04	156.328

Table 41: Max Excitation in Shear Wall

Compression Check:

For all the walls belonging to CD “A”, the demand of axial compression force should not exceed the 35% of the maximum capacity in compression of the concrete section, for all the involved combinations.

$$v_d = \frac{N_{ed}}{A_c f_{cd}} \leq 0.35$$

Shear Walls	l_w (m)	b_w (m)	N_{max} (kN)	v_d	Status
1	4	0.3	577.454	0.02	OK
2	4	0.3	602.009	0.03	OK
3	3	0.3	946.667	0.05	OK
4	3	0.3	314.351	0.02	OK
5	3	0.3	542.099	0.03	OK
6	3	0.3	918.657	0.05	OK
7	1.4	0.3	574.62	0.07	OK
8	1.4	0.3	603.04	0.07	OK

Table 42: Compression Check

5.3.3. Vertical Reinforcement Checks:

Critical Region:

$$0.5\% \leq \rho_{min} \leq 4\%$$

Considered Mechanical steel ratio, $\rho = 2\%$

Shear Walls	l_{cr} (m)	b_w (m)	A_c (critical Region) (m^2)	Area of Steel Reinf. (mm^2)	No. of $\phi 20$	Spacing (mm)
1	0.6	0.3	0.18	3600	12	120
2	0.6	0.3	0.18	3600	12	120
3	0.45	0.3	0.135	2700	10	112.5
4	0.45	0.3	0.135	2700	10	112.5
5	0.45	0.3	0.135	2700	10	112.5
6	0.45	0.3	0.135	2700	10	112.5
7	0.45	0.3	0.135	2700	10	112.5
8	0.45	0.3	0.135	2700	10	112.5

Table 43: Reinforcement in Critical Regions (shear wall)

Non-Critical Region:

Shear Walls	l_0 (m)	b_0 (m)	A_c (Non Critical Reg.) (m^2)	Area of Steel Reinf. (mm^2)	No. of $\phi 20$	Spacing (mm)
1	2.79	0.3	0.837	8370	28	215
2	2.79	0.3	0.837	8370	28	215
3	2.09	0.3	0.627	6270	20	232
4	2.09	0.3	0.627	6270	20	232
5	2.09	0.3	0.627	6270	20	232
6	2.09	0.3	0.627	6270	20	232
7	0.49	0.3	0.147	1470	6	245
8	0.49	0.3	0.147	1470	6	245

Table 44: Reinforcement outside Critical Regions (shear wall)

Checks

Min Dia of Vertical bars = 8mm (selected #20, **OK**)

Max Dia of Vertical bars = $b_{wo}/8 = 37.5$ (OK)

Spacing $S_v \leq \min(25d_b, 250\text{mm})$

$S_v \leq 250$ mm (OK)

5.3.4. Horizontal Reinforcement Calculations and Checks:**Critical Region (Confining Hoops):**

$$d_{bw} \geq (6\text{mm}, 0.4(f_{yd}/f_{wyd})^{1/2}d_{bL})$$

$$d_{bw} \geq (6\text{mm}, 8\text{mm})$$

$$d_{bw} = 8\text{mm}$$

$$\text{spacing}, S_w \leq (6d_{bL}, \frac{b_o}{3}, 125\text{mm})$$

$$\text{spacing}, S_w \leq (6 * 20, \frac{300}{3}, 125\text{mm})$$

$$\text{spacing}, S_w \leq 100$$

Outside Critical Region (Confining Hoops):

$$\text{hoops with } d_{bw} \geq \left(6mm, \frac{d_{bl}}{4}\right) = 8mm$$

Spacing up to a distance of $4b_w$ (1.2m) above or below floor beams or slabs,

$$\text{spacing } s_w \leq (12d_{bl}, 0.6b_{wo}, 240mm) = 180mm$$

Beyond that distance (centre of inter-story height = 0.9 m)

$$s_w \leq (20d_{bl}, b_{wo}, 400mm) = 300mm$$

Local Ductility Check:

The local ductility check to perform for the ductile walls in the critical region is ensured by the following relation

$$\alpha w_{wd} \geq \frac{30 \mu_\phi (v_d + w_v) b}{b_0} - 0.035$$

Where,

- $\alpha = \alpha_s \cdot \alpha_n$ confinement effectiveness factor in L_c end zoned

$$\alpha_n = 1 - \sum_n \frac{b_i^2}{6 * b_0 * h_0}$$

$$\alpha_s = \left(1 - \frac{s}{2 * b_0}\right) * \left(1 - \frac{s}{2 * h_0}\right)$$

- $w_{wd} = (A_{st, x} / (s * b_0) + A_{st, y} / (s * h_0)) * f_{yd} / f_{cd} = w_x + w_y$
- μ_ϕ required curvature ductility factor to calculate with (5.4) or (5.5) EC8 where q_0 is amplified with the factor M_{Ed} / M_{Rd}
- w_v = mechanical ratio of vertical web reinforcement ($\rho_v f_{yd} / f_{cd}$).
- $w_{wd} \geq 0.08$ in CDM class and 0.12 in CDH class.

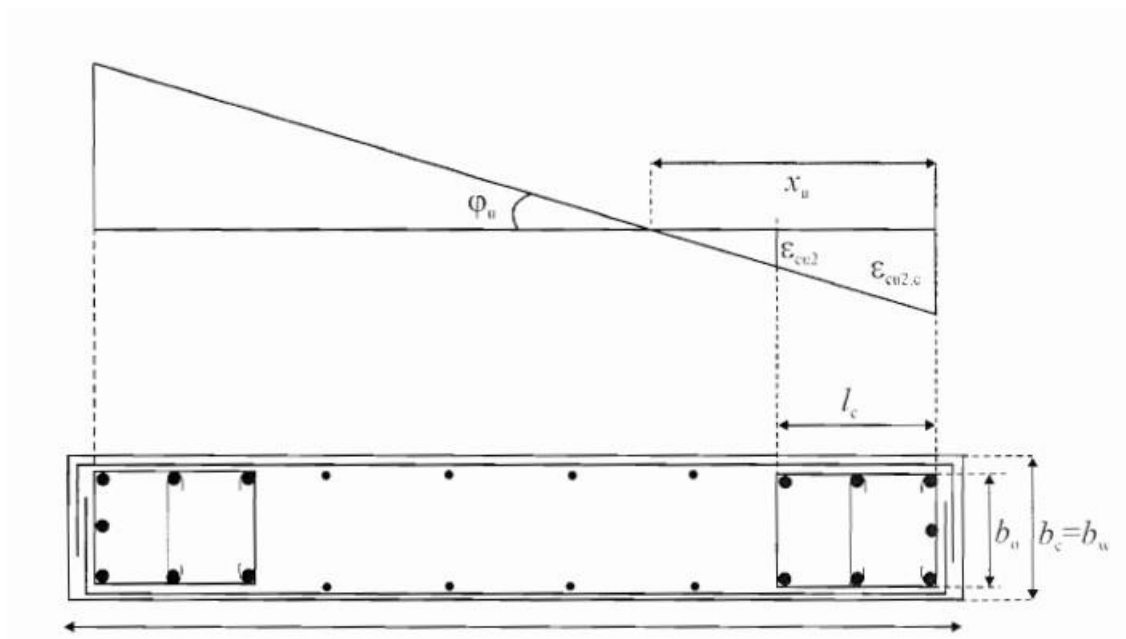


Figure 49: Confined boundary element of free-edge wall end

Shear Wall	$\varnothing_s$ (mm)	s(mm)	α	ω_{wd}	$\alpha * \omega_{wd}$	Check
1	8	120	0.3325	0.1322	0.0440	ok
2	8	120	0.3325	0.1322	0.0440	ok
3	8	112.5	0.3190	0.1433	0.0457	ok
4	8	112.5	0.3190	0.1433	0.0457	ok
5	8	112.5	0.3190	0.1433	0.0457	ok
6	8	112.5	0.3190	0.1433	0.0457	ok
7	8	112.5	0.3190	0.1433	0.0457	ok
8	8	112.5	0.3190	0.1433	0.0457	ok

Table 45: Local Ductility Check

Horizontal bars Outside Critical Region:

$$\rho_{h,min} = 0.2\%$$

$$8 \leq d_{bh} \leq \frac{b_{wo}}{8} \quad (d_{bh} = 12\text{mm})$$

$$\text{Spacing } S_h \leq \min(25d_{bh}, 250 \text{ mm}) = 250\text{mm} \quad (\text{Spacing} = 200\text{mm})$$

5.3.5. Moment Capacity Check:

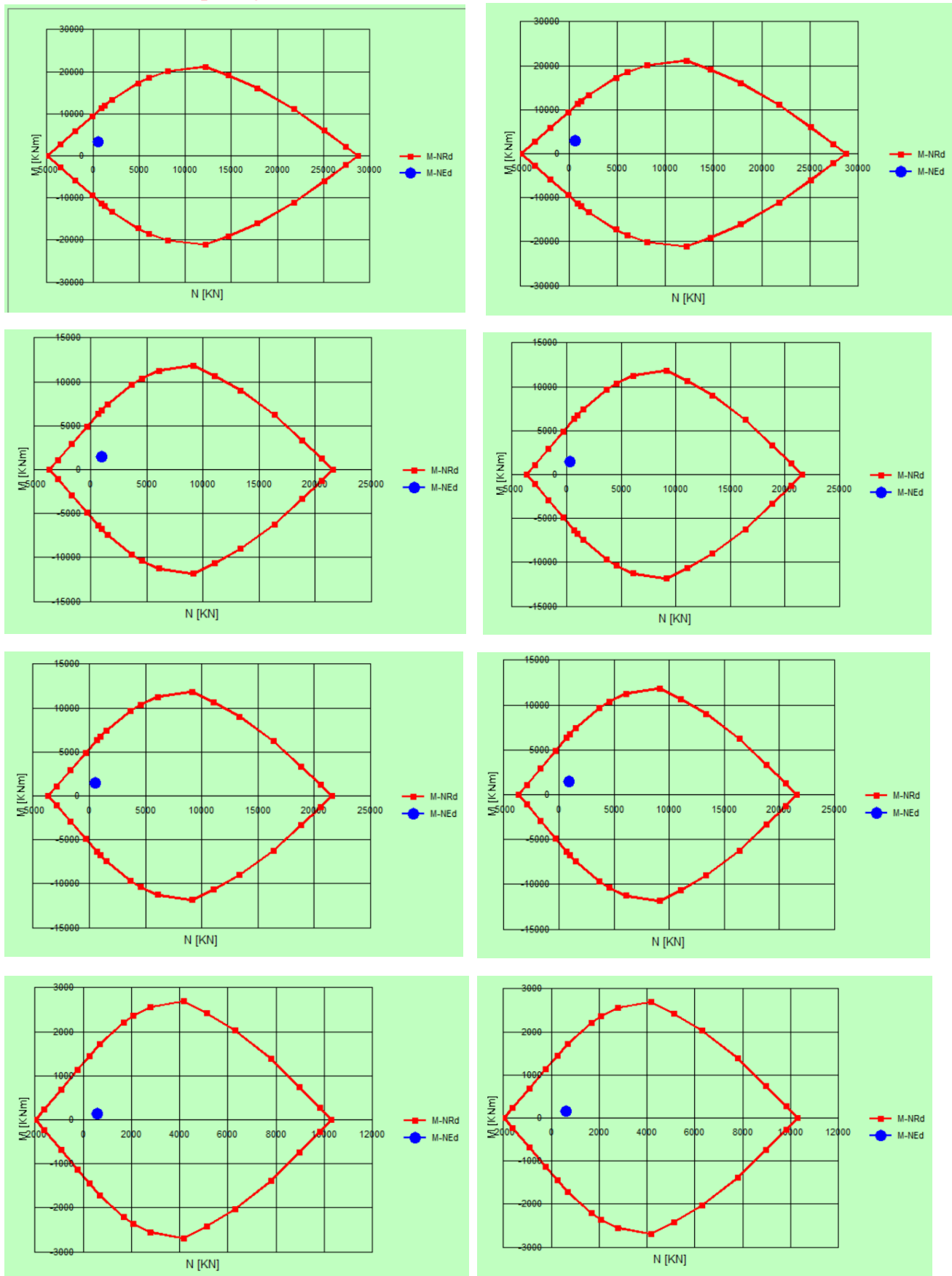
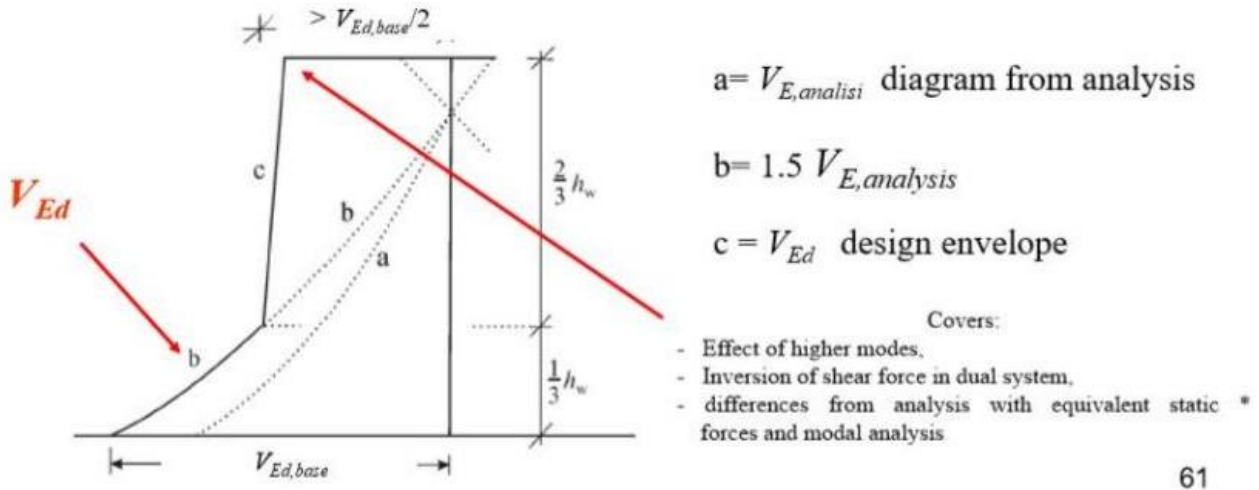


Figure 50: MN Interaction Diagrams for shear walls

5.3.6. Shear Check:

Using amplification factors for shear forces is an appropriate simplified method that may be used to take into account the uncertainties in the analysis and post-elastic dynamic effects.



61

The design shear action is amplified such that it is greater than the action provided by the seismic analysis.

- These assumptions consider not only the influence of the higher modes (more important than the influence of the bending moments) but also
- Considering the bending resistance at the base of the wall, which is generally greater than the value obtained by the analysis and leads to greater values of shear actions.
- Finally, this amplification also accounts for the interaction between the wall and the frame of the structure, where at lower story levels the wall limits the frame displacements, the opposite is true at higher stories.

These requirements are satisfied if the following simplified procedure is applied, using the capacity design rule (suggested by EC8) for a slender wall under high ductility conditions (DCH) (point 5.5.2.4.1 (7)), the design shear forces V_{Ed} must be derived by the expression:

$$V_{Ed} = \varepsilon \cdot V_{Ed}'$$

$$\varepsilon = q(((\gamma_{Rd}/q) (M_{Rd}/M_{Ed}))^2 + 0.1(Se(T_c)/Se(T_1))^2)^{1/2} \leq q$$

Where:

- q is the behavior factor used in the design
- M_{Ed} is the design bending moment at the base of the wall
- M_{Rd} is the design flexural resistance at the base of the wall
- γ_{Rd} is the factor to account for over strength due to steel strain-hardening, taken equal to 1.2
- T_1 is the fundamental period of vibration of the building in the direction of shear force V_{Ed}
- TC is the upper limit period of the constant spectral acceleration region of the spectrum

- $S_e(T)$ is the ordinate of the elastic response spectrum.

Shear Wall	M_{max} (KN.m)	M_{rd} (KN.m)	$S_e(T_c)$	$S_e(T_1)$	Amplification factor
1	3400.64	10706	0.315	0.504	1.5
2	2992.77	10753	0.315	0.504	1.5
3	1499.67	6688	0.315	0.504	1.5
4	1479.91	5804	0.315	0.504	1.5
5	1513.75	6131	0.315	0.504	1.5
6	1469.62	6652	0.315	0.504	1.5
7	133.538	1652	0.315	0.504	2.8
8	155.951	1669	0.315	0.504	2.4

Figure 51: Shear Amplification coefficient

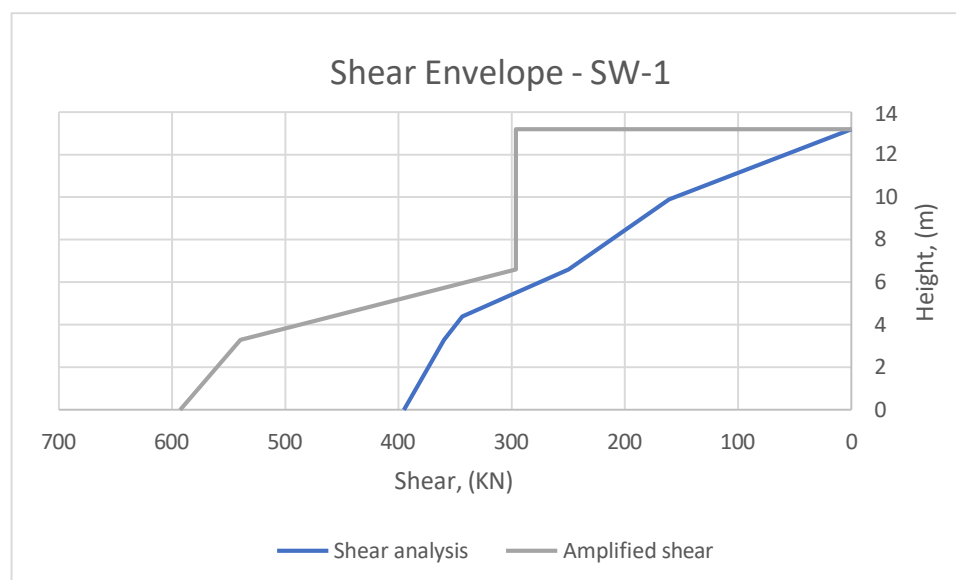


Figure 52: Shear Envelope for SW-1

Rest of the shear walls were also in accordance with the above shear envelope.

SHEAR RESISTANCE, SHEAR TENSION AND SHEAR COMPRESSION FAILURE:

Design shear force in walls of dual systems with $H_w/l_w > 2$, for z between $H_w/3$ and H_w . ⁽⁷⁾	$V_{Ed}(z) = \left(\frac{0.75z}{H_w} - \frac{1}{4} \right) \varepsilon V_{Ed}(0) + \left(1.5 - \frac{1.5z}{H_w} \right) \varepsilon V_{Ed} \left(\frac{H_w}{3} \right)$		from analysis for design seismic action & gravity
$V_{Rd,max}$ outside critical region	As in EC2: $V_{Rd,max} = 0.3(1 - f_{ck}(\text{MPa})/250)b_{wo}(0.8l_w)f_{cd}\sin 2\delta$, with $1 \leq \cot \delta \leq 2.5$		
$V_{Rd,max}$ in critical region	40% of EC2 value	As in EC2	
$V_{Rd,s}$ in critical region; web reinforcement ratios: ρ_h, ρ_v			
(i) if $\alpha_s = M_{Ed}/V_{Ed}l_w \geq 2$: $\rho_v = \rho_{v,min}$, ρ_h from $V_{Rd,s}$:	$V_{Rd,s} = b_{wo}(0.8l_w)\rho_h f_{ywd}$	As in EC2: $V_{Rd,s} = b_{wo}(0.8l_w)\rho_h f_{ywd} \cot \delta$, $1 \leq \cot \delta \leq 2.5$	
(ii) if $\alpha_s < 2$: $V_{Rd,s}$. ⁽⁸⁾ ρ_h from	$V_{Rd,s} = V_{Rd,c} + b_{wo}\alpha_s(0.75l_w)\rho_h f_{yhd}$	As in EC2: $V_{Rd,s} = b_{wo}(0.8l_w)\rho_h f_{ywd} \cot \delta$, $1 \leq \cot \delta \leq 2.5$	
ρ_v from: ⁽⁹⁾	$\rho_v f_{ywd} \geq \rho_h f_{yhd} - N_{Ed}/(0.8l_w b_{wo})$		

Shear Wall	Shear design	Shear Resistance V_{Rd}	α_s	$V_{Rd,c}$ (KN)	$V_{Rd,s}$ (KN)	CHECK
1	593.08	1002.32	0.65	987.00	4428.99	OK
2	595.51	1002.32	0.57	993.40	4010.21	OK
3	381.91	1002.32	0.45	613.63	2381.51	OK
4	504.53	1002.32	0.33	753.61	2074.21	OK
5	502.68	1002.32	0.34	747.51	2103.26	OK
6	351.85	1002.32	0.47	578.91	2459.38	OK
7	200.45	1002.32	0.08	403.59	543.56	OK
8	234.49	1002.32	0.08	473.59	613.32	OK

Table 46: Shear tension and shear compression failure

5.3.7. Shear Wall Reinforcement Details:

Shear Wall	N_{max} (KN)	M_{max} (KN.m)	M_{rd} (KN.m)	Vertical Reinforcement		Horizontal Reinforcement	Confining hoops		
				Critical Zones	Non-Critical		Critical Region	Near Beams (1.2m)	Other (Mid-storeys) (0.9m)
1	577.454	3400.64	10706	Ø20@120	Ø20@210	Ø12@200	Ø8@100	Ø8@180	Ø8@300
2	602.009	2992.77	10753	Ø20@120	Ø20@210	Ø12@200	Ø8@100	Ø8@180	Ø8@300
3	946.667	1499.67	6688	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300
4	314.351	1479.91	5804	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300
5	542.099	1513.75	6131	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300
6	918.657	1469.62	6652	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300
7	574.62	133.538	1652	Ø20@110	Ø20@240	Ø12@200	Ø8@100	Ø8@180	Ø8@300
8	603.04	155.951	1669	Ø20@110	Ø20@240	Ø12@200	Ø8@100	Ø8@180	Ø8@300

Table 47: Reinforcement Details of Shear Walls

CHAPTER 6

WEIGHT OF REINFORCEMENT:

As we must find weight of reinforcement for bending, shear in all the members that we designed. It is showed in the following tables.

6.1 BEAMS:

longitudinal reinforcement								
Beams	Spans	No. of bars	Cross Section Area(m^2)	Length (m)	Dia.(mm)	A_s (mm^2)	Volume of Conc.(m^3)	Weight (kg/m^3)
Beam 1	1	4	0.2	5	16	201	1	31.557
	2	5	0.2	5	16	201	1	39.44625
	3	5	0.2	5	16	201	1	39.44625
Beam 2	1	4	0.2	3	16	201	0.6	31.557
	2	5	0.2	5	16	201	1	39.44625
	3	4	0.2	3	16	201	0.6	31.557
	4	5	0.2	5	16	201	1	39.44625
	5	4	0.2	3	16	201	0.6	31.557

Table 48- Beam Longitudinal Reinforcement Weight

A_s We have beams with sizes of 5 and 3 meters, so we only find shear reinforcement for those 2 sizes.

Shear Reinforcement								
Beams (m)	Critical Region	Non Critical Region	No. of total stirrups	Length of Link (m)	Dia. Of stirrups (mm)	$A_{sw}(mm^2)$	Volume of Conc.(m^3)	Weight (kg/m^3)
5	18	18	36	1.8	8	50.24	1	25.55608
3	18	8	26	1.8	8	50.24	0.6	30.76195

Table 49- Beam Shear Reinforcement Weight

6.2 COLUMNS:

longitudinal reinforcement							
Column	Length(m)	Cross-Section Area(m^2)	No. of bars	Dia.(mm)	A_s (mm^2)	Volume of Conc.(m^3)	Weight (kg/m^3)
1	13.2	0.25	8	20	314	3.3	78.8768
2	13.2	0.25	8	20	314	3.3	78.8768
3	13.2	0.25	8	20	314	3.3	78.8768

Table 50- Column Longitudinal Reinforcement Weight

Shear Reinforcement								
Column	Critical Region	Non Critical Region	No. of total stirrups	Length of Link (m)	Dia. Of stirrups (mm)	$A_{sw} (mm^2)$	Volume of Conc. (m^3)	Weight (kg/ m^3)
1	64	36	100	1.8	8	50.24	3.3	21.51185
2	64	36	100	1.8	8	50.24	3.3	21.51185
9	64	36	100	1.8	8	50.24	3.3	21.51185

Table 51- Column Shear Reinforcement Weight

6.3 SHEAR WALLS:

longitudinal reinforcement								
Wall No.	Length(m)	Height(m)	No. of bars critical	No. of bars Non critical	Dia.(mm)	$A_s (mm^2)$	Volume of conc. (m^3)	Weight (kg/ m^3)
1	4	13.2	24	28	20	314	15.84	106.812
2	4	13.2	24	28	20	314	15.84	106.812
3	3	13.2	20	20	20	314	11.88	109.551
4	3	13.2	20	20	20	314	11.88	109.551
5	3	13.2	20	20	20	314	11.88	109.551
6	3	13.2	20	20	20	314	11.88	109.551
7	1.2	13.2	20	6	20	314	4.752	178.021
8	1.2	13.2	20	6	20	314	4.752	178.021

Table 52- Shear Wall Longitudinal Reinforcement Weight

shear wall shear reinforcement on all Floors								
Wall No.	Reinforcement type		Length(m)	No. of bars	Dia.(mm)	$A_s(mm^2)$	Volume of conc.(m^3)	Weight (kg/ m^3)
1	Horizontal reinforcement		8.2	66	12	113.04	15.84	30.318
	Confining hoops	Critical region	3.2	33	8	50.24		2.629
		Near Beam region(1.2m)	3.2	40				3.187
		mid storey region(0.9m)	3.2	9				0.717
2	Horizontal reinforcement		8.2	66	12	113.04	15.84	30.318
	Confining hoops	Critical region	3.2	33	8	50.24		2.629
		Near Beam region(1.2m)	3.2	40				3.187
		mid storey region(0.9m)	3.2	9				0.717
3	Horizontal reinforcement		6.2	66	12	113.04	11.88	30.565
	Confining hoops	Critical region	2.6	33	8	50.24		2.848
		Near Beam region(1.2m)	2.6	40				3.453
		mid storey region(0.9m)	2.6	9				0.777
4	Horizontal reinforcement		6.2	66	12	113.04	11.88	30.565
	Confining hoops	Critical region	2.6	33	8	50.24		2.848
		Near Beam region(1.2m)	2.6	40				3.453
		mid storey region(0.9m)	2.6	9				0.777
5	Horizontal reinforcement		6.2	66	12	113.04	11.88	30.565
	Confining hoops	Critical region	2.6	33	8	50.24		2.848
		Near Beam region(1.2m)	2.6	40				3.453
		mid storey region(0.9m)	2.6	9				0.777
6	Horizontal reinforcement		6.2	66	12	113.04	11.88	30.565
	Confining hoops	Critical region	2.6	33	8	50.24		2.848
		Near Beam region(1.2m)	2.6	40				3.453
		mid storey region(0.9m)	2.6	9				0.777
7	Horizontal reinforcement		3	66	12	113.04	4.752	36.974
	Confining hoops	Critical region	2.6	33	8	50.24		7.121
		Near Beam region(1.2m)	2.6	40				8.631
		mid storey region(0.9m)	2.6	9				1.942
8	Horizontal reinforcement		3	66	12	113.04	4.752	36.974
	Confining hoops	Critical region	2.6	33	8	50.24		7.121
		Near Beam region(1.2m)	2.6	40				8.631

		mid storey region(0.9m)	2.6	9				1.942
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Table 53- Shear Wall Shear Reinforcement Weight

6.4 SAFETY FACTOR:

Finally, the safety factor on the ground floor was computed. It is defined as the ratio between the shear resistance of the building at the ground level and the seismic base shear. The shear resistance of the building was computed as the sum of column shear resistance at the base level. Obtained results in directions X and Y are summarized in the Table below.

Direction	X	Y
V_{ED} Base (KN)	1241.44	1241.44
V_{RD} Base (KN)	11116.52	11116.52
Safety Factor	11.17	11.17

Table 54- Safety Factor Evaluation

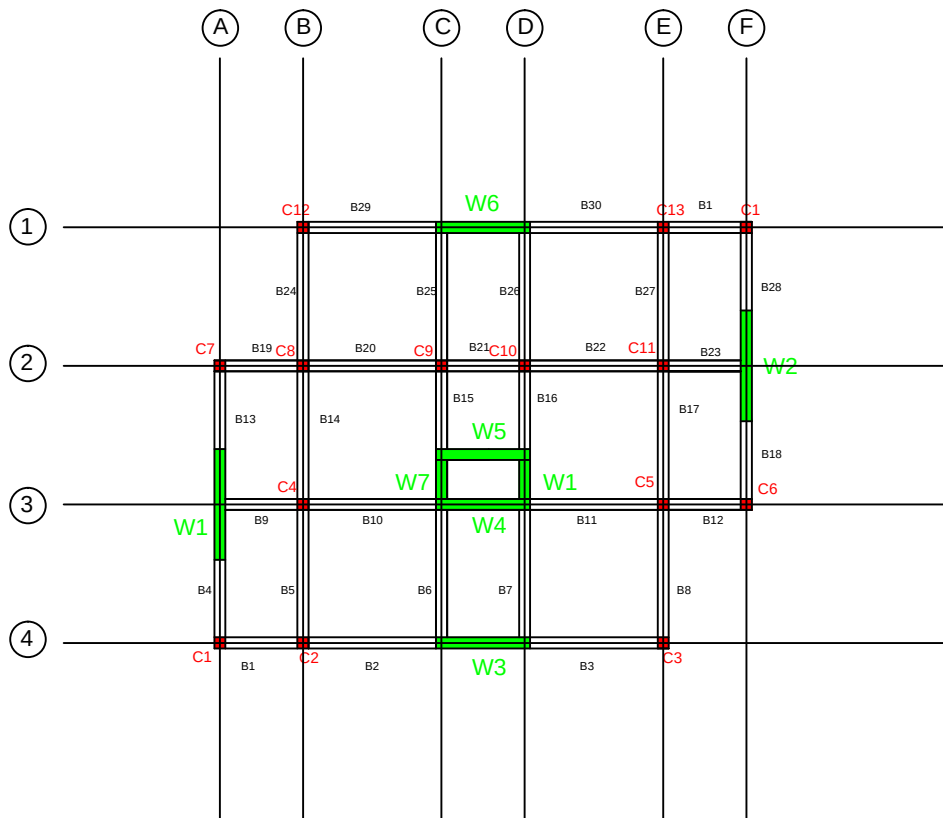
CONCLUSION:

In this project, we designed 4 story building with a dual system in the Udine region with soil type C with exposure coefficient XC1 under different loading conditions. Our site is in seismic zone II. All the members are designed under earthquake-resistant conditions. Results were carried out by MIDAS software sections are verified by VCALSU Software by Gelfi later the same model is designed again on ETABS to check the result and it is almost the same. We designed two strips of beams one in a horizontal direction other in vertical direction, 3 Columns one, corner, one external, and one internal column were designed and 8 shear walls which include 4 on the periphery and 4 for the core of the building which will be used for elevators. All the Eurocodes provisioned are followed and steel reinforcement for bending and shear are calculated with unit volume for all the members. Detailed steel reinforcement is shown in the drawing attached below.

REFERENCES:

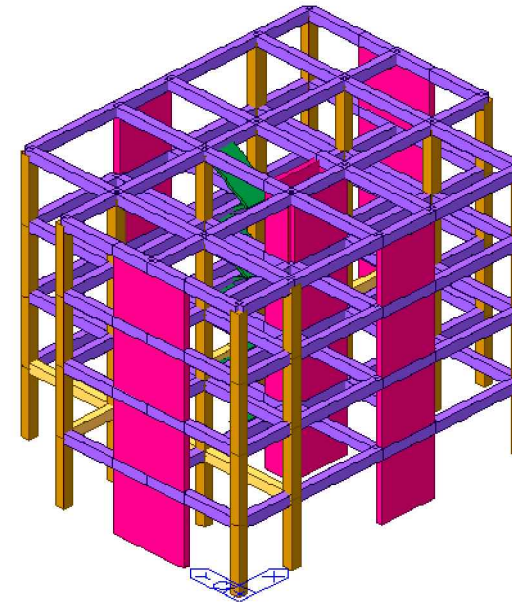
- “sismicita_motore_out @ italia.indettaglio.it.” [Online]. Available: http://italia.indettaglio.it/eng/sismicita/sismicita_motore_out.html?sigla_p=UD&valore=1
- “EUROCODE 8,” vol. 1, no. 2005, 2011.
- I. M. of I. and NTC, “Norme Tecniche per le Costruzioni. DM 17/1/2018,” *Gazz. Uff. della Repubb. Ital.*, pp. 1–198, 2018.
- MIDAS Information Technology Co., MIDAS Gen On-Line Manual. CSPFEA Engineering Solutions.
- EN 1998-1,3
- Atti della Giunta Regionale, Regione Siciliana, Deliberazione n. 408 del 19 Dicembre 2003
- Marco DePisapia, Articoli Risorse Applicazioni per progettisti strutturali, “Gerarchia delle resistenze: qual è il suo scopo (e perché manda in crisi i pilastri piatti)

FLOOR PLAN



GENERAL CONDITIONS

- CONCRETE CLASS: C35/45
- STEEL TYPE: B450C
- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: ii
- SOIL TYPE : 'C'



3D Model

M.S. CIVIL ENGINEERING
DESIGN OF NEW STRUCTURES



**POLITECNICO
MILANO 1863**

EARTHQUAKE ENGINEERING
ACADEMIC YEAR

(2023-2024)

ARCHITECTURAL DRAWINGS

ELEV : -----

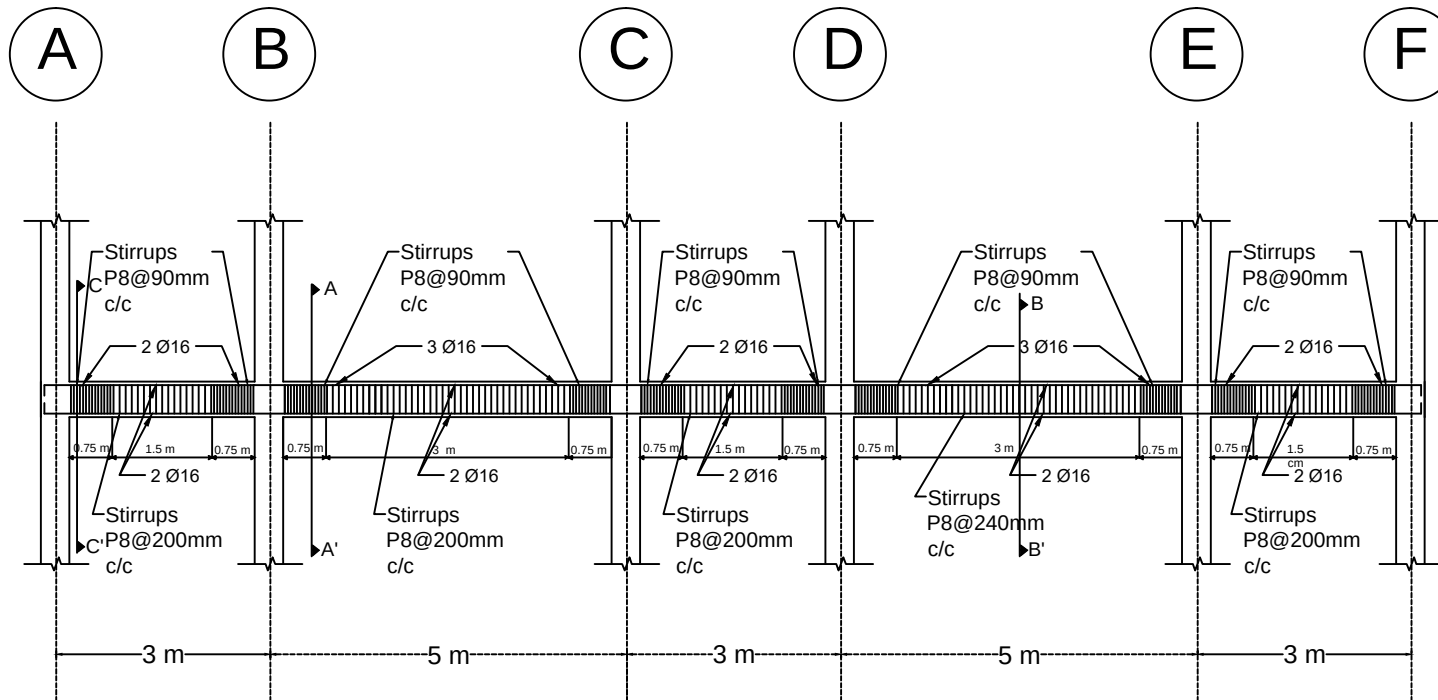
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HASEEB ALI
222834

ZIL E HASNAIN
995969

VASU SHARMA
217927

SUBMITTED TO:
PROF. LUCA MARTINELLI
PROF. CLAUDIO CHESI

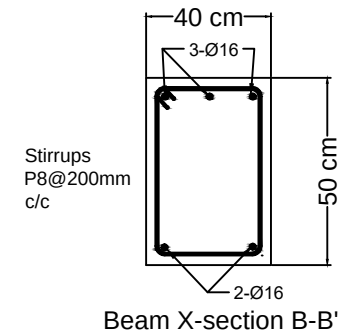
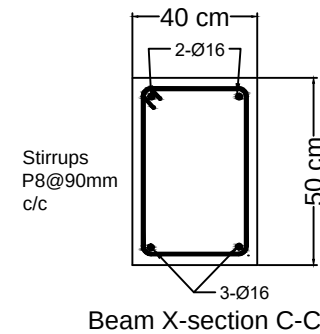
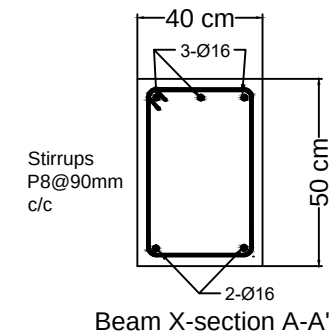
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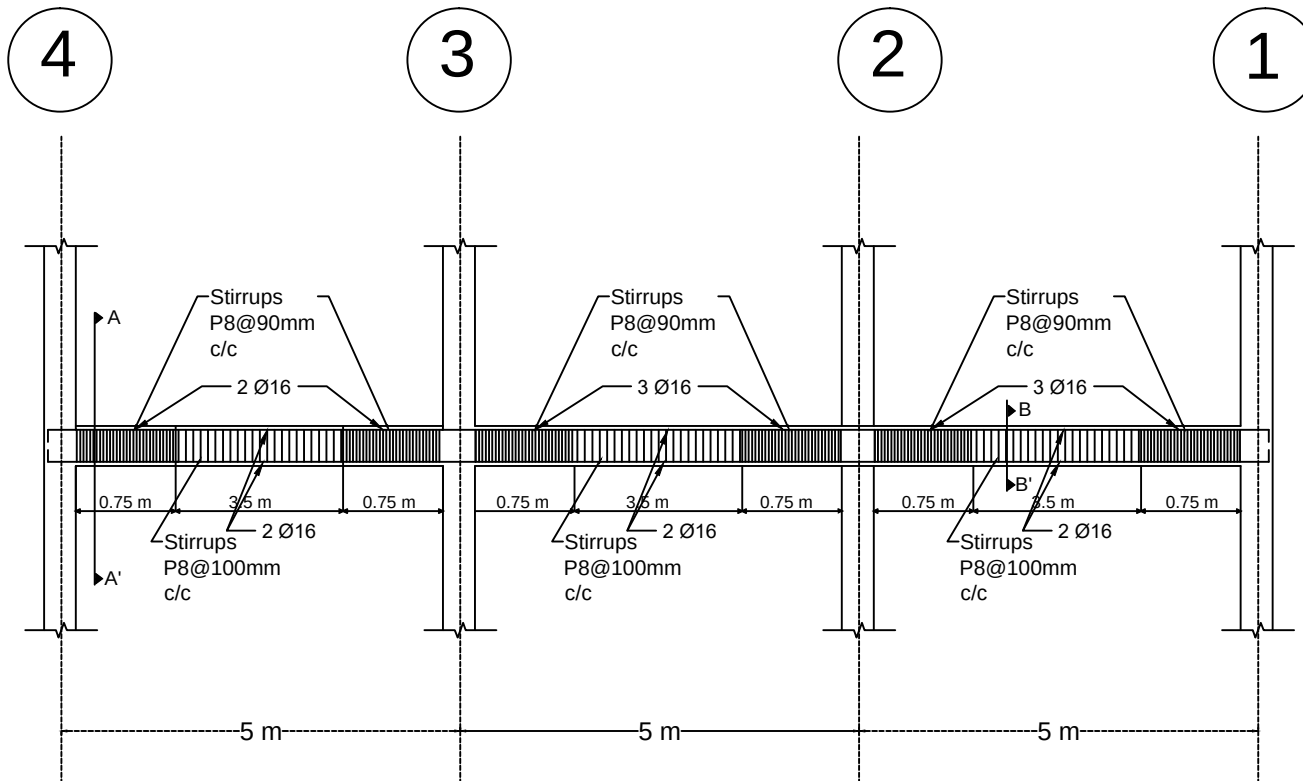


LONGITUDINAL BEAM

GENERAL CONDITIONS

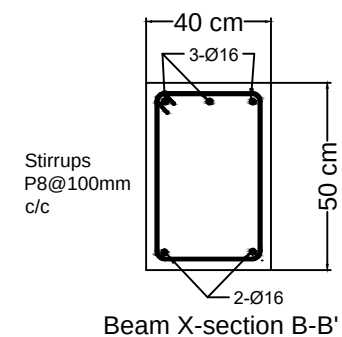
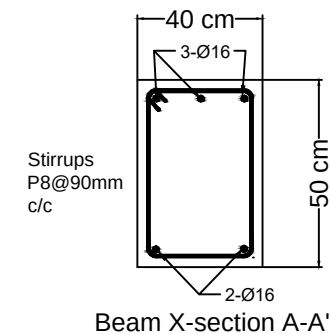
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- STEEL TYPE: B450C
- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: ii
- SOIL TYPE : 'C'





GENERAL CONDITIONS

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STRUCTURAL DRAWINGS
TRANSVERSE BEAMS
ELEV : ----

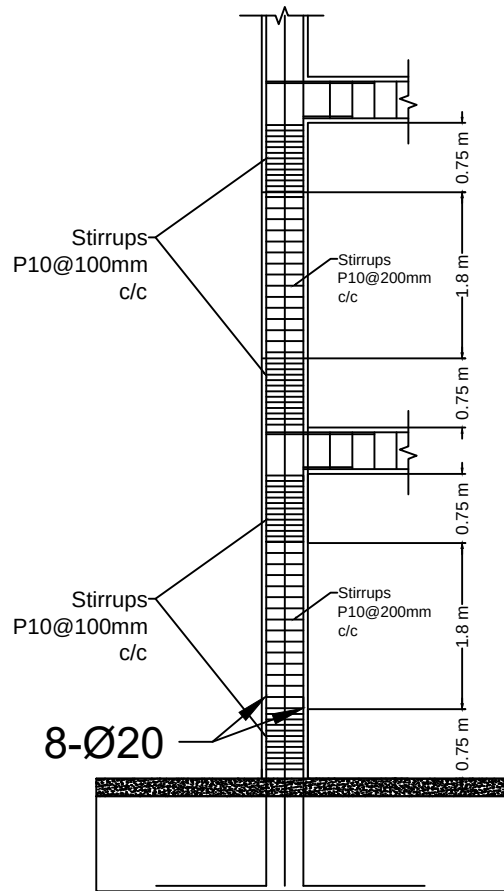
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222834

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995969

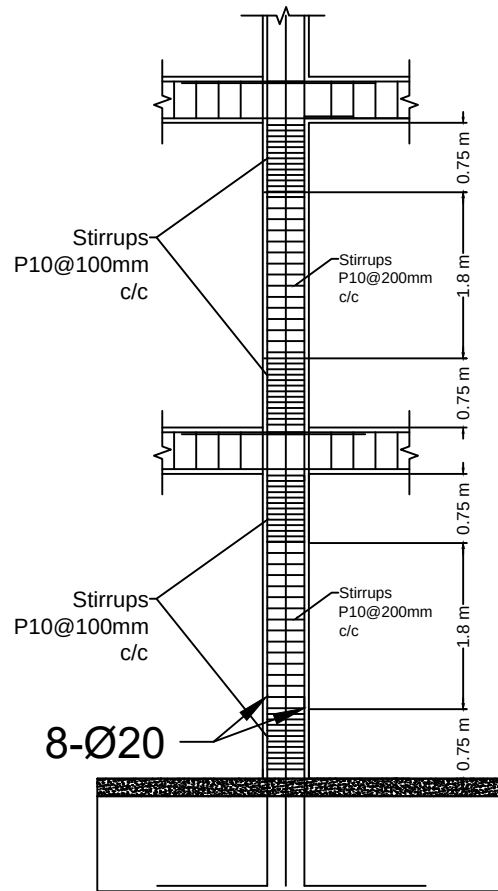
VASU SHARMA
217927

SUBMITTED TO:
PROF. LUCA MARTINELLI
PROF. CLAUDIO CHESI

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7



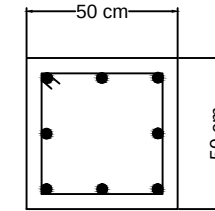
Column 1



Column 2,9

GENERAL CONDITIONS

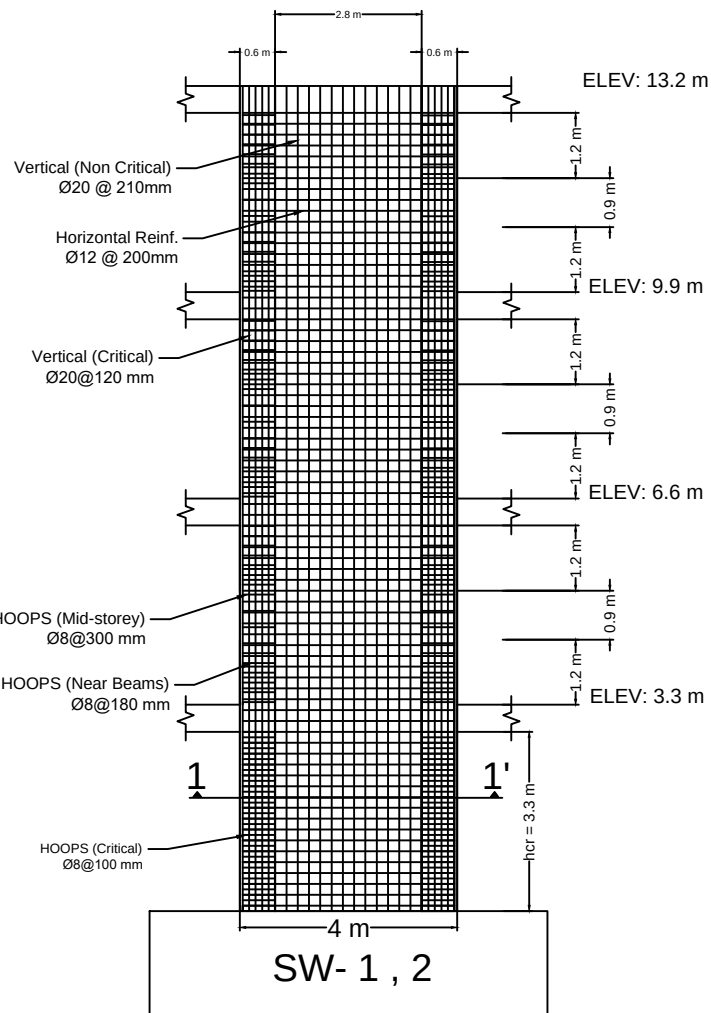
- CONCRETE CLASS: C35/45
- STEEL TYPE: B450C
- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: ii
- SOIL TYPE : 'C'



COLUMN X-SECTION
(1,2,9)

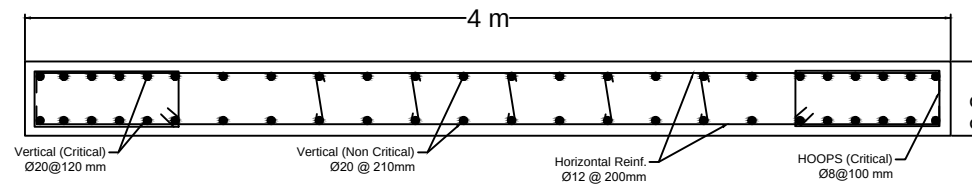
Columns	Nrd (KN)	Mrd (Col), x (KNm)	Mrd (Col), y (KNm)	Vertical Reinf.	Stirrups	
					Critical Region (lcr = 0.75m)	Non Critical Region
1	5940.45	190.00	126.30	8Ø20	Ø8@100	Ø8@200
2	5940.45	196.90	104.70	8Ø20	Ø8@100	Ø8@200
3	5940.45	202.30	99.58	8Ø20	Ø8@100	Ø8@200





GENERAL CONDITIONS

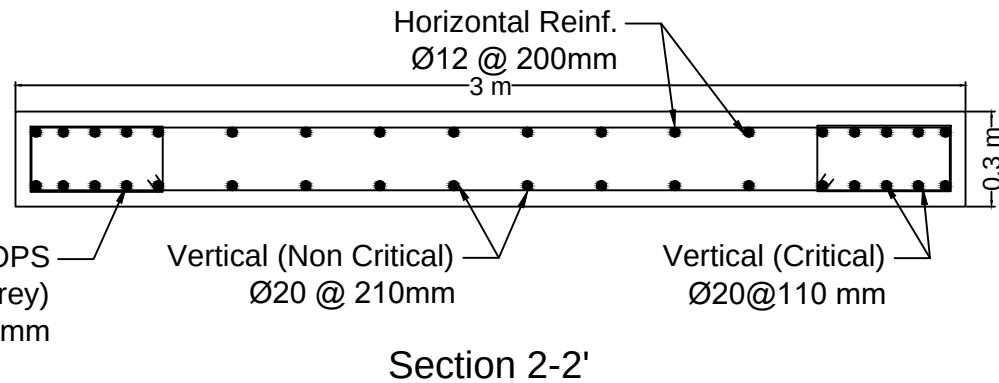
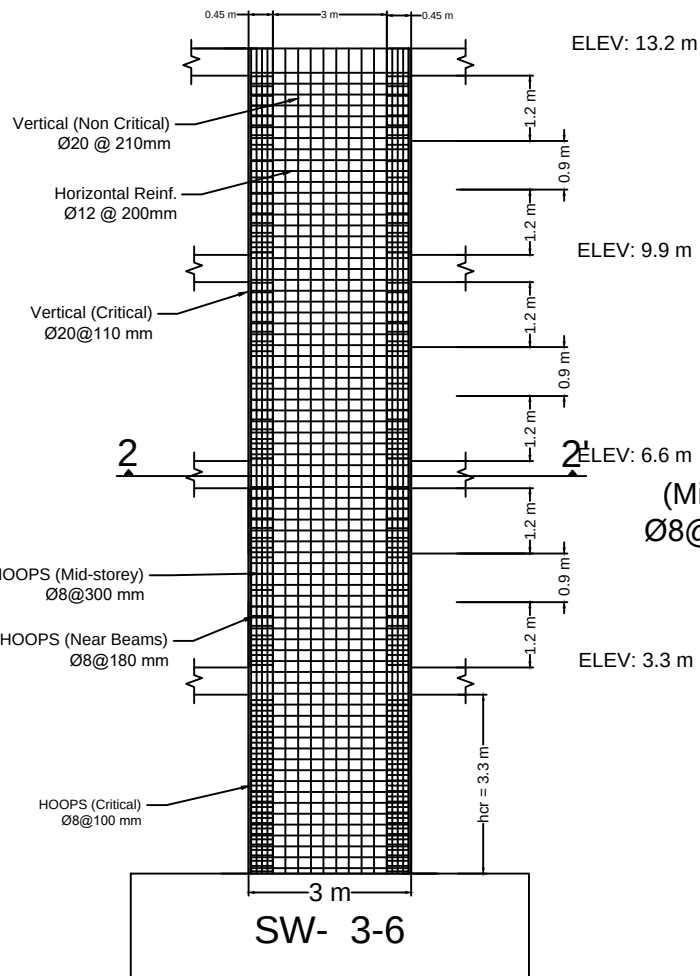
- CONCRETE CLASS: C35/45
- STEEL TYPE: B450C
- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: ii
- SOIL TYPE : 'C'



section 1-1'

Shear Wall	Nmax (KN)	Mmax (KNm)	Mrd (KNm)	Vertical Reinf.		Horizontal Reinf.	Confining hoops		
				Critical Zones	Non-Critical		Critical Region	Near Beams (1.2m)	Other (Mid-storeys) (0.9m)
1	577.454	3400.64	10706	Ø20@120	Ø20@210	Ø12@200	Ø8@100	Ø8@180	Ø8@300
2	602.009	2992.77	10753	Ø20@120	Ø20@210	Ø12@200	Ø8@100	Ø8@180	Ø8@300



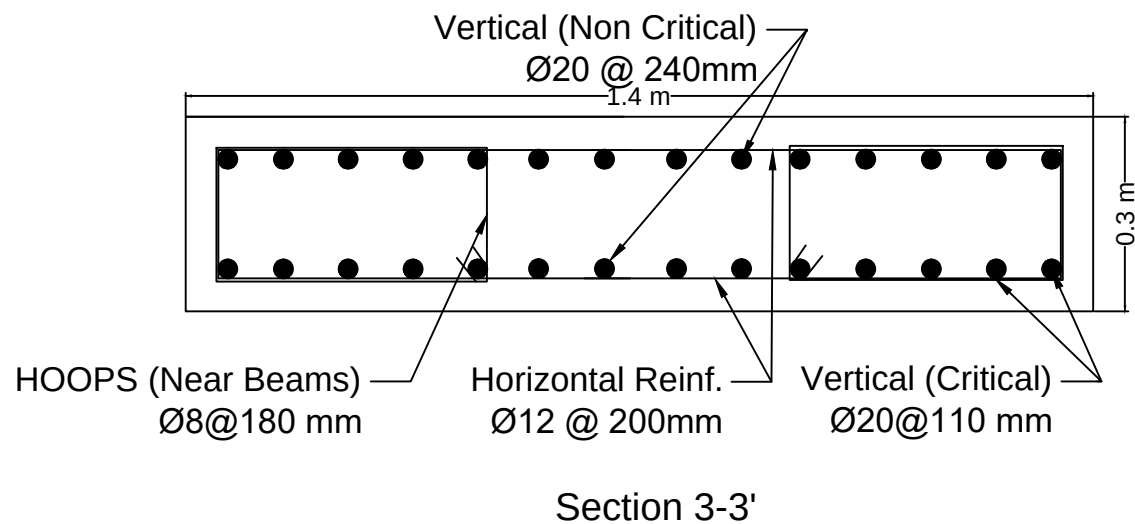
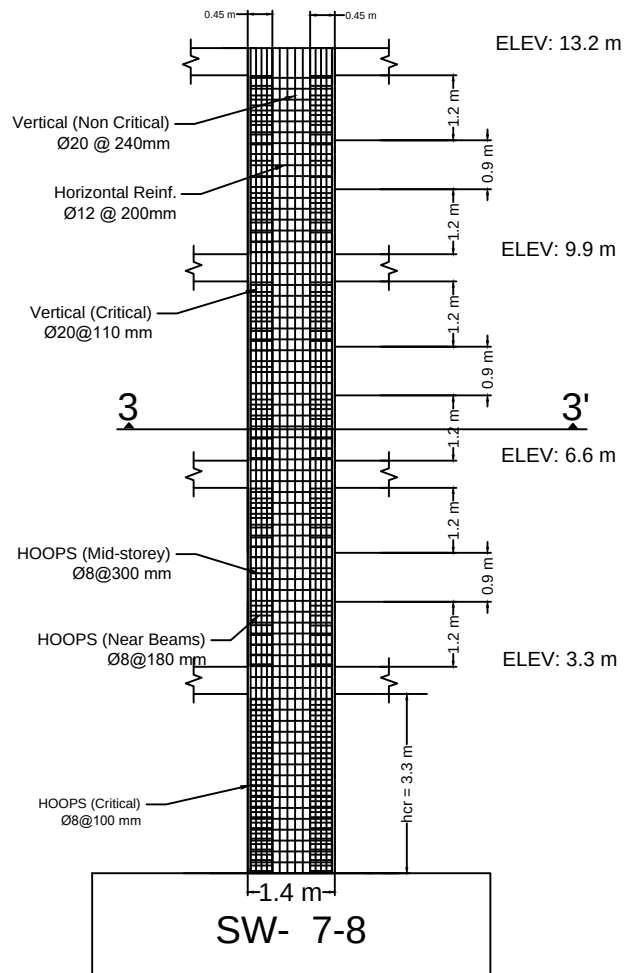


GENERAL CONDITIONS

- CONCRETE CLASS: C35/45
- STEEL TYPE: B450C
- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: ii
- SOIL TYPE : 'C'

Shear Wall	Nmax (KN)	Mmax (KNm)	Mrd (KNm)	Vertical Reinf.		Horizontal Reinf.	Confining hoops		
				Critical Zones	Non-Critical		Critical Region	Near Beams (1.2m)	Other (Mid-storeys) (0.9m)
3	946.667	1499.67	6688	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300
4	314.351	1479.91	5804	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300
5	542.099	1513.75	6131	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300
6	918.657	1469.62	6652	Ø20@110	Ø20@230	Ø12@200	Ø8@100	Ø8@180	Ø8@300





GENERAL CONDITIONS

- CONCRETE CLASS: C35/45
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- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: ii
- SOIL TYPE : 'C'

Shear Wall	Nmax (KN)	Mmax (KNm)	Mrd (KNm)	Vertical Reinf.		Horizontal Reinf.	Confining hoops		
				Critical Zones	Non-Critical		Critical Region	Near Beams (1.2m)	Other (Mid-storeys) (0.9m)
7	574.62	133.538	1652	Ø20@110	Ø20@240	Ø12@200	Ø8@100	Ø8@180	Ø8@300
8	603.04	155.951	1669	Ø20@110	Ø20@240	Ø12@200	Ø8@100	Ø8@180	Ø8@300

